

upvoted 1 times

pentium75 2 years, 5 months ago

Requirement is backup, not migration.

upvoted 3 times

gold4otas 3 years, 2 months ago

Selected Answer: A

A. Create a backup vault in AWS Backup to retain RDS backups. Create a new backup plan with a daily schedule and an expiration period of 2 years after creation. Assign the RDS DB instances to the backup plan.

upvoted 2 times

techhb 3 years, 4 months ago

Selected Answer: A

A is right choice

upvoted 4 times

Aninina 3 years, 4 months ago

Selected Answer: A

A A A A A A

upvoted 3 times

mhmt4438 3 years, 4 months ago

Selected Answer: A

Correct answer is A

upvoted 3 times

A company's compliance team needs to move its file shares to AWS. The shares run on a Windows Server SMB file share. A self-managed on-premises Active Directory controls access to the files and folders.

The company wants to use Amazon FSx for Windows File Server as part of the solution. The company must ensure that the on-premises Active Directory groups restrict access to the FSx for Windows File Server SMB compliance shares, folders, and files after the move to AWS. The company has created an FSx for Windows File Server file system.

Which solution will meet these requirements?

- A. Create an Active Directory Connector to connect to the Active Directory. Map the Active Directory groups to IAM groups to restrict access.
- B. Assign a tag with a Restrict tag key and a Compliance tag value. Map the Active Directory groups to IAM groups to restrict access.
- C. Create an IAM service-linked role that is linked directly to FSx for Windows File Server to restrict access.
- D. Join the file system to the Active Directory to restrict access. **Most Voted**

Correct Answer: D

Community vote distribution

D (90%)

10%

Comments

mhmt4438 **Highly Voted** 3 years, 4 months ago

Selected Answer: D

D. Join the file system to the Active Directory to restrict access.

Joining the FSx for Windows File Server file system to the on-premises Active Directory will allow the company to use the existing Active Directory groups to restrict access to the file shares, folders, and files after the move to AWS. This option allows the company to continue using their existing access controls and management structure, making the transition to AWS more seamless.

upvoted 23 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: D

D. allows the file system to leverage the existing AD infrastructure for authentication and access control.

Option A is incorrect because mapping the AD groups to IAM groups is not applicable in this scenario. IAM is primarily used for managing access to AWS resources, while the requirement is to integrate with the on-premises AD for access control.

Option B is incorrect because assigning a tag with a Restrict tag key and a Compliance tag value does not provide the necessary integration with the on-premises AD for access control. Tags are used for organizing and categorizing resources and do not provide authentication or access control mechanisms.

Option C is incorrect because creating an IAM service-linked role linked directly to FSx for Windows File Server does not integrate with the on-premises AD. IAM roles are used within AWS for managing permissions and do not provide the necessary integration with external AD systems.

upvoted 9 times

zdi561 **Most Recent** 1 year, 4 months ago

Selected Answer: D

<https://aws.amazon.com/blogs/storage/using-amazon-fsx-for-windows-file-server-with-an-on-premises-active-directory/#:~:text=Perform%20the%20following%20steps;shown%20in%20the%20following%20screenshot>

upvoted 1 times

Lin878 2 years ago

Selected Answer: D

When you create a file system with Amazon FSx, you join it to your Active Directory domain to provide user authentication and file- and folder-level access control.

upvoted 4 times

bujuman 2 years, 3 months ago

Selected Answer: D

D is relevant and accurate answer when we consider this:

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/creating-joined-ad-file-systems.html>

"When you create a new FSx for Windows File Server file system, you can configure Microsoft Active Directory integration so that it joins to your self-managed Microsoft Active Directory domain. To do this, provide the following information for your Microsoft Active Directory"

upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

The on-premise AD already has restrictions via group in place so D makes no sense as the groups are already linked to file system.

"The company must ensure that the on-premises Active Directory groups restrict access to the FSx for Windows File Server SMB compliance shares, folders, and files after the move to AWS."

The question is about linking the on-prem permissions to the new FSx server on AWS and this can only be done by A

upvoted 1 times

awsgeek75 2 years, 4 months ago

Actually neither A nor D make sense.

"A self-managed on-premises Active Directory controls access to the files and folders." This makes D sound useless and at the same time does not allow the on-prem AD to control file access on FSx.

A uses IAM roles which is irrelevant to this setup.

BC are totally wrong

upvoted 1 times

meowruki 2 years, 6 months ago

Selected Answer: D

Option A: Creating an Active Directory Connector and mapping groups to IAM groups is more relevant for AWS Directory Service, such as AWS Managed Microsoft AD, and not for integrating with existing on-premises Active Directory.

Option B: Using tags is typically not used for access control purposes. Tags are metadata and are not directly involved in user authentication and authorization.

Option C: Creating an IAM service-linked role directly linked to FSx for Windows File Server is not the standard approach for integrating with existing on-premises Active Directory.

upvoted 4 times

wrmari 2 years, 8 months ago

Selected Answer: D

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/self-managed-AD.html>

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

This allows the on-premises Active Directory to manage permissions to the FSx file shares, meeting the key requirement to use existing AD groups to control access after migrating to AWS.

Joining FSx to the AD domain allows the native file system permissions, users, and groups to be applied from Active Directory. Access is handled seamlessly via the trust relationship between FSx and AD.

The other options would not leverage the existing AD identities and groups

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

The other options would not leverage the existing AD identities and groups:

A) AD Connector and IAM groups would require re-mapping AD groups to IAM, adding complexity. Native AD integration is simpler.

B) Tags and IAM groups also don't use native AD semantics.

C) Service-linked roles are not applicable for managing end user access.

So D is the correct option to meet the requirements using the native Active Directory integration built into FSx for Windows.

upvoted 3 times

mtmayer 2 years, 10 months ago

Selected Answer: A

The AD is on-premises... You need the connector.

upvoted 2 times

Mia2009687 2 years, 11 months ago

Selected Answer: D

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/aws-ad-integration-fsxW.html>

upvoted 2 times

kraken21 3 years, 2 months ago

Selected Answer: D

Other options are referring to IAM based control which is not possible. Existing AD should be used without IAM.

upvoted 3 times

Abhineet9148232 3 years, 2 months ago

Selected Answer: D

<https://aws.amazon.com/blogs/storage/using-amazon-fsx-for-windows-file-server-with-an-on-premises-active-directory/>

upvoted 3 times

somsundar 3 years, 3 months ago

Answer D. Amazon FSx does not support Active Directory Connector .

upvoted 3 times

JA2018 1 year, 6 months ago

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/aws-ad-integration-fsxW.html>

upvoted 1 times

Abhineet9148232 3 years, 3 months ago

Selected Answer: D

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/self-managed-AD.html>

upvoted 4 times

Yelizaveta 3 years, 3 months ago

Selected Answer: D

Note:

Amazon FSx does not support Active Directory Connector and Simple Active Directory.

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/aws-ad-integration-fsxW.html>

upvoted 4 times

aakashkumar1999 3 years, 4 months ago

Selected Answer: A

The answer will be AD connector so : A, it will create a proxy between your onpremises AD which you can use to restrict access

upvoted 2 times

JA2018 1 year, 6 months ago

from: <https://docs.aws.amazon.com/fsx/latest/WindowsGuide/aws-ad-integration-fsxW.html>

Amazon FSx does not support Active Directory Connector and Simple Active Directory.

upvoted 1 times

A company recently announced the deployment of its retail website to a global audience. The website runs on multiple Amazon EC2 instances behind an Elastic Load Balancer. The instances run in an Auto Scaling group across multiple Availability Zones.

The company wants to provide its customers with different versions of content based on the devices that the customers use to access the website.

Which combination of actions should a solutions architect take to meet these requirements? (Choose two.)

A. Configure Amazon CloudFront to cache multiple versions of the content. **Most Voted**

B. Configure a host header in a Network Load Balancer to forward traffic to different instances.

C. Configure a Lambda@Edge function to send specific objects to users based on the User-Agent header. **Most Voted**

D. Configure AWS Global Accelerator. Forward requests to a Network Load Balancer (NLB). Configure the NLB to set up host-based routing to different EC2 instances.

E. Configure AWS Global Accelerator. Forward requests to a Network Load Balancer (NLB). Configure the NLB to set up path-based routing to different EC2 instances.

Correct Answer: AC

Community vote distribution

AC (100%)

Comments

Parsons **Highly Voted** 2 years, 11 months ago

Selected Answer: AC

A, C is correct.

NLB listener rule only supports Protocol & Port (Not host/based routing like ALB) => D, E is incorrect.
NLB just works layer 4 (TCP/UDP) instead of Layer 7 (HTTP) => B is incorrect.

After eliminating, AC should be the answer.

upvoted 19 times

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: AC

A. allows customers to receive the appropriate version of the content based on their location and device type.

C. By creating a Lambda@Edge, you can inspect the User-Agent header of incoming requests and determine the type of device being used. Based on this information, you can customize the response and send the appropriate version of the content to the user.

upvoted 11 times

Ruffyt **Most Recent** 2 years ago

A C

Configure Amazon CloudFront to cache multiple versions of the content.

Configure a function to send specific objects to users based on the User-Agent header.

upvoted 3 times

sunhouse 2 years, 1 month ago

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/header-caching.html>

upvoted 3 times

rrbrish73 2 years, 2 months ago

<https://medium.com/swlh/serve-different-content-based-on-user-agent-in-aws-cloudfront-using-lambda-edge-28877294340b>

upvoted 2 times

cookieMr 2 years, 5 months ago

Selected Answer: AC

A. allows customers to receive the appropriate version of the content based on their location and device type.

C. By creating a Lambda@Edge, you can inspect the User-Agent header of incoming requests and determine the type of device being used. Based on this information, you can customize the response and send the appropriate version of the content to the user.

B. does not address the requirement of serving different content versions based on device types.

D. & E. do not address the device-specific content requirement.

Therefore, options A and C are the correct combination of actions to meet the requirement of providing different versions of content based on the devices that customers use to access the website.

upvoted 6 times

Yadav_Sanjay 2 years, 6 months ago

Selected Answer: AC

NLB does not supports routing

upvoted 3 times

omoakin 2 years, 7 months ago

A C

Configure Amazon CloudFront to cache multiple versions of the content.

Configure a function to send specific objects to users based on the User-Agent header.

upvoted 2 times

omoakin 2 years, 7 months ago

C

Configure a function to send specific objects to users based on the User-Agent header.

upvoted 1 times

GalileoEC2 2 years, 8 months ago

Using a Directory Connector to connect the on-premises Active Directory to AWS is one way to enable access to AWS resources, including Amazon FSx for Windows File Server. However, joining the Amazon FSx for Windows File Server file system to the on-premises Active Directory is a separate step that allows you to control access to the file shares using the same Active Directory groups that are used on-premises.

upvoted 1 times

LoXeras 2 years, 8 months ago

I guess this belongs to the question before #260

upvoted 3 times

wors 2 years, 10 months ago

So will this mean the entire architecture needs to move to lambda in order to leverage off lambda edge? This doesn't make sense as the question outlines the architecture already in ec2, asg and elb?

Just looking for clarification if I am missing something

upvoted 2 times

pentium75 1 year, 11 months ago

No, Lamba function will just do something like "if user-agent is 'iOS' then send content from elb-01 otherwise send content from elb-02".

upvoted 2 times

devonwho 2 years, 10 months ago

Selected Answer: AC

AC are the correct answers.

For C:

IMPROVED USER EXPERIENCE

Lambda@Edge can help improve your users' experience with your websites and web applications across the world, by letting you personalize content for them without sacrificing performance.

Real-time Image Transformation

You can customize your users' experience by transforming images on the fly based on the user characteristics. For example, you can resize images based on the viewer's device type—mobile, desktop, or tablet. You can also cache the transformed images at CloudFront Edge locations to further improve performance when delivering images.

<https://aws.amazon.com/lambda/edge/>

upvoted 3 times

mhmt4438 2 years, 11 months ago

Selected Answer: AC

Correct answer is A,C

upvoted 4 times

Aninina 2 years, 11 months ago

Selected Answer: AC

C. Configure a Lambda@Edge function to send specific objects to users based on the User-Agent header.

Lambda@Edge allows you to run a Lambda function in response to specific CloudFront events, such as a viewer request, an origin request, a response, or a viewer response.

upvoted 3 times

Morinator 2 years, 11 months ago

Selected Answer: AC

<https://www.examttopics.com/discussions/amazon/view/67881-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 4 times

A company plans to use Amazon ElastiCache for its multi-tier web application. A solutions architect creates a Cache VPC for the ElastiCache cluster and an App VPC for the application's Amazon EC2 instances. Both VPCs are in the us-east-1 Region.

The solutions architect must implement a solution to provide the application's EC2 instances with access to the ElastiCache cluster.

Which solution will meet these requirements MOST cost-effectively?

A. Create a peering connection between the VPCs. Add a route table entry for the peering connection in both VPCs. Configure an inbound rule for the ElastiCache cluster's security group to allow inbound connection from the application's security group.

Most Voted

B. Create a Transit VPC. Update the VPC route tables in the Cache VPC and the App VPC to route traffic through the Transit VPC. Configure an inbound rule for the ElastiCache cluster's security group to allow inbound connection from the application's security group.

C. Create a peering connection between the VPCs. Add a route table entry for the peering connection in both VPCs. Configure an inbound rule for the peering connection's security group to allow inbound connection from the application's security group.

D. Create a Transit VPC. Update the VPC route tables in the Cache VPC and the App VPC to route traffic through the Transit VPC. Configure an inbound rule for the Transit VPC's security group to allow inbound connection from the application's security group.

Correct Answer: A

Community vote distribution

A (100%)

Comments

mhmt4438 **Highly Voted** 2 years, 4 months ago

Selected Answer: A

A. Create a peering connection between the VPCs. Add a route table entry for the peering connection in both VPCs. Configure an inbound rule for the ElastiCache cluster's security group to allow inbound connection from the application's security group.

Creating a peering connection between the VPCs allows the application's EC2 instances to communicate with the ElastiCache cluster directly and efficiently. This is the most cost-effective solution as it does not involve creating additional resources such as a Transit VPC, and it does not incur additional costs for traffic passing through the Transit VPC. Additionally, it is also more secure as it allows you to configure a more restrictive security group rule to allow inbound connection from only the application's security group.

upvoted 17 times

Ruffyt **Most Recent** 1 year, 7 months ago

A. Create a peering connection between the VPCs. Add a route table entry for the peering connection in both VPCs. Configure an inbound rule for the ElastiCache cluster's security group to allow inbound connection from the application's security group.

Creating a peering connection between the VPCs allows the application's EC2 instances to communicate with the ElastiCache cluster directly and efficiently. This is the most cost-effective solution as it does not involve creating additional resources such as a Transit VPC, and it does not incur additional costs for traffic passing through the Transit VPC. Additionally, it is also more secure as it allows you to configure a more restrictive security group rule to allow inbound connection from only the application's security group.

upvoted 3 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: A

Create a peering connection between the VPCs. Add a route table entry for the peering connection in both VPCs. Configure an inbound rule for the ElastiCache cluster's security group to allow inbound connection from the application's security group.

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

Create a VPC peering connection between the Cache VPC and App VPC. This allows private IP connectivity between the VPCs. Add route table entries in each VPC to route traffic destined to the other VPC via the peering connection. This enables network routing. Configure security groups to allow inbound connections from the application instances to the ElastiCache cluster.

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: A

Creating a peering connection between the VPCs is a cost-effective way to establish connectivity. By adding a route table entry for the peering connection in both VPCs, traffic can flow between them. Configuring an inbound rule in the ElastiCache cluster's security group allows inbound connections from the application's security group, enabling access to the ElastiCache cluster from the EC2 instances in the App VPC.

Option B suggests creating a Transit VPC, which adds unnecessary complexity and cost for this scenario.

Option C suggests configuring an inbound rule for the peering connection's security group, which is not necessary as the security group for the ElastiCache cluster should be used to control inbound connections.

Option D suggests configuring an inbound rule for the Transit VPC's security group, which is not needed in this case and adds unnecessary complexity.

Therefore, option A is the most cost-effective solution to provide the application's EC2 instances with access to the ElastiCache cluster.

upvoted 4 times

smartegnine 1 year, 12 months ago

Selected Answer: A

A is correct,

1. VPC transit is used for more complex architecture and can do VPCs to VPCs connectivity. But for simple VPC 2 VPC can use peer connection.
2. To enable private IPv4 traffic between instances in peered VPCs, you must add a route to the route tables associated with the subnets for both instances.

So base on 1, B and D are out, base on 2 C is out

upvoted 2 times

wRhIH 2 years ago

Why not C ? any explanation?

upvoted 1 times

smartegnine 1 year, 11 months ago

Application read from ElasticCache, not viseversa, so inbound rule should be ElasticCach

upvoted 3 times

Cor5in 1 year, 11 months ago

Thank you Sir!

upvoted 1 times

smartegnine 1 year, 12 months ago

To enable private IPv4 traffic between instances in peered VPCs, you must add a route to the route tables associated with the subnets for both instances.

<https://docs.aws.amazon.com/vpc/latest/peering/vpc-peering-routing.html>

upvoted 3 times

nder 2 years, 3 months ago

Selected Answer: A

Cost Effectively!

upvoted 2 times

A company is building an application that consists of several microservices. The company has decided to use container technologies to deploy its software on AWS. The company needs a solution that minimizes the amount of ongoing effort for maintenance and scaling. The company cannot manage additional infrastructure.

Which combination of actions should a solutions architect take to meet these requirements? (Choose two.)

- A. Deploy an Amazon Elastic Container Service (Amazon ECS) cluster. **Most Voted**
- B. Deploy the Kubernetes control plane on Amazon EC2 instances that span multiple Availability Zones.
- C. Deploy an Amazon Elastic Container Service (Amazon ECS) service with an Amazon EC2 launch type. Specify a desired task number level of greater than or equal to 2.
- D. Deploy an Amazon Elastic Container Service (Amazon ECS) service with a Fargate launch type. Specify a desired task number level of greater than or equal to 2. **Most Voted**
- E. Deploy Kubernetes worker nodes on Amazon EC2 instances that span multiple Availability Zones. Create a deployment that specifies two or more replicas for each microservice.

Correct Answer: AD

Community vote distribution

AD (100%)

Comments

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: AD

Options B and E suggest deploying the Kubernetes control plane and worker nodes on EC2 instances, which would require managing the infrastructure and add ongoing maintenance overhead, contrary to the requirement of minimizing effort.

Option C suggests using the Amazon EC2 launch type for ECS, which still requires managing EC2 instances and is not as cost-effective and scalable as using Fargate.

Therefore, the combination of deploying an Amazon ECS cluster and an ECS service with a Fargate launch type (options A and D) is the most suitable for minimizing maintenance and scaling effort without managing additional infrastructure.

upvoted 9 times

AlessandraSAA **Highly Voted** 3 years, 3 months ago

Selected Answer: AD

ECS has 2 launch type, EC2 (you maintain the infra) and Fargate (serverless). Since the question asks for no additional infra to manage it should be Fargate.

upvoted 5 times

TariqKipkemei **Most Recent** 2 years, 8 months ago

Selected Answer: AD

Company needs a solution that minimizes the amount of ongoing effort for maintenance and scaling = Serverless = ECS with Fargate.

upvoted 4 times

JA2018 1 year, 6 months ago

Rule of thumbs for AWS exams, always opt for AWS managed services

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AD

ECS allows deploying and managing containers without having to provision the underlying infrastructure. This minimizes maintenance effort. Using Fargate launch type means ECS will handle provisioning and scaling the infrastructure automatically. This removes the management overhead for the company.

Running multiple tasks and specifying desired count ≥ 2 will provide high availability across Availability Zones. Together, ECS plus Fargate provide a fully managed container platform. The company doesn't need to provision or manage servers.

upvoted 4 times

LoXeras 3 years, 2 months ago

Selected Answer: AD

AWS Fargate is server less solution to use on ECS: https://docs.aws.amazon.com/AmazonECS/latest/developerguide/AWS_Fargate.html

upvoted 3 times

lambda15 3 years, 2 months ago

why is c is incorrect ?

upvoted 1 times

Julio98 3 years, 2 months ago

Because in the question says, "minimizes the amount of ongoing effort for maintenance and scaling", and EC2 instances you need effort to maintain the infrastructure unlike fargate that is serverless.

upvoted 3 times

WheracanIstart 3 years, 3 months ago

Selected Answer: AD

Amazon Fargate is a service that is fully manageable by Amazon; it offers provisioning, configuration and scaling feature. It is "serverless"..

upvoted 2 times

devonwho 3 years, 4 months ago

Selected Answer: AD

AWS Fargate is a technology that you can use with Amazon ECS to run containers without having to manage servers or clusters of Amazon EC2 instances. With Fargate, you no longer have to provision, configure, or scale clusters of virtual machines to run containers.

<https://docs.aws.amazon.com/AmazonECS/latest/userguide/what-is-fargate.html>

upvoted 4 times

Aninina 3 years, 4 months ago

A D is the correct answer

upvoted 2 times

mhmt4438 3 years, 4 months ago

Selected Answer: AD

A,D is correct answer

upvoted 3 times

AHUI 3 years, 4 months ago

AD:

<https://www.examttopics.com/discussions/amazon/view/60032-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 4 times

Morinator 3 years, 4 months ago

Selected Answer: AD

AD - EC2 out for this, cluster + fargate is the right answer

upvoted 5 times

A company has a web application hosted over 10 Amazon EC2 instances with traffic directed by Amazon Route 53. The company occasionally experiences a timeout error when attempting to browse the application. The networking team finds that some DNS queries return IP addresses of unhealthy instances, resulting in the timeout error.

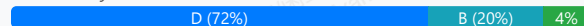
What should a solutions architect implement to overcome these timeout errors?

- A. Create a Route 53 simple routing policy record for each EC2 instance. Associate a health check with each record.
- B. Create a Route 53 failover routing policy record for each EC2 instance. Associate a health check with each record.
- C. Create an Amazon CloudFront distribution with EC2 instances as its origin. Associate a health check with the EC2 instances.
- D. Create an Application Load Balancer (ALB) with a health check in front of the EC2 instances. Route to the ALB from Route 53.

Most Voted

Correct Answer: D

Community vote distribution



Comments

Guru4Cloud Highly Voted 2 years, 9 months ago

Selected Answer: D

ALB performs health checks on the EC2 instances, so it will only route traffic to healthy instances. This avoids the timeout errors. ALB provides load balancing across the instances, improving performance and availability. Route 53 routes to the ALB DNS name, so you don't have to manage records for each EC2 instance. This is a standard and robust architecture for public-facing web applications. The ALB acts as the entry point and handles health checks and scaling.

upvoted 17 times

jlteunissen Highly Voted 2 years, 9 months ago

Selected Answer: B

It is not clear from the question whether the 10 EC2s are running within the same region. ALB can only direct traffic within region, while route 53 can route traffic to multiple locations, hence C and D are wrong.

upvoted 10 times

pentium75 2 years, 5 months ago

But B has one primary record and 9 failover records. A is correct, simple policy with health checks, that makes sure that only IPs of healthy instances are returned.

upvoted 3 times

JA2018 1 year, 6 months ago

err, can someone advise which is the correct answer?

upvoted 1 times

LeonSauveterre Most Recent 1 year, 6 months ago

Selected Answer: D

At first sight I thought option D is too complicated, but come to think of it, this should be the final answer. An ALB is specifically designed for this use case.

Option B is not suited for scenarios with multiple EC2 instances in the same environment because it works on a primary-secondary model, not load balancing across multiple instances.

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: D

D for sure

upvoted 2 times

Hrishi_707 2 years, 3 months ago

Those who are confused between A and D, A is wrong as you can't associate a health check with Simple routing policy record.
upvoted 2 times

MrPCarrot 2 years, 3 months ago

D is the best answer
upvoted 2 times

awsgeek75 2 years, 4 months ago

If you focus on the question, both A and D seems to be correct.

A is correct because simple routing policy for health check is doable BUT it is also wrong because we don't know how to determine the health of instance.

D is correct because "The company occasionally experiences a timeout error when attempting to browse the application" which suggest application is being accessed by a browser with means it's HTTP based and ALB is better for HTTP based healthchecks.

A web application timing out is not necessarily unhealthy instance, strictly speaking. It's just bad web application running on a healthy instance! So A may not be correct also.

upvoted 4 times

farnamjam 2 years, 5 months ago

Selected Answer: D

Although B can work as well, but it's not a professional choice to associate the healthcheck with 10 EC2 instances, ALB is better option here.

A is incorrect: Simple Routing Policies Can't be associated with Health Checks

C is incorrect: Cloudfront is for caching content which is irrelevant.

upvoted 4 times

awsgeek75 2 years, 5 months ago

Simple routing policies can be associated with Health Checks <https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/dns-failover-simple-configs.html>

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: A

A meets the requirement ("overcome these timeout errors") without any other changes.

"If you configure health checking for all the records in a group of records that have the same name, the same type (such as A or AAAA), and the same routing policy (such as weighted or failover), Route 53 responds to DNS queries by choosing a healthy record and returning the applicable value from that record. (<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/health-checks-how-route-53-chooses-records.html>)

upvoted 3 times

pentium75 2 years, 5 months ago

"You can use Route 53 health checking to configure active-active and active-passive failover configurations. You configure active-active failover using any routing policy (or combination of routing policies) other than failover. ... Use this failover configuration when you want all of your resources to be available the majority of the time. When a resource becomes unavailable, Route 53 can detect that it's unhealthy and stop including it when responding to queries.

upvoted 2 times

pentium75 2 years, 5 months ago

In active-active failover, all the records that have the same name, the same type (such as A or AAAA), and the same routing policy (such as weighted or latency) are active unless Route 53 considers them unhealthy. Route 53 can respond to a DNS query using any healthy record." (<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/dns-failover-types.html#dns-failover-types-active-passive>)

upvoted 2 times

pentium75 2 years, 5 months ago

B - No, "you configure active-active failover using any routing policy (or combination of routing policies) OTHER THAN FAILOVER". With B, all traffic would go one primary instance unless it is unhealthy.

C - Not sure how to configure multiple EC2 instances as the origin without an LB. Even if that would be possible it would introduce more changes and complexity, which is not asked for.

D - Would work if all EC2 instances are in the same region, which we don't know. But it would also incur additional cost and potentially have other effects.

upvoted 2 times

Ruffyt 2 years, 7 months ago

B is wrong.

The DNS cache in clients could drive to timeouts. With ALB this issue won't happen since the DNS register will be the same and ALB will take care of unhealthy nodes.

upvoted 3 times

rlamberti 2 years, 7 months ago

Selected Answer: D

B is wrong.

The DNS cache in clients could drive to timeouts. With ALB this issue won't happen since the DNS register will be the same and ALB will take care of unhealthy nodes.

upvoted 3 times

daniel1 2 years, 8 months ago

Selected Answer: D

D. ****Application Load Balancer (ALB) with Health Checks, Routed via Route 53****:

- Creating an ALB in front of the EC2 instances and configuring health checks on the ALB will ensure that only healthy instances receive traffic. Route 53 can then direct traffic to the ALB, which in turn, routes traffic to healthy instances based on the health check results.

Among the provided options, the one that directly addresses the issue of routing traffic only to healthy instances is:

****D. Create an Application Load Balancer (ALB) with a health check in front of the EC2 instances. Route to the ALB from Route 53.****

upvoted 4 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: B

Clearly the question is all about Amazon Route 53 that has Failover routing policy that is used when you want to configure active-passive failover.

upvoted 1 times

slackbot 2 years, 9 months ago

i was looking at A, but indeed D is the best option, because the usually the TTL of the records is at least 60 seconds (nobody sets lower unless testing something ,because there is a charge per number of unique requests. ALB health check can be set as low as desired, which helps exclude the problematic ec2 faster than the DNS TTL expires

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: D

By creating an ALB and configuring health checks, the architect ensures that only healthy instances receive traffic. The ALB periodically checks the health of the EC2 instances based on the configured health check settings.

Routing traffic to the ALB from Route 53 ensures that DNS queries return the IP address of the ALB instead of individual instances. This allows the ALB to distribute traffic only to healthy instances, avoiding timeouts caused by unhealthy instances.

A & B: While associating health checks with each record can help identify unhealthy instances, it does not provide automatic load balancing and distribution of traffic to healthy instances.

C: While CloudFront can improve performance and availability, it is primarily a CDN and may not directly address the issue of load balancing and distributing traffic to healthy instances.

Therefore, option D is the most appropriate solution to overcome the timeout errors by implementing an ALB with health checks and routing traffic through Route 53.

upvoted 5 times

joechen2023 2 years, 11 months ago

Selected Answer: C

I believe both C and D will work, but C seems less complex.

hopefully somebody here is more advanced(not an old student learning AWS like me) to explain why not C.

upvoted 3 times

Abrar2022 3 years ago

Selected Answer: D

Option D allows for the creation of an Application Load Balancer which can detect unhealthy instances and redirect traffic away from them.

upvoted 3 times

A solutions architect needs to design a highly available application consisting of web, application, and database tiers. HTTPS content delivery should be as close to the edge as possible, with the least delivery time.

Which solution meets these requirements and is MOST secure?

- A. Configure a public Application Load Balancer (ALB) with multiple redundant Amazon EC2 instances in public subnets. Configure Amazon CloudFront to deliver HTTPS content using the public ALB as the origin.
- B. Configure a public Application Load Balancer with multiple redundant Amazon EC2 instances in private subnets. Configure Amazon CloudFront to deliver HTTPS content using the EC2 instances as the origin.
- C. Configure a public Application Load Balancer (ALB) with multiple redundant Amazon EC2 instances in private subnets. Configure Amazon CloudFront to deliver HTTPS content using the public ALB as the origin. **Most Voted**
- D. Configure a public Application Load Balancer with multiple redundant Amazon EC2 instances in public subnets. Configure Amazon CloudFront to deliver HTTPS content using the EC2 instances as the origin.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Aninina **Highly Voted** 3 years, 4 months ago

C. Configure a public Application Load Balancer (ALB) with multiple redundant Amazon EC2 instances in private subnets. Configure Amazon CloudFront to deliver HTTPS content using the public ALB as the origin.

This solution meets the requirements for a highly available application with web, application, and database tiers, as well as providing edge-based content delivery. Additionally, it maximizes security by having the ALB in a private subnet, which limits direct access to the web servers, while still being able to serve traffic over the Internet via the public ALB. This will ensure that the web servers are not exposed to the public Internet, which reduces the attack surface and provides a secure way to access the application.

upvoted 18 times

JA2018 1 year, 6 months ago

err, how do we know there's already an ALB in the public subnet(s)?

upvoted 1 times

JA2018 1 year, 6 months ago

This is *NOT* stated in the sample question listed here.

upvoted 1 times

JA2018 1 year, 6 months ago

sorry missed out on this "Configure Amazon CloudFront to deliver HTTPS content using the public ALB as the origin."

upvoted 1 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: C

A. exposes the EC2 instances directly to the public internet, which may compromise security.

B. lacks a load balancer in the public subnet, which is required for efficient load distribution and high availability.

D. provides load balancing and HTTPS content delivery, it exposes the EC2 instances directly to the public internet, which may pose security risks.

C. provides high availability, secure access through private subnets, and optimized HTTPS content delivery using CloudFront with a public ALB as the origin.

upvoted 11 times

meowruki Most Recent 2 years, 6 months ago

C. Configure a public Application Load Balancer (ALB) with multiple redundant Amazon EC2 instances in private subnets. Configure Amazon CloudFront to deliver HTTPS content using the public ALB as the origin.

Here's the reasoning:

Public ALB in Private Subnets: Placing the ALB in private subnets enhances security by preventing direct access from the internet. The ALB in private subnets can communicate with the application instances in the same private subnets.

CloudFront with ALB as Origin: Configuring CloudFront to deliver HTTPS content using the public ALB as the origin allows for content to be cached and distributed globally, reducing latency for end users.

upvoted 2 times

Ruffyt 2 years, 7 months ago

C. Configure a public Application Load Balancer (ALB) with multiple redundant Amazon EC2 instances in private subnets. Configure Amazon CloudFront to deliver HTTPS content using the public ALB as the origin.

This solution meets the requirements for a highly available application with web, application, and database tiers, as well as providing edge-based content delivery. Additionally, it maximizes security by having the ALB in a private subnet, which limits direct access to the web servers, while still being able to serve traffic over the Internet via the public ALB. This will ensure that the web servers are not exposed to the public Internet, which reduces the attack surface and provides a secure way to access the application.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

Keyword: Instances in private, ALB in public, point cloudfront to the public ALB

upvoted 2 times

mhmt4438 3 years, 4 months ago

Selected Answer: C

Answer is C

upvoted 4 times

AHUI 3 years, 4 months ago

ans: C

<https://www.examttopics.com/discussions/amazon/view/46401-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

Morinator 3 years, 4 months ago

Selected Answer: C

Instances in private, ALB in public, point cloudfront to the public ALB

upvoted 5 times

A company has a popular gaming platform running on AWS. The application is sensitive to latency because latency can impact the user experience and introduce unfair advantages to some players. The application is deployed in every AWS Region. It runs on Amazon EC2 instances that are part of Auto Scaling groups configured behind Application Load Balancers (ALBs). A solutions architect needs to implement a mechanism to monitor the health of the application and redirect traffic to healthy endpoints.

Which solution meets these requirements?

- A. Configure an accelerator in AWS Global Accelerator. Add a listener for the port that the application listens on, and attach it to a Regional endpoint in each Region. Add the ALB as the endpoint. **Most Voted**
- B. Create an Amazon CloudFront distribution and specify the ALB as the origin server. Configure the cache behavior to use origin cache headers. Use AWS Lambda functions to optimize the traffic.
- C. Create an Amazon CloudFront distribution and specify Amazon S3 as the origin server. Configure the cache behavior to use origin cache headers. Use AWS Lambda functions to optimize the traffic.
- D. Configure an Amazon DynamoDB database to serve as the data store for the application. Create a DynamoDB Accelerator (DAX) cluster to act as the in-memory cache for DynamoDB hosting the application data.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Aninina **Highly Voted** 2 years, 11 months ago

Selected Answer: A

A. Configure an accelerator in AWS Global Accelerator. Add a listener for the port that the application listens on, and attach it to a Regional endpoint in each Region. Add the ALB as the endpoint.

AWS Global Accelerator directs traffic to the optimal healthy endpoint based on health checks, it can also route traffic to the closest healthy endpoint based on geographic location of the client. By configuring an accelerator and attaching it to a Regional endpoint in each Region, and adding the ALB as the endpoint, the solution will redirect traffic to healthy endpoints, improving the user experience by reducing latency and ensuring that the application is running optimally. This solution will ensure that traffic is directed to the closest healthy endpoint and will help to improve the overall user experience.

upvoted 24 times

michellemeloc **Highly Voted** 2 years, 6 months ago

Selected Answer: A

Delivery gaming content --> AWS GLOBAL ACCELERATOR

upvoted 18 times

pentium75 **Most Recent** 1 year, 11 months ago

Selected Answer: A

Would have selected A just because B, C and D don't make any sense or have nothing to do with the requirements. But now learned that Global Accelerator checks health of resources BEHIND ALB/NLB, so it meets the requirements.

upvoted 6 times

mwwt2022 1 year, 11 months ago

gaming platform -> Can't be CloudFront. Probably go for global accelerator

upvoted 5 times

Ruffyt 2 years ago

A. Configure an accelerator in AWS Global Accelerator. Add a listener for the port that the application listens on, and attach it to a Regional endpoint in each Region. Add the ALB as the endpoint.

AWS Global Accelerator directs traffic to the optimal healthy endpoint based on health checks, it can also route traffic to the closest healthy endpoint based on geographic location of the client. By configuring an accelerator and attaching it to a Regional endpoint in each Region, and adding the ALB as the endpoint, the solution will redirect traffic to healthy endpoints, improving the user experience by reducing latency and ensuring that the application is running optimally. This solution will ensure that traffic is directed to the closest healthy endpoint and will help to improve the overall user experience.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

A. Configure an accelerator in AWS Global Accelerator. Add a listener for the port that the application listens on, and attach it to a Regional endpoint in each Region. Add the ALB as the endpoint

upvoted 2 times

bjexamprep 2 years, 4 months ago

Is any answer relevant to the question?

upvoted 3 times

cookieMr 2 years, 5 months ago

Selected Answer: A

B. While CloudFront can help with caching and content delivery, it does not provide the mechanism to monitor the health of the application or perform traffic redirection based on health checks.

C. This configuration is suitable for static content delivery but does not address the health monitoring and traffic redirection requirements of the application.

D. While this can enhance performance, it does not monitor the health of the application or redirect traffic based on health checks.

Therefore, option A is the most suitable solution as it leverages AWS Global Accelerator to monitor application health, route traffic to healthy endpoints, and optimize the user experience while addressing latency concerns.

upvoted 6 times

antropaws 2 years, 6 months ago

Selected Answer: A

Agree with A

upvoted 2 times

Bhrino 2 years, 9 months ago

Selected Answer: A

Global accelerators can be used for non http cases such as UDP, tcp , gaming , or voip

upvoted 9 times

pentium75 1 year, 11 months ago

Though we seem to have http/https here, otherwise they could not use ALBs

upvoted 2 times

mhmt4438 2 years, 11 months ago

Selected Answer: A

Correct answer is A

upvoted 2 times

AHUI 2 years, 11 months ago

A:

<https://www.examttopics.com/discussions/amazon/view/46403-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

alanp 2 years, 11 months ago

A. When you have an Application Load Balancer or Network Load Balancer that includes multiple target groups, Global Accelerator considers the load balancer endpoint to be healthy only if each target group behind the load balancer has at least one healthy target. If any single target group for the load balancer has only unhealthy targets, Global Accelerator considers the endpoint to be unhealthy.

<https://docs.aws.amazon.com/global-accelerator/latest/dg/about-endpoint-groups-health-check-options.html>

upvoted 9 times

Morinator 2 years, 11 months ago

Selected Answer: A

<https://docs.aws.amazon.com/global-accelerator/latest/dg/about-endpoint-groups-health-check-options.html>

upvoted 2 times

A company has one million users that use its mobile app. The company must analyze the data usage in near-real time. The company also must encrypt the data in near-real time and must store the data in a centralized location in Apache Parquet format for further processing.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create an Amazon Kinesis data stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data. Invoke an AWS Lambda function to send the data to the Kinesis Data Analytics application.
- B. Create an Amazon Kinesis data stream to store the data in Amazon S3. Create an Amazon EMR cluster to analyze the data. Invoke an AWS Lambda function to send the data to the EMR cluster.
- C. Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon EMR cluster to analyze the data.
- D. Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

mhmt4438 **Highly Voted** 2 years, 11 months ago

Selected Answer: D

D. Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data.

This solution will meet the requirements with the least operational overhead as it uses Amazon Kinesis Data Firehose, which is a fully managed service that can automatically handle the data collection, data transformation, encryption, and data storage in near-real time. Kinesis Data Firehose can automatically store the data in Amazon S3 in Apache Parquet format for further processing. Additionally, it allows you to create an Amazon Kinesis Data Analytics application to analyze the data in near real-time, with no need to manage any infrastructure or invoke any Lambda function. This way you can process a large amount of data with the least operational overhead.

upvoted 60 times

jainparag1 2 years, 10 months ago

Nicely explained. Thanks.

upvoted 3 times

LuckyAro 2 years, 10 months ago

Apache Parquet format processing was not mentioned in the answer options. Strange.

upvoted 7 times

WheracanIstart 2 years, 9 months ago

Thanks for the explanation!

upvoted 2 times

antropaws 2 years, 6 months ago

<https://aws.amazon.com/blogs/big-data/analyzing-apache-parquet-optimized-data-using-amazon-kinesis-data-firehose-amazon-athena-and-amazon-redshift/>

upvoted 2 times

cookieMr **Highly Voted** 2 years, 5 months ago

Selected Answer: D

A. requires invoking a Lambda to send the data to the analytics application. This introduces additional operational overhead and complexity.

B. While EMR is a powerful tool for big data processing, it requires more operational management and configuration compared to Kinesis Data Analytics.

C. introduces unnecessary complexity by involving EMR for data analysis when Kinesis Data Analytics can perform the analysis in a more streamlined and automated manner.

Therefore, option D is the most suitable solution as it leverages Kinesis Data Firehose for data ingestion, stores the data in S3, and utilizes Kinesis Data Analytics for near-real-time analysis, providing a low operational overhead solution for data usage analysis and encryption.

upvoted 8 times

farnamjam Most Recent 1 year, 10 months ago

Selected Answer: D

A and B are out.

Kinesis Data Streams cannot directly send data to S3 by itself

upvoted 3 times

Ruffyit 2 years ago

D. Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data.

This solution will meet the requirements with the least operational overhead as it uses Amazon Kinesis Data Firehose, which is a fully managed service that can automatically handle the data collection, data transformation, encryption, and data storage in near-real time. Kinesis Data Firehose can automatically store the data in Amazon S3 in Apache Parquet format for further processing. Additionally, it allows you to create an Amazon Kinesis Data Analytics application to analyze the data in near real-time, with no need to manage any infrastructure or invoke any Lambda function. This way you can process a large amount of data with the least operational overhead.

upvoted 3 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: D

Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

D. Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data

upvoted 2 times

AHUI 2 years, 11 months ago

D:

<https://www.examtopycs.com/discussions/amazon/view/82022-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 1 times

Aninina 2 years, 11 months ago

Selected Answer: D

D. Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data.

Amazon Kinesis Data Firehose can automatically encrypt and store the data in Amazon S3 in Apache Parquet format for further processing, which reduces the operational overhead. It also allows for near-real-time data analysis using Kinesis Data Analytics, which is a fully managed service that makes it easy to analyze streaming data using SQL. This solution eliminates the need for setting up and maintaining an EMR cluster, which would require more operational overhead.

upvoted 3 times

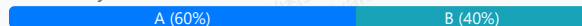
A gaming company has a web application that displays scores. The application runs on Amazon EC2 instances behind an Application Load Balancer. The application stores data in an Amazon RDS for MySQL database. Users are starting to experience long delays and interruptions that are caused by database read performance. The company wants to improve the user experience while minimizing changes to the application's architecture.

What should a solutions architect do to meet these requirements?

- A. Use Amazon ElastiCache in front of the database. **Most Voted**
- B. Use RDS Proxy between the application and the database.
- C. Migrate the application from EC2 instances to AWS Lambda.
- D. Migrate the database from Amazon RDS for MySQL to Amazon DynamoDB.

Correct Answer: A

Community vote distribution



Comments

Steve_4542636 **Highly Voted** 3 years, 3 months ago

Selected Answer: A

Rds proxy is for too many connections, not for performance
upvoted 36 times

Yadav_Sanjay 3 years ago

Can't use cache as score gates updated. If data would have been static then definitely can go with A. But here score is dynamic...
upvoted 10 times

rfelipem 3 years ago

Users are starting to experience long delays and interruptions caused by the "read performance" of the database... While the score is dynamic, there is also read activity in the DB that is causing the delays and outages and this can be improved with Elastic Cache.
upvoted 5 times

Mkenya08 2 years, 4 months ago

ElastiCache stores data in memory, which means it may not always have the most up-to-date information. This might be acceptable for certain use cases where slightly stale data is acceptable, but for applications like gaming scores, real-time accuracy is often crucial.
upvoted 5 times

Maru86 2 years, 3 months ago

"Data in the cache is never stale.
Because the data in the cache is updated every time it's written to the database, the data in the cache is always current and updated whenever data is written to the database." <https://docs.aws.amazon.com/AmazonElastiCache/latest/mem-ug/Strategies.html#Strategies.WriteThrough>
upvoted 4 times

JackyCCK 2 years, 2 months ago

RDS Proxy: It helps applications improve scalability and performance by managing database connections and pooling, which can significantly reduce the load on the database
upvoted 3 times

vipyodha 2 years, 11 months ago

to use elasticsearch, you need to perform heavy code change, and also elasticsearch does caching that can improve read performance but will not provide scalability
upvoted 9 times

pentium75 2 years, 5 months ago

We should "minimize", not "avoid", code changes.

upvoted 3 times

chickenmf 2 years, 2 months ago

minimize *architectural changes, NOT code changes

upvoted 3 times

kraken21 **Highly Voted** 3 years, 2 months ago

Selected Answer: B

RDX proxy will : "improve the user experience while minimizing changes".

upvoted 26 times

pentium75 2 years, 5 months ago

.. but not address issues with "database read performance".

upvoted 3 times

Charlesvg **Most Recent** 11 months, 3 weeks ago

Selected Answer: A

RDS proxy won't help for read performance issues

upvoted 2 times

thiahthura 1 year, 6 months ago

Selected Answer: A

users are experiencing long delays. This is the priority issue , so we need to use cache service urgently to solve this issues. If we need performance , we can use proxy. But in this case , latency is our priority. So Answer is A.

upvoted 2 times

LeonSauveterre 1 year, 6 months ago

Selected Answer: A

A: Correct. Not suitable for write-heavy applications though.

B: Reduces pressure on the database by reusing connections, but cannot address the slow-read problem.

C: Why? Who would do this?

D: It completely changes the architecture.

upvoted 1 times

Mish 1 year, 7 months ago

Selected Answer: A

Use Amazon ElastiCache in front of the database.

upvoted 2 times

pujithacg8 1 year, 10 months ago

A. Use Amazon ElastiCache in front of the database.

Explanation:

Amazon ElastiCache is a managed caching service that can be placed in front of the database to cache frequently accessed data. By caching the most common queries, it can significantly reduce the load on the database, leading to faster response times and an improved user experience.

This solution minimizes changes to the existing architecture because it doesn't require modifications to the application's core logic or database schema. Instead, it optimizes performance by reducing the need for repetitive database reads.

ElastiCache supports both Redis and Memcached, which are widely used for caching in web applications.

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: A

A for sure

upvoted 2 times

Mayank0502 1 year, 11 months ago

Selected Answer: B

answer should be B

upvoted 1 times

Manjeet_Kumar 1 year, 11 months ago

Selected Answer: A

Not B because this can help with connection management and improve scalability and availability but won't directly address read performance issue: caused by high read traffic.

upvoted 2 times

MomenAWS 1 year, 12 months ago

Selected Answer: A

Another vague question from AWS: I would prefer A over B
upvoted 2 times

rohitph 2 years ago

Selected Answer: B

ask is - minimum changes.
RDS Proxy is a feature of Amazon RDS.
This would be easiest option to try.
upvoted 1 times

MandAsh 2 years ago

Selected Answer: A

Elasticache - reduce load due to read operations
RDS proxy - reduce load due to lot of connections
Here problem is read operations thus A is solution
upvoted 3 times

hb0011 2 years ago

Selected Answer: A

I vote for Elasticache
upvoted 2 times

rjjkc 2 years ago

Selected Answer: B

B - "Use RDS Proxy between the application and the database."
B because it improves the user experience while minimizing the changes.
If A is used, you have to modify your application to get the data from the cache first, if it is not there, then get from db; and also to invalidate the cache if there is a db update.
upvoted 1 times

osmk 2 years, 2 months ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonElastiCache/latest/APIReference/Welcome.html>
upvoted 2 times

Uzbekistan 2 years, 2 months ago

Selected Answer: B

RDS Proxy: It helps applications improve scalability and performance by managing database connections and pooling, which can significantly reduce the load on the database. By using RDS Proxy, you can offload connection management tasks from your application and optimize database access, thereby improving read performance without making significant changes to the application's architecture.

Minimizing Changes: Implementing RDS Proxy does not require architectural changes to the application. It works as a proxy layer between the application and the RDS database, so the application code remains unchanged. This minimizes the effort and risk associated with making modifications to the application.
upvoted 2 times

An ecommerce company has noticed performance degradation of its Amazon RDS based web application. The performance degradation is attributed to an increase in the number of read-only SQL queries triggered by business analysts. A solutions architect needs to solve the problem with minimal changes to the existing web application.

What should the solutions architect recommend?

- A. Export the data to Amazon DynamoDB and have the business analysts run their queries.
- B. Load the data into Amazon ElastiCache and have the business analysts run their queries.
- C. Create a read replica of the primary database and have the business analysts run their queries. **Most Voted**
- D. Copy the data into an Amazon Redshift cluster and have the business analysts run their queries.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Aninina **Highly Voted** 2 years, 11 months ago

Selected Answer: C

C. Create a read replica of the primary database and have the business analysts run their queries.

Creating a read replica of the primary RDS database will offload the read-only SQL queries from the primary database, which will help to improve the performance of the web application. Read replicas are exact copies of the primary database that can be used to handle read-only traffic, which will reduce the load on the primary database and improve the performance of the web application. This solution can be implemented with minimal changes to the existing web application, as the business analysts can continue to run their queries on the read replica without modifying the code.

upvoted 6 times

awsgeek75 **Most Recent** 1 year, 10 months ago

Selected Answer: C

"A solutions architect needs to solve the problem with minimal changes to the existing web application."

ABD all require major changes to the application.

A: DynamoDB is NoSQL so big change

B: ElastiCache is a caching layer which require code change normally significant code change

D: Redshift is analytics so not a solution

upvoted 4 times

Ruffyit 2 years ago

. While DynamoDB is a scalable NoSQL database, it requires changes to the application's data model and query patterns.

B. ElastiCache is an in-memory data store that can improve query performance, but it is primarily used for caching rather than running complex queries.

D. Redshift is a powerful data warehousing solution, but migrating the data and adapting the queries to Redshift's columnar architecture would require significant changes to the application and query logic.

Therefore, option C is the most appropriate recommendation as it leverages read replicas in RDS to offload read-only query traffic from the primary database, allowing the business analysts to run their queries without impacting the performance of the web application. It provides a scalable and efficient solution with minimal changes to the existing web application.

upvoted 2 times

nileeka97 2 years, 2 months ago

Selected Answer: C

C. Create a read replica of the primary database and have the business analysts run their queries

upvoted 2 times

cookieMr 2 years, 5 months ago

Selected Answer: C

A. While DynamoDB is a scalable NoSQL database, it requires changes to the application's data model and query patterns.

B. ElastiCache is an in-memory data store that can improve query performance, but it is primarily used for caching rather than running complex queries.

D. Redshift is a powerful data warehousing solution, but migrating the data and adapting the queries to Redshift's columnar architecture would require significant changes to the application and query logic.

Therefore, option C is the most appropriate recommendation as it leverages read replicas in RDS to offload read-only query traffic from the primary database, allowing the business analysts to run their queries without impacting the performance of the web application. It provides a scalable and efficient solution with minimal changes to the existing web application.

upvoted 3 times

antropaws 2 years, 6 months ago

Selected Answer: C

C, no doubt.

upvoted 3 times

mhmt4438 2 years, 11 months ago

Selected Answer: C

C is correct answer

upvoted 3 times

bamishr 2 years, 11 months ago

Selected Answer: C

Create a read replica of the primary database and have the business analysts run their queries.

upvoted 2 times

A company is using a centralized AWS account to store log data in various Amazon S3 buckets. A solutions architect needs to ensure that the data is encrypted at rest before the data is uploaded to the S3 buckets. The data also must be encrypted in transit.

Which solution meets these requirements?

- A. Use client-side encryption to encrypt the data that is being uploaded to the S3 buckets. **Most Voted**
- B. Use server-side encryption to encrypt the data that is being uploaded to the S3 buckets.
- C. Create bucket policies that require the use of server-side encryption with S3 managed encryption keys (SSE-S3) for S3 uploads.
- D. Enable the security option to encrypt the S3 buckets through the use of a default AWS Key Management Service (AWS KMS) key.

Correct Answer: A

Community vote distribution

A (93%)

7%

Comments

techhb **Highly Voted** 3 years, 4 months ago

Selected Answer: A

here keyword is "before" "the data is encrypted at rest before the data is uploaded to the S3 buckets."
upvoted 29 times

mhmt4438 **Highly Voted** 3 years, 4 months ago

Selected Answer: A

<https://www.examttopics.com/discussions/amazon/view/53840-exam-aws-certified-solutions-architect-associate-saa-c02/>
upvoted 5 times

Neeraja2023 **Most Recent** 7 months, 1 week ago

Selected Answer: C

C. Create bucket policies that require the use of server-side encryption with S3 managed encryption keys (SSE-S3) for S3 uploads.

Ensures all uploaded objects are encrypted at rest with SSE-S3.

Can also enforce HTTPS (in-transit encryption) by adding a bucket policy denying non-SSL requests (aws:SecureTransport: false).

Enforces encryption automatically and centrally, with minimal operational overhead.

While encryption happens after upload, this is the standard AWS best practice for at-rest encryption and policy enforcement.

Best practical and enforceable approach.

upvoted 1 times

Dantecito 1 year, 4 months ago

Selected Answer: A

For those who are confused with the data also must be encrypted in transit, Amazon S3 TLS (Transport Layer Security) is used by default and you can't disable it.

upvoted 3 times

LeonSauveterre 1 year, 6 months ago

Selected Answer: A

After a little googling, for those who are confused by "at rest":

1. *Encryption at Rest* refers to the encryption applied to the stored data. Encryption may be implemented at the source, where data is generated and stored at the origin.

2. *Encryption in Transit* refers to encrypting data that is transferred between two nodes of the network.

3. *End-to-End Encryption* refers to the combination of the encryption at rest and encryption in transit.

upvoted 5 times

babayomi 1 year, 7 months ago

A

I believe the question is crafted to cause some confusion. At the same time it is simple to answer, since client side encryption answers the the requirements.

upvoted 2 times

reviewmine 2 years, 3 months ago

Selected Answer: A

Answer is A. Encrypt it first before uploading to S3.

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: C

I think the many votes for A are caused by misunderstanding the wording as

"Ensure that

the data is encrypted at rest before the data is uploaded"

But that doesn't make sense, it means

"Ensure that the data is encrypted at rest
before the data is uploaded"

So, before you allow people to upload data, make sure that it gets encrypted.

upvoted 3 times

pentium75 2 years, 5 months ago

On second thought, C would not enforce encryption in transit. Thus must be A indeed.

upvoted 3 times

awsgeek75 2 years, 4 months ago

For a moment I bought into your reasoning for C assuming that maybe the question is missing some grammar construct but realised that C does not really solve the encryption in transit issue like I originally thought. BUT good work!

upvoted 2 times

Cyberkayu 2 years, 5 months ago

BCD, data not yet encrypted before landing on S3 bucket

upvoted 3 times

palthainon 2 years, 7 months ago

Selected Answer: C

HTTPs would encrypt in transe, SSE3 managed keys fulfills requirement for at rest. This is an aws exam, not a best practices exam.

upvoted 1 times

awsgeek75 2 years, 4 months ago

No. HTTPS is not enough for encryption in transit when it comes to S3.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/UsingClientSideEncryption.html>

"Client-side encryption is the act of encrypting your data locally to help ensure its security in transit and at rest. "

upvoted 4 times

petertang224 2 years, 7 months ago

Its_SaKar

upvoted 1 times

prabhjot 2 years, 8 months ago

Ans is B - Server-Side Encryption (SSE): ensure data is encrypted at rest and also Encryption in Transit: When you upload data to Amazon S3 using standard HTTPS requests.

upvoted 4 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: A

Use client-side encryption to encrypt the data that is being uploaded to the S3 buckets

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

A. Use client-side encryption to encrypt the data that is being uploaded to the S3 buckets.
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

A. Use client-side encryption to encrypt the data that is being uploaded to the S3 buckets.
upvoted 2 times

Abobaloyi 2 years, 11 months ago

Selected Answer: A

data must be encrypted before uploaded , which means the client need to do it before uploading the data to S3
upvoted 4 times

datz 3 years, 2 months ago

Selected Answer: A

A, would meet requirements.
upvoted 2 times

A solutions architect observes that a nightly batch processing job is automatically scaled up for 1 hour before the desired Amazon EC2 capacity is reached. The peak capacity is the 'same every night and the batch jobs always start at 1 AM. The solutions architect needs to find a cost-effective solution that will allow for the desired EC2 capacity to be reached quickly and allow the Auto Scaling group to scale down after the batch jobs are complete.

What should the solutions architect do to meet these requirements?

- A. Increase the minimum capacity for the Auto Scaling group.
- B. Increase the maximum capacity for the Auto Scaling group.
- C. Configure scheduled scaling to scale up to the desired compute level. **Most Voted**
- D. Change the scaling policy to add more EC2 instances during each scaling operation.

Correct Answer: C

Community vote distribution

C (100%)

Comments

ManOnTheMoon **Highly Voted** 2 years, 4 months ago

GOOD LUCK EVERYONE :) YOU CAN DO THIS

upvoted 48 times

david76x **Highly Voted** 2 years, 4 months ago

Selected Answer: C

C is correct. Goodluck everybody!

upvoted 19 times

Guru4Cloud **Most Recent** 1 year, 9 months ago

Selected Answer: C

Configuring scheduled scaling actions allows the Auto Scaling group to scale up to the desired capacity at a scheduled time (1 AM in this case) when the batch jobs start. This ensures the desired compute capacity is reached immediately.

The Auto Scaling group can then scale down based on metrics after the batch jobs complete.

upvoted 10 times

hsinchang 1 year, 10 months ago

Selected Answer: C

The time is given, use scheduled for optimal cost

upvoted 4 times

qacollin 2 years, 1 month ago

just scheduled my exam :)

upvoted 8 times

awscerts023 2 years, 4 months ago

Reached here ! Did anyone schedule the real exam now ? How was it ?

upvoted 6 times

pal40sg 2 years, 4 months ago

Thanks to everyone who contributed with answers :)

upvoted 6 times

ProfXsamson 2 years, 4 months ago

Selected Answer: C

C. I'm here at the end, leaving this here for posterity sake 02/01/2023.

upvoted 6 times

dedline 2 years, 4 months ago

GL ALL!

upvoted 5 times

mhmt4438 2 years, 4 months ago

Selected Answer: C

<https://www.examttopics.com/discussions/amazon/view/27868-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

Aninina 2 years, 4 months ago

Selected Answer: C

C. Configure scheduled scaling to scale up to the desired compute level.

By configuring scheduled scaling, the solutions architect can set the Auto Scaling group to automatically scale up to the desired compute level at a specific time (1AM) when the batch job starts and then automatically scale down after the job is complete. This will allow the desired EC2 capacity to be reached quickly and also help in reducing the cost.

upvoted 7 times

bamishr 2 years, 5 months ago

Selected Answer: C

Configure scheduled scaling to scale up to the desired compute level.

upvoted 2 times

Morinator 2 years, 5 months ago

Selected Answer: C

predictable = schedule scaling

upvoted 5 times

A company serves a dynamic website from a fleet of Amazon EC2 instances behind an Application Load Balancer (ALB). The website needs to support multiple languages to serve customers around the world. The website's architecture is running in the us-west-1 Region and is exhibiting high request latency for users that are located in other parts of the world.

The website needs to serve requests quickly and efficiently regardless of a user's location. However, the company does not want to recreate the existing architecture across multiple Regions.

What should a solutions architect do to meet these requirements?

- A. Replace the existing architecture with a website that is served from an Amazon S3 bucket. Configure an Amazon CloudFront distribution with the S3 bucket as the origin. Set the cache behavior settings to cache based on the Accept-Language request header.
- B. Configure an Amazon CloudFront distribution with the ALB as the origin. Set the cache behavior settings to cache based on the Accept-Language request header. **Most Voted**
- C. Create an Amazon API Gateway API that is integrated with the ALB. Configure the API to use the HTTP integration type. Set up an API Gateway stage to enable the API cache based on the Accept-Language request header.
- D. Launch an EC2 instance in each additional Region and configure NGINX to act as a cache server for that Region. Put all the EC2 instances and the ALB behind an Amazon Route 53 record set with a geolocation routing policy.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Yechi **Highly Voted** 3 years, 3 months ago

Selected Answer: B

Configuring caching based on the language of the viewer

If you want CloudFront to cache different versions of your objects based on the language specified in the request, configure CloudFront to forward the Accept-Language header to your origin.

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/header-caching.html>

upvoted 13 times

Trains **Most Recent** 2 years, 6 months ago

Isn't CloudFront for static websites though? Question specifically states the content is dynamic

upvoted 3 times

Cloud_A 2 years, 4 months ago

Cloudfront serves for both static and dynamic. If it was just static, then you can consider AWS S3.

upvoted 6 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

By caching content based on the Accept-Language request header, CloudFront can serve the appropriate version of the website to users based on their language preferences. This solution allows the company to improve the website's performance for users around the world without having to recreate the existing architecture in multiple Regions.

upvoted 4 times

A1975 2 years, 10 months ago

Selected Answer: B

CloudFront allows you to customize cache behavior based on various request headers. By setting the cache behavior to cache based on the Accept-Language request header, CloudFront can store and serve language-specific versions of the website content, reducing the need to repeatedly fetch

the content from the ALB for users with the same language preference.

upvoted 3 times

kraken21 3 years, 2 months ago

Selected Answer: B

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/header-caching.html#header-caching-web-language>

upvoted 1 times

vherman 3 years, 3 months ago

Selected Answer: B

B is correct

upvoted 1 times

Steve_4542636 3 years, 3 months ago

Selected Answer: B

I think it's b

upvoted 2 times

LuckyAro 3 years, 3 months ago

Selected Answer: B

B is the correct answer

upvoted 1 times

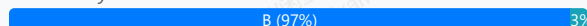
A rapidly growing ecommerce company is running its workloads in a single AWS Region. A solutions architect must create a disaster recovery (DR) strategy that includes a different AWS Region. The company wants its database to be up to date in the DR Region with the least possible latency. The remaining infrastructure in the DR Region needs to run at reduced capacity and must be able to scale up if necessary.

Which solution will meet these requirements with the LOWEST recovery time objective (RTO)?

- A. Use an Amazon Aurora global database with a pilot light deployment.
- B. Use an Amazon Aurora global database with a warm standby deployment. **Most Voted**
- C. Use an Amazon RDS Multi-AZ DB instance with a pilot light deployment.
- D. Use an Amazon RDS Multi-AZ DB instance with a warm standby deployment.

Correct Answer: B

Community vote distribution



Comments

Yechi **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Note: The difference between pilot light and warm standby can sometimes be difficult to understand. Both include an environment in your DR Region with copies of your primary Region assets. The distinction is that pilot light cannot process requests without additional action taken first, whereas warm standby can handle traffic (at reduced capacity levels) immediately. The pilot light approach requires you to "turn on" servers, possibly deploy additional (non-core) infrastructure, and scale up, whereas warm standby only requires you to scale up (everything is already deployed and running). Use your RTO and RPO needs to help you choose between these approaches.

<https://docs.aws.amazon.com/whitepapers/latest/disaster-recovery-workloads-on-aws/disaster-recovery-options-in-the-cloud.html>
upvoted 28 times

nickolaj **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Option A is incorrect because while Amazon Aurora global database is a good solution for disaster recovery, pilot light deployment provides only a minimalistic setup and would require manual intervention to make the DR Region fully operational, which increases the recovery time.

Option B is a better choice than Option A as it provides a warm standby deployment, which is an automated and more scalable setup than pilot light deployment. In this setup, the database is replicated to the DR Region, and the standby instance can be brought up quickly in case of a disaster.

Option C is incorrect because Multi-AZ DB instances provide high availability, not disaster recovery.

Option D is a good choice for high availability, but it does not meet the requirement for DR in a different region with the least possible latency.
upvoted 23 times

awsgeek75 **Most Recent** 1 year, 10 months ago

Selected Answer: B

B: Warm Standby is better when it comes to LOWEST RTO.

<https://docs.aws.amazon.com/whitepapers/latest/disaster-recovery-workloads-on-aws/disaster-recovery-options-in-the-cloud.html>
upvoted 2 times

pentium75 1 year, 11 months ago

Selected Answer: B

"Different Region" rules out C and D ("Multi-AZ" is within a region)

"Run at reduced capacity" = warm standby (while "pilot light" means that DR resources are shut down and are started manually in case of failover)
upvoted 4 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: B

The warm standby approach involves ensuring that there is a scaled down, but fully functional, copy of your production environment in another Region.

With the pilot light approach, you replicate your data from one Region to another and provision a copy of your core workload infrastructure. Resources required to support data replication and backup, such as databases and object storage, are always on. Other elements, such as application servers, are loaded with application code and configurations, but are "switched off".

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

An Amazon Aurora global database with a warm standby deployment provides continuous replication from one AWS Region to another, keeping the DR database up-to-date with minimal latency.

upvoted 2 times

A1975 2 years, 4 months ago

Selected Answer: B

In a Pilot Light scenario, only an EC2 Instance and a DB may be running. In Warm Standby, however, everything is running — in a much smaller capacity. This means the load balancer, gateways, databases, all subnets, and everything else are ready to go on a moment's notice.

with reference to below statement Option B is a better choice than Option A.

"The remaining infrastructure in the DR Region needs to run at reduced capacity and must be able to scale up if necessary".

upvoted 2 times

krisfromtw 2 years, 9 months ago

Selected Answer: D

should be D.

upvoted 2 times

leoattf 2 years, 9 months ago

No, my friend. The question asks for deployment in another Region. Hence, it cannot be C or D.

The answer is B because is Global (different regions) and Ward Standby has faster RTO than Pilot Light.

upvoted 10 times

pentium75 1 year, 11 months ago

"Multi-AZ" = multiple AZs in same region, but requirement is "a different AWS Region".

upvoted 3 times

A company runs an application on Amazon EC2 instances. The company needs to implement a disaster recovery (DR) solution for the application. The DR solution needs to have a recovery time objective (RTO) of less than 4 hours. The DR solution also needs to use the fewest possible AWS resources during normal operations.

Which solution will meet these requirements in the MOST operationally efficient way?

- A. Create Amazon Machine Images (AMIs) to back up the EC2 instances. Copy the AMIs to a secondary AWS Region. Automate infrastructure deployment in the secondary Region by using AWS Lambda and custom scripts.
- B. Create Amazon Machine Images (AMIs) to back up the EC2 instances. Copy the AMIs to a secondary AWS Region. Automate infrastructure deployment in the secondary Region by using AWS CloudFormation. **Most Voted**
- C. Launch EC2 instances in a secondary AWS Region. Keep the EC2 instances in the secondary Region active at all times.
- D. Launch EC2 instances in a secondary Availability Zone. Keep the EC2 instances in the secondary Availability Zone active at all times.

Correct Answer: B

Community vote distribution

B (100%)

Comments

NolaHOLA **Highly Voted** 3 years, 3 months ago

Guys, sorry but I don't really have time to deep dive as my exam is soon. Based on chatGPT and my previous study the answer should be B "Create Amazon Machine Images (AMIs) to back up the EC2 instances. Copy the AMIs to a secondary AWS Region. Automate infrastructure deployment in the secondary Region by using AWS CloudFormation," would likely be the most suitable solution for the given requirements.

This option allows for the creation of Amazon Machine Images (AMIs) to back up the EC2 instances, which can then be copied to a secondary AWS region to provide disaster recovery capabilities. The infrastructure deployment in the secondary region can be automated using AWS CloudFormation, which can help to reduce the amount of time and resources needed for deployment and management.

upvoted 13 times

NBone 2 years, 10 months ago

please how do you use chatGPT to study for these questions?

upvoted 5 times

nickolaj **Highly Voted** 3 years, 3 months ago

Selected Answer: B

Option B would be the most operationally efficient solution for implementing a DR solution for the application, meeting the requirement of an RTO of less than 4 hours and using the fewest possible AWS resources during normal operations.

By creating Amazon Machine Images (AMIs) to back up the EC2 instances and copying them to a secondary AWS Region, the company can ensure that they have a reliable backup in the event of a disaster. By using AWS CloudFormation to automate infrastructure deployment in the secondary Region, the company can minimize the amount of time and effort required to set up the DR solution.

upvoted 9 times

djgodzilla **Most Recent** 2 years, 5 months ago

Option E : Automate infrastructure deployment in the secondary Region by using terraform and ditch AWS CloudFormation 🤔🤔.

upvoted 6 times

pentium75 2 years, 5 months ago

Selected Answer: B

A is not "most operationally efficient"

C and D do not meet the "use the fewest possible AWS resources during normal operations" requirement

upvoted 3 times

vijaykamal 2 years, 8 months ago

Selected Answer: B

Option D suggests launching EC2 instances in a secondary Availability Zone (AZ), but AZs are not separate AWS Regions. While it provides high availability within a Region, it doesn't offer geographic redundancy, which is essential for disaster recovery.

upvoted 3 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: B

needs to use the fewest possible AWS resources during normal operations = backup & restore

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

Create Amazon Machine Images (AMIs) to back up the EC2 instances. Copy the AMIs to a secondary AWS Region. Automate infrastructure deployment in the secondary Region by using AWS CloudFormation

upvoted 2 times

AMYYMY 2 years, 9 months ago

B SHOULD BE RIGHT

upvoted 2 times

A1975 2 years, 10 months ago

Selected Answer: B

Option A: Add complexity and management overhead.

Option B: Creating AMIs for backup and using AWS CloudFormation for infrastructure deployment in the secondary Region is a more streamlined and automated approach. CloudFormation allows you to define and provision resources in a declarative manner, making it easier to maintain and update your infrastructure. This solution is more operationally efficient compared to Option A.

Option C: could be expensive and not fully aligned with the requirement of using the fewest possible AWS resources during normal operations.

Option D: might not be sufficient for meeting the DR requirements, as Availability Zones are still within the same AWS Region and might be subject to the same regional-level failures.

upvoted 2 times

NBone 2 years, 10 months ago

Please I would really appreciate clarification with this question. The community has voted 100% that the right answer is B. However, option D is shown to be the correct answer. So, who sets the correct answer? Which one should new comers like myself believe? the community's or the other (which am guessing is set by the moderators???) Please help.

upvoted 2 times

Kaula 2 years, 2 months ago

From Examtopics support:

In very few cases a conflict is raised between the provided and most-voted answers.

In case of conflict between the provided & most voted answers, we suggest our customers rely on the most voted answers and consider them most correct.

upvoted 2 times

SimiTik 3 years, 1 month ago

C may satisfy the requirement of using the fewest possible AWS resources during normal operations, it may not be the most operationally efficient or cost-effective solution in the long term.

upvoted 3 times

AlmeroSenior 3 years, 3 months ago

So Weird , they have product for this > Elastic Disaster Recovery , but option is not given .

upvoted 2 times

Yechi 3 years, 3 months ago

Selected Answer: B

https://docs.aws.amazon.com/zh_cn/whitepapers/latest/disaster-recovery-workloads-on-aws/disaster-recovery-options-in-the-cloud.html#backup-and-restore

upvoted 5 times

Joan111edu 3 years, 3 months ago

Selected Answer: B

the answer should be B

----> recovery time objective (RTO) of less than 4 hours.

https://docs.aws.amazon.com/zh_cn/whitepapers/latest/disaster-recovery-workloads-on-aws/disaster-recovery-options-in-the-cloud.html#backup-and-restore

upvoted 4 times

JA2018 1 year, 6 months ago

English version: <https://docs.aws.amazon.com/whitepapers/latest/disaster-recovery-workloads-on-aws/disaster-recovery-options-in-the-cloud.html>

upvoted 1 times

A company runs an internal browser-based application. The application runs on Amazon EC2 instances behind an Application Load Balancer. The instances run in an Amazon EC2 Auto Scaling group across multiple Availability Zones. The Auto Scaling group scales up to 20 instances during work hours, but scales down to 2 instances overnight. Staff are complaining that the application is very slow when the day begins, although it runs well by mid-morning.

How should the scaling be changed to address the staff complaints and keep costs to a minimum?

- A. Implement a scheduled action that sets the desired capacity to 20 shortly before the office opens.
- B. Implement a step scaling action triggered at a lower CPU threshold, and decrease the cooldown period.
- C. Implement a target tracking action triggered at a lower CPU threshold, and decrease the cooldown period. **Most Voted**
- D. Implement a scheduled action that sets the minimum and maximum capacity to 20 shortly before the office opens.

Correct Answer: C

Community vote distribution

C (66%)

A (34%)

Comments

asoli **Highly Voted** 3 years, 2 months ago

Selected Answer: C

At first, I thought the answer is A. But it is C.

It seems that there is no information in the question about CPU or Memory usage.

So, we might think the answer is A. why? because what we need is to have the required (desired) number of instances. It already has scheduled scaling that works well in this scenario. Scale down after working hours and scale up in working hours. So, it just needs to adjust the desired number to start from 20 instances.

But here is the point it shows A is WRONG!!!

If it started with desired 20 instances, it will keep it for the whole day. What if the load is reduced? We do not need to keep the 20 instances always. That 20 is the MAXIMUM number we need, not the DESIRE number. So it is against COST that is the main objective of this question.

So, the answer is C

upvoted 36 times

c10356a 2 years, 5 months ago

There is no cooldown period in target tracking, but warm-up time.

upvoted 2 times

pentium75 2 years, 5 months ago

There is a cooldown period in the auto-scaling group, which helps 'keeping costs to a minimum' as instances would be removed sooner.

upvoted 4 times

mandragon 3 years, 1 month ago

If it starts with 20 instances it will not keep it all day. It will scale down based on demand. The scheduled action in Option A simply ensures that there are enough instances running to handle the increased traffic when the day begins, while still allowing the Auto Scaling group to scale up or down based on demand during the rest of the day. <https://docs.aws.amazon.com/autoscaling/ec2/userguide/scale-your-group.html>

upvoted 10 times

xdkonorek2 2 years, 6 months ago

This is right, setting desired capacity doesn't turn off autoscaling policies

upvoted 3 times

TheFivePips 2 years, 3 months ago

From what I can tell, you must specify an end time, or else it will run indefinitely. So I think A would be right, if they specified an end time. Otherwise C is more cost effective

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-scheduled-scaling.html>

upvoted 3 times

meowruki Highly Voted 2 years, 6 months ago

Selected Answer: C

C. Implement a target tracking action triggered at a lower CPU threshold, and decrease the cooldown period.

Here's the reasoning:

Target Tracking Scaling Policy: With a target tracking scaling policy, you can set a target value for a specific metric, such as CPU utilization. The Auto Scaling group then adjusts the capacity to maintain that target.

Lower CPU Threshold: By triggering the target tracking action at a lower CPU threshold, the Auto Scaling group can proactively add instances when the workload increases, helping to address the slowness at the beginning of the day.

Decrease Cooldown Period: Reducing the cooldown period allows the Auto Scaling group to scale in and out more rapidly, making adjustments quicker in response to changing demand.

upvoted 7 times

meowruki 2 years, 6 months ago

Options A and D involve scheduled actions, which are time-based and may not be as responsive to immediate changes in demand. They also do not dynamically respond to varying workloads.

upvoted 2 times

Neeraja2023 Most Recent 7 months, 1 week ago

Selected Answer: A

A. Implement a scheduled action that sets the desired capacity to 20 shortly before the office opens.

This proactively scales up before the morning rush.

- Ensures instances are ready and warm when traffic starts.
- Keeps costs low because at night, it still scales down to 2.
- Exactly solves the problem with minimal cost and operational effort.
- Best practice for predictable workload patterns.

upvoted 1 times

In2718 9 months, 4 weeks ago

Selected Answer: A

Chat says answer A - Pre-scaling with a scheduled action warms up instances before the morning surge, fixing the slow start while still letting the group scale down to 2 overnight. Options B/C wait for load to rise (still slow at start). D fixes capacity at 20 (not cost-effective).

upvoted 1 times

Clouddon 11 months ago

Selected Answer: A

A. Implement a scheduled action that sets the desired capacity to 20 shortly before the office opens.

 **Explanation:**

The issue is that Auto Scaling reacts to demand. When the day begins, the Auto Scaling group starts with only 2 instances. It takes time to scale up to 20, and in the meantime, users experience slow performance.

To fix this:

You want to proactively scale up before demand starts.

This ensures that the full capacity is already in place by the time staff begin using the application.

Scheduled scaling is ideal for this use case because the workload is predictable (e.g., work hours are known).

upvoted 3 times

srcntpc 1 year ago

Selected Answer: A

Why A is Correct:

Scheduled scaling action increases capacity to 20 instances shortly before office hours.

Ensures instances are already running and warm when employees start using the app.

This avoids delays caused by dynamic scaling and instance warm-up.

upvoted 3 times

zdi561 1 year, 4 months ago

Selected Answer: A

Decreasing cpu threshold could cause more running instance(might have more than two at night) therefore higher cost. Schedule at desired to 20 minutes before the day will solve the issue that the app is very slow at beginning of the day

upvoted 2 times

LeonSauveterre 1 year, 6 months ago

Selected Answer: A

Exam-wise, you gotta choose A.

The problem is explicitly described as slowness at the start of the day, which implies a predictable traffic pattern. Option C argues for a responsive approach (reacting dynamically to real-time demand changes) rather than a predictive approach (pre-scaling based on a known schedule).

Also, option A can minimize costs compared to option D as well, because the desired capacity is a flexible number that the Auto Scaling group targets at specific times but doesn't enforce as a hard minimum.

If the emphasis is on handling varying traffic patterns throughout the day, Option C could be considered viable.

upvoted 1 times

AbhiBK 1 year, 9 months ago

Question is about cost effectiveness hence Use target tracking scaling policies to maintain a specific metric, such as CPU utilization or request count per target. This allows the Auto Scaling group to dynamically adjust the number of instances based on real-time demand. We do not need to have 20 instances up and running during start of the day.

upvoted 2 times

maryam_sh 1 year, 9 months ago

Selected Answer: A

Not C because: target tracking scaling attempts to maintain a target metric (like average CPU utilization). While more responsive, it would still react to increased load rather than pre-scaling. The initial slow period would persist as the scaling reacts to the increased demand rather than anticipating it.

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: A

A for sure.

upvoted 3 times

foha2012 2 years, 4 months ago

Answer is A. Makes more sense to me.

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: C

A is not cost effective, it would set the number of instances to maximum even before the first employee arrives.

D is not cost effective, it would cause the permanent use of 20 instances

B could almost work, but if you configure small steps then it scales too slowly in the morning; if you configure big steps (like "add 8 instances at a time") it would scale in the morning but not be cost-efficient during the day.

C would address the requirement, it would scale to meet a certain CPU utilization. Decreasing the cooldown period (which is not possible for the scaling policy itself but for the auto-scaling group) would help 'keeping costs to a minimum'.

upvoted 4 times

Mikado211 2 years, 5 months ago

Selected Answer: A

The question 369 is exactly the same problem,

Since a scheduled scaling doesn't disable the autoscaling later in the day the A works perfectly well.

upvoted 3 times

Cyberkayu 2 years, 5 months ago

A. since only a boot storm issue at 9am and settle down in mid morning, 20 instance is enough to support the workload

NOT C. Reduce threshold to trigger (lets say 50% from 80% utilization) and lower cool down period, will still take time to ramp up to max 20 instance

upvoted 1 times

pentium75 2 years, 5 months ago

How would scaling up to the maximum number of instances "keep costs to a minimum"?

upvoted 2 times

wearrexdzw3123 2 years, 6 months ago

My mistake, I should have chosen c. A lower threshold can expand in advance, and lowering cooling can increase the expansion frequency.

upvoted 3 times

wearrexdzw3123 2 years, 7 months ago

Selected Answer: A

I choose option A because the root of the problem is the inability of the scaling speed in the morning to meet the demand, rather than what criteria to use for scaling.

upvoted 1 times

pentium75 2 years, 5 months ago

How would scaling up to the maximum number of instances "keep costs to a minimum"?

upvoted 2 times

A company has a multi-tier application deployed on several Amazon EC2 instances in an Auto Scaling group. An Amazon RDS for Oracle instance is the application's data layer that uses Oracle-specific PL/SQL functions. Traffic to the application has been steadily increasing. This is causing the EC2 instances to become overloaded and the RDS instance to run out of storage. The Auto Scaling group does not have any scaling metrics and defines the minimum healthy instance count only. The company predicts that traffic will continue to increase at a steady but unpredictable rate before leveling off.

What should a solutions architect do to ensure the system can automatically scale for the increased traffic? (Choose two.)

- A. Configure storage Auto Scaling on the RDS for Oracle instance. **Most Voted**
- B. Migrate the database to Amazon Aurora to use Auto Scaling storage.
- C. Configure an alarm on the RDS for Oracle instance for low free storage space.
- D. Configure the Auto Scaling group to use the average CPU as the scaling metric. **Most Voted**
- E. Configure the Auto Scaling group to use the average free memory as the scaling metric.

Correct Answer: AD

Community vote distribution

AD (88%)

6%

Comments

klayytech **Highly Voted** 3 years, 2 months ago

Selected Answer: AD

A) Configure storage Auto Scaling on the RDS for Oracle instance.

= Makes sense. With RDS Storage Auto Scaling, you simply set your desired maximum storage limit, and Auto Scaling takes care of the rest.

B) Migrate the database to Amazon Aurora to use Auto Scaling storage.

= Scenario specifies application's data layer uses Oracle-specific PL/SQL functions. This rules out migration to Aurora.

C) Configure an alarm on the RDS for Oracle instance for low free storage space.

= You could do this but what does it fix? Nothing. The CW notification isn't going to trigger anything.

D) Configure the Auto Scaling group to use the average CPU as the scaling metric.

= Makes sense. The CPU utilization is the precursor to the storage outage. When the ec2 instances are overloaded, the RDS instance storage hits its limits, too.

upvoted 30 times

In2718 **Most Recent** 11 months, 1 week ago

Selected Answer: AC

This is a very poorly written question. The correlation between CPU and storage is in direct and not given the reason why. If you do A then just monitor for C. Choice A will cover all of this, and the real world you would not assume CPU is directly related to data storage. Remember this for the real world.

upvoted 1 times

BombArat 1 year, 11 months ago

Selected Answer: AE

B mean db migration and only oracle specific commands are allowed

C is just notification not High availability

Now the toss up is between D and E since D is measuring CPU % and E is measuring memory and the question states "This is causing the EC2 instances to become overloaded and the RDS instance to run out of storage" i will err on the side of option E

upvoted 1 times

Uzbekistan 2 years, 2 months ago

Selected Answer: AC

To ensure the system can automatically scale for the increased traffic, you can take the following steps:
A. Configure storage Auto Scaling on the RDS for Oracle instance.

By enabling storage Auto Scaling on the RDS instance, you ensure that additional storage is provisioned automatically when the existing storage reaches capacity. This helps prevent the RDS instance from running out of storage due to increased traffic and data growth.

C. Configure an alarm on the RDS for Oracle instance for low free storage space.

Setting up an alarm for low free storage space on the RDS instance allows you to receive notifications when the storage capacity is approaching its limits. This proactive monitoring helps you take necessary actions, such as adding more storage or scaling resources, before it affects the application's performance.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: AD

B is not possible because the application "uses Oracle-specific PL/SQL functions"

C does not meet the "automatically scale" requirement

E would require an agent on the hosts which we might not have, plus CPU is a better indicator than memory

upvoted 2 times

meowruki 2 years, 6 months ago

Selected Answer: AD

A. Configure storage Auto Scaling on the RDS for Oracle instance.

This option allows the RDS instance to automatically scale its storage based on the actual storage usage, ensuring that you don't run out of storage.

D. Configure the Auto Scaling group to use the average CPU as the scaling metric.

By using CPU utilization as a scaling metric, the Auto Scaling group can dynamically adjust the number of EC2 instances based on the application's demand. This helps in handling increased traffic and preventing overload on existing instances.

upvoted 2 times

meowruki 2 years, 6 months ago

Option B (Migrate the database to Amazon Aurora): While Amazon Aurora provides benefits such as auto-scaling storage and high performance, it involves migrating from Oracle to Aurora, which might require application changes and data migration efforts.

Option C (Configure an alarm on the RDS for Oracle instance for low free storage space): While it's good to have an alarm for low storage space, configuring storage Auto Scaling (Option A) is a more proactive solution that automatically adjusts storage before reaching a critical point.

Option E (Configure the Auto Scaling group to use the average free memory as the scaling metric): While monitoring memory is important for application performance, CPU utilization is often a more direct and responsive metric for auto-scaling in many scenarios.

upvoted 3 times

JA2018 1 year, 6 months ago

STEM mentioned the application "uses Oracle-specific PL/SQL functions"

upvoted 1 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: AD

Configure storage Auto Scaling on the RDS for Oracle instance and Configure the Auto Scaling group to use the average CPU as the scaling metric to accommodate the increased traffic automatically.

upvoted 2 times

vijaykamal 2 years, 8 months ago

Selected Answer: AD

Option B (Migrate the database to Amazon Aurora) may be a good long-term solution, but it involves database migration, which can be complex and time-consuming. For immediate scalability and to address the storage issue, configuring storage Auto Scaling on the existing RDS instance is a more immediate and straightforward solution.

Option C (Configure an alarm on the RDS for Oracle instance for low free storage space) is useful for monitoring, but it doesn't proactively address the storage issue by automatically expanding storage as needed.

Option E (Configure the Auto Scaling group to use the average free memory as the scaling metric) is less common as a scaling metric for EC2 instances compared to CPU utilization. While memory can be an important factor for application performance, CPU utilization is typically a more commonly used metric for scaling decisions. It also doesn't directly address the RDS storage issue.

upvoted 2 times

JA2018 1 year, 6 months ago

the application "uses Oracle-specific PL/SQL functions"

Rules out Option B in my mind

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AD

A. By enabling storage Auto Scaling on the RDS for Oracle instance, it will automatically add more storage when the existing storage is running out, ensuring the application's data layer can handle the increased data storage requirements.

D. By configuring the Auto Scaling group to use the average CPU utilization as the scaling metric, it can automatically add more EC2 instances to the Auto Scaling group when the CPU utilization exceeds a certain threshold. This will help handle the increased traffic and workload on the EC2 instances in the multi-tier application.

upvoted 2 times

A1975 2 years, 10 months ago

Selected Answer: AD

A. By enabling storage Auto Scaling on the RDS for Oracle instance, it will automatically add more storage when the existing storage is running out, ensuring the application's data layer can handle the increased data storage requirements.

D. By configuring the Auto Scaling group to use the average CPU utilization as the scaling metric, it can automatically add more EC2 instances to the Auto Scaling group when the CPU utilization exceeds a certain threshold. This will help handle the increased traffic and workload on the EC2 instances in the multi-tier application.

upvoted 2 times

kruasan 3 years, 1 month ago

Selected Answer: AD

These options will allow the system to scale both the compute tier (EC2 instances) and the data tier (RDS storage) automatically as traffic increases:

A. Storage Auto Scaling will allow the RDS for Oracle instance to automatically increase its allocated storage when free storage space gets low. This ensures the database does not run out of capacity and can continue serving data to the application.

D. Configuring the EC2 Auto Scaling group to scale based on average CPU utilization will allow it to launch additional instances automatically as traffic causes higher CPU levels across the instances. This scales the compute tier to handle increased demand.

upvoted 3 times

kraken21 3 years, 2 months ago

Selected Answer: AD

Auto scaling storage RDS will ease storage issues and migrating Oracle PL/SQL to Aurora is cumbersome. Also Aurora has auto storage scaling by default.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_PIOPS.StorageTypes.html#USER_PIOPS.Autoscaling

upvoted 3 times

Nel8 3 years, 3 months ago

Selected Answer: BD

My answer is B & D...

B. Migrate the database to Amazon Aurora to use Auto Scaling Storage. --- Aurora storage is also self-healing. Data blocks and disks are continuously scanned for errors and repaired automatically.

D. Configure the Auto Scaling group to use the average CPU as the scaling metric. -- Good choice.

I believe either A & C or B & D options will work.

upvoted 3 times

FourOfAKind 3 years, 3 months ago

In this question, you have Oracle DB, and Amazon Aurora is for MySQL/PostgreSQL. A and D are the correct choices.

upvoted 5 times

[Removed] 3 years, 2 months ago

You can migrate Oracle PL/SQL to Aurora:

<https://docs.aws.amazon.com/dms/latest/oracle-to-aurora-mysql-migration-playbook/chap-oracle-aurora-mysql.sql.html>

upvoted 1 times

pentium75 2 years, 5 months ago

But the application "uses Oracle-specific PL/SQL functions".

upvoted 2 times

Ja13 3 years, 3 months ago

Selected Answer: AD

a and d

upvoted 4 times

KZM 3 years, 3 months ago

A and D.

upvoted 4 times

GwonLEE 3 years, 3 months ago

Selected Answer: AD

a and d

upvoted 4 times

LuckyAro 3 years, 3 months ago

Selected Answer: AD

A and D

upvoted 3 times

A company provides an online service for posting video content and transcoding it for use by any mobile platform. The application architecture uses Amazon Elastic File System (Amazon EFS) Standard to collect and store the videos so that multiple Amazon EC2 Linux instances can access the video content for processing. As the popularity of the service has grown over time, the storage costs have become too expensive.

Which storage solution is MOST cost-effective?

- A. Use AWS Storage Gateway for files to store and process the video content.
- B. Use AWS Storage Gateway for volumes to store and process the video content.
- C. Use Amazon EFS for storing the video content. Once processing is complete, transfer the files to Amazon Elastic Block Store (Amazon EBS).
- D. Use Amazon S3 for storing the video content. Move the files temporarily over to an Amazon Elastic Block Store (Amazon EBS) volume attached to the server for processing. **Most Voted**

Correct Answer: D

Community vote distribution

D (80%)

A (20%)

Comments

bdp123 **Highly Voted** 3 years, 3 months ago

Selected Answer: D

Storage gateway is not used for storing content - only to transfer to the Cloud

upvoted 28 times

pentium75 2 years, 5 months ago

A doesn't say "store content ON the gateway", it says "use AWS Storage Gateway for files" (which is the product) "to store the video content" [on S3].

upvoted 3 times

jaswanth 2 years, 4 months ago

Creating storage gateway for files will mount S3 bucket as an NFS volume that can be shared among EC2 Instances in the same manner as EFS but more cost effectively.

upvoted 1 times

JustEugen 9 months, 1 week ago

AWS Storage Gateway is a hybrid cloud storage service that connects on-premises environments with AWS cloud storage

upvoted 1 times

kraken21 **Highly Voted** 3 years, 2 months ago

Selected Answer: D

There is no on-prem/non Aws infrastructure to create a gateway. Also, EFS+EBS is more expensive than EFS and S3. So D is the best option.

upvoted 12 times

pentium75 2 years, 5 months ago

But how do you attach "an EBS volume" to all the servers, and how will you use the files on it then to serve customers.

upvoted 3 times

In2718 **Most Recent** 11 months, 1 week ago

Selected Answer: A

Answer A is closer to the following principle and D is near impossible to implement: "

upvoted 1 times

ln2718 11 months, 1 week ago

Selected Answer: A

I like this answer below -- A
upvoted 1 times

FlyingHawk 1 year, 6 months ago

Selected Answer: D

Option D (S3 + EBS) is the most cost-effective and scalable choice for a service that involves storing video content and performing I/O-intensive transcoding for mobile platforms.
Option A (S3 File Gateway) could work for applications with strict file system requirements but is less optimal for high-performance transcoding which is required by this online service
upvoted 2 times

Uzbekistan 2 years, 2 months ago

Selected Answer: D

D. Use Amazon S3 for storing the video content. Move the files temporarily over to an Amazon Elastic Block Store (Amazon EBS) volume attached to the server for processing.

Amazon S3 (Simple Storage Service) is highly durable, scalable, and cost-effective for storing large volumes of data, such as video content. It offers lower storage costs compared to Amazon EFS and is suitable for storing large files like video content.

For processing the video content, you can temporarily move the files from Amazon S3 to an Amazon EBS volume attached to the EC2 instances. This approach allows you to leverage the high-performance storage of Amazon EBS for processing, while still benefiting from the cost-effectiveness of Amazon S3 for long-term storage. Once processing is complete, you can remove the temporary files from the EBS volume and store the final results back in S3.
upvoted 4 times

bujuman 2 years, 3 months ago

Selected Answer: A

Answer is closer to the following principle and D is near impossible to implement:
"Amazon S3 File Gateway – Amazon S3 File Gateway supports a file interface into Amazon Simple Storage Service (Amazon S3) and combines a service and a virtual software appliance. By using this combination, you can store and retrieve objects in Amazon S3 using industry-standard file protocols such as Network File System (NFS) and Server Message Block (SMB). You deploy the gateway into your on-premises environment as a virtual machine (VM) running on VMware ESXi, Microsoft Hyper-V, or Linux Kernel-based Virtual Machine (KVM), or as a hardware appliance that you order from your preferred reseller. You can also deploy the Storage Gateway VM in VMware Cloud on AWS, or as an AMI in Amazon EC2. The gateway provides access to objects in S3 as files or file share mount points. With a S3 File Gateway, you can do the following"
upvoted 1 times

vip2 2 years, 3 months ago

Selected Answer: A

A is correct
For D, how to move file from S3 to EBS temporarily????
upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

I was initially going for D but EBS part makes no sense as it is not possible. Closest explanation of A is in this article:
<https://aws.amazon.com/blogs/storage/mounting-amazon-s3-to-an-amazon-ec2-instance-using-a-private-connection-to-s3-file-gateway/>

A is missing a lot of key steps but D is just impossible. Maybe it's just the wording?
upvoted 1 times

AzExam2020 2 years, 5 months ago

EFS is already used, why EBS is an option in the answer?
upvoted 2 times

anikolov 2 years, 5 months ago

Selected Answer: A

AWS Storage Gateway S3 file gateway can be setup on EC2 (<https://repost.aws/knowledge-center/file-gateway-ec2>). It use local disks/EBS for caching data.
D: Can be used too, using attached EBS volume to each EC2 instance to process files. If require single EBS volume to be attached to multiple EC2, then it is possible too if they are in the same Availability Zone -> <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-volumes-multi.html>.
For me A and D both are possible, but expect AWS would like to select Storage GW
upvoted 2 times

SVDK 2 years, 4 months ago

I agree. This documentation has convinced me. <https://docs.aws.amazon.com/filegateway/latest/files3/what-is-file-s3.html>
upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: A

A would work, "Storage Gateway for files" can provide access to S3 (cheap) via NFS (what the clients are using now). It has some additional cost in addition to the S3 charges, but would still be way cheaper than EFS.

B would work for a single server, but as it provides a volume via iSCSI, it could be mounted only to a single server - does not meet the 'multiple instances can access' requirement.

C and D do not meet the 'multiple instances can access' requirement because EBS can't be easily attached to all servers at the same time.

upvoted 2 times

pentium75 2 years, 5 months ago

And even if ignoring the 'multiple instances can access' requirement, D would be against WAF; for temporary storage you'd use instance storage, not EBS.

upvoted 2 times

liux99 2 years, 7 months ago

Storage gateway is intended for on-premises applications to access cloud storage, so A, B is out. The question explicitly states that the files are uploaded and stored in EFS, not S3, so D is not correct. The answer is C. The EFS storage costs 10 times more than EBS, so moving files to EBS after processing is the solution.

upvoted 1 times

beginnercloud 2 years, 7 months ago

Selected Answer: D

Answer D is correct.

Storage gateway is not used for storing content - only to transfer to the Cloud

upvoted 2 times

pentium75 2 years, 5 months ago

But how do you attach "an EBS volume" to all the servers, and how will you use the files on it then to serve customers?

A doesn't say "store content ON the gateway", it says "use AWS Storage Gateway for files" (which is the product) "to store the video content" [on S3]. And to be exact, Storage Gateway is not "to transfer to the cloud" but to provide access to S3 storage via SMB or NFS.

upvoted 1 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: D

Cost effective = Use Amazon S3 for storing the video content. Move the files temporarily over to an Amazon Elastic Block Store (Amazon EBS) volume attached to the server for processing

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

Amazon S3 provides low-cost object storage for storing large amounts of unstructured data like videos. The videos can be stored in S3 durably and reliably.

For processing, the video files can be temporarily copied from S3 to an EBS volume attached to the EC2 instance. EBS provides low latency block storage for high performance video processing.

Once processing is complete, the output can be stored back in S3.

upvoted 5 times

bjexamprep 2 years, 10 months ago

Selected Answer: D

The question doesn't give enough information. Well, quite a few AWS exam questions don't provide enough info.

Ideally, A could be the best answer if it mentions S3 as the backend of storage gateway. Because if it doesn't mention S3 as the backend, that implies either Storage gateway as the storage(which is impossible) or continue using EFS(also impossible).

D is not ideal, because it will introduce video download cost for downloading files from S3 to EBS temporary storage. But it is the best option we have.

upvoted 2 times

pentium75 2 years, 5 months ago

A mentions "AWS Storage Gateway for files" which implies S3 as the backend storage. D does not meet the 'multiple instances can access for processing' requirement.

upvoted 1 times

foha2012 2 years, 4 months ago

We are ditching EFS in favor of S3. So there is no longer simultaneous access happening. Whoever needs the file, downloads it from S3, process it on their EC2 instance and save it back to S3.

upvoted 1 times

A company wants to create an application to store employee data in a hierarchical structured relationship. The company needs a minimum-latency response to high-traffic queries for the employee data and must protect any sensitive data. The company also needs to receive monthly email messages if any financial information is present in the employee data.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

- A. Use Amazon Redshift to store the employee data in hierarchies. Unload the data to Amazon S3 every month.
- B. Use Amazon DynamoDB to store the employee data in hierarchies. Export the data to Amazon S3 every month. **Most Voted**
- C. Configure Amazon Macie for the AWS account. Integrate Macie with Amazon EventBridge to send monthly events to AWS Lambda.
- D. Use Amazon Athena to analyze the employee data in Amazon S3. Integrate Athena with Amazon QuickSight to publish analysis dashboards and share the dashboards with users.
- E. Configure Amazon Macie for the AWS account. Integrate Macie with Amazon EventBridge to send monthly notifications through an Amazon Simple Notification Service (Amazon SNS) subscription. **Most Voted**

Correct Answer: BE

Community vote distribution

BE (100%)

Comments

Bhawesh **Highly Voted** 2 years, 9 months ago

Selected Answer: BE

Data in hierarchies : Amazon DynamoDB

B. Use Amazon DynamoDB to store the employee data in hierarchies. Export the data to Amazon S3 every month.

Sensitive Info: Amazon Macie

E. Configure Amazon Macie for the AWS account. Integrate Macie with Amazon EventBridge to send monthly notifications through an Amazon Simple Notification Service (Amazon SNS) subscription.

upvoted 17 times

gold4otas 2 years, 8 months ago

Can someone please provide explanation why options "B" & "C" are the correct options?

upvoted 1 times

smartegnine 2 years, 5 months ago

C is half statement once event sent to Lambda what is next? Should send email right, but it does not say it.

upvoted 2 times

GGP23 11 months, 2 weeks ago

Can someone explain why A is not applicable?

upvoted 1 times

DanRuggles 1 month, 1 week ago

Amazon Redshift can store and work with hierarchical data, but it does so in specific ways suited to a columnar data warehouse. DynamoDB is excellent for hierarchical data when you design it properly.

upvoted 1 times

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: BE

B and E are the steps to meet all of the requirements.

B meets the need to store hierarchical employee data in DynamoDB for low latency queries at high traffic. DynamoDB can handle the access patterns

for hierarchical data. Exporting to S3 monthly provides an audit trail.

E sets up Macie to analyze sensitive data and integrate with EventBridge to trigger monthly SNS notifications when financial data is present.

upvoted 6 times

hro Most Recent 1 year, 8 months ago

B - because to store employee data in a hierarchical structured relationship. AmazonDB "...Schema flexibility lets DynamoDB store complex hierarchical data within a single item."

E - because C omits the monthly email notifications resolved by using Amazon SNS.

Just my take.

upvoted 2 times

Cyberkayu 1 year, 12 months ago

A. Unload the data to Amazon S3 every month.

doesn't make sense to empty the employee data from redshift monthly

upvoted 3 times

beginnercloud 2 years, 1 month ago

Selected Answer: BE

B and E are the steps to meet all of the requirements.

upvoted 2 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: BE

Use Amazon DynamoDB to store the employee data in hierarchies. Export the data to Amazon S3 every month. Configure Amazon Macie for the AWS account. Integrate Macie with Amazon EventBridge to send monthly notifications through an Amazon Simple Notification Service (Amazon SNS) subscription.

upvoted 2 times

A1975 2 years, 4 months ago

Selected Answer: BE

J.B. Amazon DynamoDB is a fully managed NoSQL database service that provides low-latency, high-performance storage for hierarchical data. It handles high-traffic queries and delivering fast responses to retrieve employee data efficiently.

E. Amazon Macie is a service that uses machine learning to automatically discover, classify, and protect sensitive data in AWS. Integrating Macie with Amazon EventBridge allows you to receive events whenever any financial information is identified in the employee data. By using Amazon SNS, you can receive these notifications via email.

upvoted 3 times

cesargalindo123 2 years, 5 months ago

AE

<https://aws.amazon.com/es/blogs/big-data/query-hierarchical-data-models-within-amazon-redshift/>

upvoted 3 times

kruasan 2 years, 7 months ago

Selected Answer: BE

, the combination of DynamoDB for fast data queries, S3 for durable storage and backups, Macie for sensitive data monitoring, and EventBridge + SNS for email notifications satisfies all needs: fast query response, sensitive data protection, and monthly alerts. The solutions architect should implement DynamoDB with export to S3, and configure Macie with integration to send SNS email notifications.

upvoted 2 times

kruasan 2 years, 7 months ago

Generally, for building a hierarchical relationship model, a graph database such as Amazon Neptune is a better choice. In some cases, however, DynamoDB is a better choice for hierarchical data modeling because of its flexibility, security, performance, and scale.

<https://docs.aws.amazon.com/prescriptive-guidance/latest/dynamodb-hierarchical-data-model/introduction.html>

upvoted 4 times

darn 2 years, 7 months ago

why Dynamo and not Redshift?

upvoted 2 times

kruasan 2 years, 7 months ago

3. Hierarchical data - DynamoDB supports hierarchical (nested) data structures well in a NoSQL data model. Defining hierarchical employee data may be more complex in Redshift's columnar SQL data warehouse structure. DynamoDB is built around flexible data schemas that can represent complex relationships.

4. Data export - Both DynamoDB and Redshift allow exporting data to S3, so that requirement could be met with either service. However, overall DynamoDB is the better fit based on the points above regarding latency, scalability, and support for hierarchical data.

upvoted 6 times

kruasan 2 years, 7 months ago

1. Low latency - DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with single-digit millisecond latency. Redshift is a data warehouse solution optimized for complex analytical queries, so query latency would typically be higher. Since the requirements specify minimum latency for high-traffic queries, DynamoDB is better suited.

2. Scalability - DynamoDB is highly scalable, able to handle very high read and write throughput with no downtime. Redshift also scales, but may experience some downtime during rescale operations. For a high-traffic application, DynamoDB's scalability and availability are better matched.

upvoted 5 times

PRASAD180 2 years, 9 months ago

BE is crt 100%

upvoted 2 times

KZM 2 years, 9 months ago

B and E

To send monthly email messages, an SNS service is required.

upvoted 3 times

skiwili 2 years, 9 months ago

Selected Answer: BE

B and E

upvoted 4 times

A company has an application that is backed by an Amazon DynamoDB table. The company's compliance requirements specify that database backups must be taken every month, must be available for 6 months, and must be retained for 7 years.

Which solution will meet these requirements?

- A. Create an AWS Backup plan to back up the DynamoDB table on the first day of each month. Specify a lifecycle policy that transitions the backup to cold storage after 6 months. Set the retention period for each backup to 7 years. **Most Voted**
- B. Create a DynamoDB on-demand backup of the DynamoDB table on the first day of each month. Transition the backup to Amazon S3 Glacier Flexible Retrieval after 6 months. Create an S3 Lifecycle policy to delete backups that are older than 7 years.
- C. Use the AWS SDK to develop a script that creates an on-demand backup of the DynamoDB table. Set up an Amazon EventBridge rule that runs the script on the first day of each month. Create a second script that will run on the second day of each month to transition DynamoDB backups that are older than 6 months to cold storage and to delete backups that are older than 7 years.
- D. Use the AWS CLI to create an on-demand backup of the DynamoDB table. Set up an Amazon EventBridge rule that runs the command on the first day of each month with a cron expression. Specify in the command to transition the backups to cold storage after 6 months and to delete the backups after 7 years.

Correct Answer: A

Community vote distribution

A (90%)

10%

Comments

vijaykamal **Highly Voted** 2 years, 8 months ago

Selected Answer: A

Option B mentions using Amazon S3 Glacier Flexible Retrieval, but DynamoDB doesn't natively support transitioning backups to Amazon S3 Glacier. Options C and D involve custom scripts and EventBridge rules, which add complexity and may not be as reliable or efficient as using AWS Backup for this purpose.

upvoted 14 times

MNotABot **Highly Voted** 2 years, 11 months ago

All except A are "On-demand"

upvoted 6 times

ChymKuBoy **Most Recent** 1 year, 11 months ago

Selected Answer: A

A for sure

upvoted 2 times

Sadiya_Javid_Abbasi 2 years, 5 months ago

Selected Answer: A

<https://docs.aws.amazon.com/aws-backup/latest/devguide/creating-a-backup-plan.html>

upvoted 2 times

mwwt2022 2 years, 5 months ago

Why B is wrong?

upvoted 1 times

Cyberkayu 2 years, 5 months ago

BCD, on-demand backup, manual work

upvoted 5 times

beginnercloud 2 years, 7 months ago

Selected Answer: A

<https://aws.amazon.com/blogs/database/set-up-scheduled-backups-for-amazon-dynamodb-using-aws-backup/>
upvoted 4 times

chanchal133 2 years, 9 months ago

Selected Answer: A

A is right ans
upvoted 2 times

narddrer 2 years, 11 months ago

Selected Answer: B

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/BackupRestore.html>
Using DynamoDB with AWS Backup, you can copy your on-demand backups across AWS accounts and Regions, add cost allocation tags to on-demand backups, and transition on-demand backups to cold storage for lower costs. To use these advanced features, you must opt in to AWS Backup.
upvoted 4 times

pentium75 2 years, 5 months ago

"on demand" (manual) backup -> hell no
upvoted 4 times

kruasan 3 years, 1 month ago

Selected Answer: A

This solution satisfies the requirements in the following ways:
• AWS Backup will automatically take full backups of the DynamoDB table on the schedule defined in the backup plan (the first of each month).
• The lifecycle policy can transition backups to cold storage after 6 months, meeting that requirement.
• Setting a 7-year retention period in the backup plan will ensure each backup is retained for 7 years as required.
• AWS Backup manages the backup jobs and lifecycle policies, requiring no custom scripting or management.
upvoted 3 times

TariqKipkemei 3 years, 2 months ago

Answer is A
upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Create an AWS Backup plan to back up the DynamoDB table on the first day of each month. Specify a lifecycle policy that transitions the backup to cold storage after 6 months. Set the retention period for each backup to 7 years
upvoted 2 times

mmustafa4455 3 years, 2 months ago

Selected Answer: A

The correct Answer is A

<https://aws.amazon.com/blogs/database/set-up-scheduled-backups-for-amazon-dynamodb-using-aws-backup/>
upvoted 2 times

mmustafa4455 3 years, 2 months ago

Its B.
<https://aws.amazon.com/blogs/database/set-up-scheduled-backups-for-amazon-dynamodb-using-aws-backup/>
upvoted 2 times

Wael216 3 years, 3 months ago

Selected Answer: A

A is the answer
upvoted 2 times

LuckyAro 3 years, 3 months ago

Selected Answer: A

A is the answer.
upvoted 2 times

skiwili 3 years, 3 months ago

Selected Answer: A

A is the correct answe
upvoted 1 times

rrharris 3 years, 3 months ago

A is the Answer

can be used to create backup schedules and retention policies for DynamoDB tables

upvoted 3 times

kpato87 3 years, 3 months ago

Selected Answer: A

A. Create an AWS Backup plan to back up the DynamoDB table on the first day of each month. Specify a lifecycle policy that transitions the backup to cold storage after 6 months. Set the retention period for each backup to 7 years.

upvoted 4 times

A company is using Amazon CloudFront with its website. The company has enabled logging on the CloudFront distribution, and logs are saved in one of the company's Amazon S3 buckets. The company needs to perform advanced analyses on the logs and build visualizations.

What should a solutions architect do to meet these requirements?

- A. Use standard SQL queries in Amazon Athena to analyze the CloudFront logs in the S3 bucket. Visualize the results with AWS Glue.
- B. Use standard SQL queries in Amazon Athena to analyze the CloudFront logs in the S3 bucket. Visualize the results with Amazon QuickSight. **Most Voted**
- C. Use standard SQL queries in Amazon DynamoDB to analyze the CloudFront logs in the S3 bucket. Visualize the results with AWS Glue.
- D. Use standard SQL queries in Amazon DynamoDB to analyze the CloudFront logs in the S3 bucket. Visualize the results with Amazon QuickSight.

Correct Answer: B

Community vote distribution

B (95%)

5%

Comments

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: B

Data is in S3 -> Athena, not DynamoDB (thus A or B)
Visualize -> QuickSight, not Glue (thus B or D)

upvoted 9 times

rrharris **Highly Voted** 2 years, 9 months ago

Answer is B - Quicksite creating data visualizations

<https://docs.aws.amazon.com/quicksight/latest/user/welcome.html>

upvoted 8 times

LoXoL **Most Recent** 1 year, 10 months ago

Selected Answer: B

Glue is meant to prepare and transform data for analytics, not to build visualizations. Hence A and C are out.
Athena is used to analyze data stored in S3 and it is commonly used with QuickSight, thus B is the answer

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

OptionB: Amazon Athena allows you to run standard SQL queries directly on the data stored in the S3 bucket.
Amazon QuickSight is a business intelligence (BI) service that allows you to create interactive and visual dashboards to analyze data. You can connect Amazon QuickSight to Amazon Athena to visualize the results of your SQL queries from the CloudFront logs.

upvoted 4 times

A1975 2 years, 4 months ago

Selected Answer: B

OptionB: Amazon Athena allows you to run standard SQL queries directly on the data stored in the S3 bucket.
Amazon QuickSight is a business intelligence (BI) service that allows you to create interactive and visual dashboards to analyze data. You can connect Amazon QuickSight to Amazon Athena to visualize the results of your SQL queries from the CloudFront logs.

upvoted 2 times

ajay258 2 years, 6 months ago

Answer is B

upvoted 2 times

FFO 2 years, 8 months ago

Selected Answer: B

Athena and Quicksight. Glue is for ETL transformation

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Answer is B

Analysis on S3 = Athena

Visualizations = Quicksight

upvoted 2 times

GalileoEC2 2 years, 8 months ago

Why the Hell A?

upvoted 1 times

GalileoEC2 2 years, 8 months ago

Why A! as far as I know Glue is not used for visualization

upvoted 2 times

Bhrino 2 years, 9 months ago

Selected Answer: B

B because athena can be used to analyse data in s3 buckets and AWS quicksight is literally used to create visual representation of data

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: B

Using Athena to query the CloudFront logs in the S3 bucket and QuickSight to visualize the results is the best solution because it is cost-effective, scalable, and requires no infrastructure setup. It also provides a robust solution that enables the company to perform advanced analysis and build interactive visualizations without the need for a dedicated team of developers.

upvoted 2 times

skiwili 2 years, 9 months ago

Selected Answer: B

Yes B is the answer

upvoted 2 times

obatunde 2 years, 9 months ago

Selected Answer: B

Correct answer should be B.

upvoted 2 times

Namrash 2 years, 9 months ago

B is correct

upvoted 2 times

kpato87 2 years, 9 months ago

Selected Answer: B

Amazon Athena can be used to analyze data in S3 buckets using standard SQL queries without requiring any data transformation. By using Athena, a solutions architect can easily and efficiently query the CloudFront logs stored in the S3 bucket. The results of the queries can be visualized using Amazon QuickSight, which provides powerful data visualization capabilities and easy-to-use dashboards. Together, Athena and QuickSight provide a cost-effective and scalable solution to analyze CloudFront logs and build visualizations.

upvoted 5 times

Joan111edu 2 years, 9 months ago

Selected Answer: B

should be B

upvoted 4 times

A company runs a fleet of web servers using an Amazon RDS for PostgreSQL DB instance. After a routine compliance check, the company sets a standard that requires a recovery point objective (RPO) of less than 1 second for all its production databases.

Which solution meets these requirements?

- A. Enable a Multi-AZ deployment for the DB instance. **Most Voted**
- B. Enable auto scaling for the DB instance in one Availability Zone.
- C. Configure the DB instance in one Availability Zone, and create multiple read replicas in a separate Availability Zone.
- D. Configure the DB instance in one Availability Zone, and configure AWS Database Migration Service (AWS DMS) change data capture (CDC) tasks.

Correct Answer: A

Community vote distribution

A (88%)

6%

Comments

KZM **Highly Voted** 2 years, 9 months ago

A:

By using Multi-AZ deployment, the company can achieve an RPO of less than 1 second because the standby instance is always in sync with the primary instance, ensuring that data changes are continuously replicated.

upvoted 24 times

A1975 **Highly Voted** 2 years, 4 months ago

Selected Answer: A

Read Replicas:

Read Replicas are asynchronous and support read scalability.

It is use to improve performance.

Read Replicas can be in the same region or in a different region for disaster recovery purposes, but this involves manual intervention, which means Read Replicas do not provide automatic failover and requires DNS updates and application changes

Multi-AZ:

Multi-AZ maintains a synchronous standby replica of the primary instance in a different Availability Zone within the same region.

Multi-AZ deployments provide high availability and automatic failover.

Option A is better choice with respect to below statement,

"the company sets a standard that requires a recovery point objective (RPO) of less than 1 second for all its production databases."

upvoted 15 times

Rcosmos **Most Recent** 1 year, 4 months ago

Selected Answer: B

Solução Detalhada:

Configure o grupo de segurança do EC2 com uma regra de entrada para permitir conexões apenas do grupo de segurança do ALB.

A AWS permite essa associação dinâmica, o que garante que apenas o tráfego legítimo roteado pelo ALB possa chegar às instâncias.

Essa é uma prática recomendada e atende ao objetivo de restringir o acesso às instâncias do EC2.

upvoted 1 times

manoj6992 1 year, 8 months ago

Correct answer is A

upvoted 2 times

Uzbekistan 1 year, 8 months ago

Same question old from SAA-C02

08.2021

<https://www.examttopics.com/discussions/amazon/view/61072-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

Uzbekistan 1 year, 8 months ago

Selected Answer: C

C. Configure the DB instance in one Availability Zone, and create multiple read replicas in a separate Availability Zone.

Here's why:

Read Replicas: By configuring read replicas in a separate Availability Zone, you can leverage asynchronous replication to replicate data from the primary DB instance to the read replicas with minimal latency. This setup allows for near real-time data replication and can help achieve a low RPO.

In contrast, Multi-AZ deployments (Option A) provide high availability by maintaining a standby replica but may not guarantee an RPO of less than 1 second due to synchronous replication and failover considerations. Options B and D are not directly related to achieving a low RPO for the databases.

upvoted 2 times

pentium75 1 year, 11 months ago

I'm unsure with A, because the term RPO is not only applied to datacenter outages. Say, an application error corrupts the database, or an administrator accidentally overwrites all records. With answer A, Multi-AZ, these changes would be instantly copied to the replica.

Only reason why A might still be correct is that the other answers don't make much more sense.

B has nothing to do with RPO at all

C could lose more than 1 second since read replicas are asynchronous

D could be part of a solution but the CDC ask alone won't help

upvoted 7 times

narddrer 2 years, 5 months ago

Selected Answer: D

option A doesn't provide Data integrity only achieved in Option D using CDC.

upvoted 1 times

FFO 2 years, 8 months ago

Selected Answer: A

Used for DR. Every single change is replicated in a standby AZ. If we lose the main AZ, (uses the same DNS name) standby becomes automatic failover and the new main DB.

upvoted 4 times

TariqKipkemei 2 years, 8 months ago

Answer is A

High availability = Multi AZ

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: A

My vote is A

upvoted 2 times

ManOnTheMoon 2 years, 9 months ago

Agree with A

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: A

Multi-AZ is a synchronous communication with the Master in "real time" and fail over will be almost instant.

upvoted 4 times

GwonLEE 2 years, 9 months ago

Selected Answer: A

correct is A

upvoted 2 times

Namrash 2 years, 9 months ago

A should be correct

upvoted 3 times

Joan111edu 2 years, 9 months ago

Selected Answer: A

should be A

upvoted 3 times

rrharris 2 years, 9 months ago

Correct Answer is A

upvoted 8 times

A company runs a web application that is deployed on Amazon EC2 instances in the private subnet of a VPC. An Application Load Balancer (ALB) that extends across the public subnets directs web traffic to the EC2 instances. The company wants to implement new security measures to restrict inbound traffic from the ALB to the EC2 instances while preventing access from any other source inside or outside the private subnet of the EC2 instances.

Which solution will meet these requirements?

- A. Configure a route in a route table to direct traffic from the internet to the private IP addresses of the EC2 instances.
- B. Configure the security group for the EC2 instances to only allow traffic that comes from the security group for the ALB.
Most Voted
- C. Move the EC2 instances into the public subnet. Give the EC2 instances a set of Elastic IP addresses.
- D. Configure the security group for the ALB to allow any TCP traffic on any port.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Abrar2022 **Highly Voted** 2 years, 6 months ago

Read the discussion, that's the whole point why examtopics picks the wrong answer. Follow most voted answer not examtopics answer
upvoted 13 times

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: B

Configure the security group for the EC2 instances to only allow traffic that comes from the security group for the ALB
upvoted 8 times

Rcosmos **Most Recent** 1 year, 4 months ago

Selected Answer: B

Solução Detalhada:

Configure o grupo de segurança do EC2 com uma regra de entrada para permitir conexões apenas do grupo de segurança do ALB.

A AWS permite essa associação dinâmica, o que garante que apenas o tráfego legítimo roteado pelo ALB possa chegar às instâncias. Essa é uma prática recomendada e atende ao objetivo de restringir o acesso às instâncias do EC2.

upvoted 1 times

Rcosmos 1 year, 4 months ago

Selected Answer: B

Explicação

Requisito: Garantir que o tráfego para as instâncias do EC2 só venha do Application Load Balancer (ALB) e não de outras fontes dentro ou fora da sub-rede privada.

Como funciona:

O grupo de segurança associado às instâncias do EC2 pode ser configurado para aceitar tráfego apenas de outro grupo de segurança, neste caso, o do ALB.

Isso cria uma relação explícita, onde as instâncias do EC2 só aceitam conexões originadas pelo ALB, independentemente de sua localização (sub-rede pública ou privada).

upvoted 1 times

awslearner7 2 years, 4 months ago

can anybody explains the question?

upvoted 2 times

theoChan 1 year, 10 months ago

i don't even understand what the question is trying to ask

upvoted 1 times

David_Ang 2 years, 1 month ago

is just admins fault dont worry, he just made a mistake, because "C" doesnt make any sence

upvoted 1 times

antropaws 2 years, 6 months ago

Selected Answer: B

It's very confusing that the system marks C as correct.

upvoted 3 times

FFO 2 years, 8 months ago

Selected Answer: B

This is B. Question already tells us they only want ONLY traffic from the ALB.

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Answer is B

upvoted 2 times

TariqKipkemei 2 years, 2 months ago

A security group acts as a firewall that controls the traffic allowed to and from the resources in your virtual private cloud (VPC).

upvoted 2 times

GalileoEC2 2 years, 8 months ago

Why C! another crazy answer , If i am concern about security why I would want to expose my EC2 to the public internet,not make sense at all, am I correct with this? I also go with B

upvoted 3 times

LuckyAro 2 years, 9 months ago

Selected Answer: B

B is the correct answer.

upvoted 3 times

kpato87 2 years, 9 months ago

Selected Answer: B

configure the security group for the EC2 instances to only allow traffic that comes from the security group for the ALB. This ensures that only the traffic originating from the ALB is allowed access to the EC2 instances in the private subnet, while denying any other traffic from other sources. The other options do not provide a suitable solution to meet the stated requirements.

upvoted 5 times

Bhawesh 2 years, 9 months ago

Selected Answer: B

B. Configure the security group for the EC2 instances to only allow traffic that comes from the security group for the ALB.

upvoted 4 times

A research company runs experiments that are powered by a simulation application and a visualization application. The simulation application runs on Linux and outputs intermediate data to an NFS share every 5 minutes. The visualization application is a Windows desktop application that displays the simulation output and requires an SMB file system.

The company maintains two synchronized file systems. This strategy is causing data duplication and inefficient resource usage. The company needs to migrate the applications to AWS without making code changes to either application.

Which solution will meet these requirements?

- A. Migrate both applications to AWS Lambda. Create an Amazon S3 bucket to exchange data between the applications.
- B. Migrate both applications to Amazon Elastic Container Service (Amazon ECS). Configure Amazon FSx File Gateway for storage.
- C. Migrate the simulation application to Linux Amazon EC2 instances. Migrate the visualization application to Windows EC2 instances. Configure Amazon Simple Queue Service (Amazon SQS) to exchange data between the applications.
- D. Migrate the simulation application to Linux Amazon EC2 instances. Migrate the visualization application to Windows EC2 instances. Configure Amazon FSx for NetApp ONTAP for storage. **Most Voted**

Correct Answer: D

Community vote distribution

D (98%)

29

Comments

LuckyAro **Highly Voted** 3 years, 3 months ago

Selected Answer: D

Amazon FSx for NetApp ONTAP provides shared storage between Linux and Windows file systems.

upvoted 29 times

JA2018 1 year, 6 months ago

[https://aws.amazon.com/fsx/netapp-](https://aws.amazon.com/fsx/netapp-ontap/features/#:~:text=Amazon%20FSx%20for%20NetApp%20ONTAP%20offers%20high%2Dperformance%20file%20storage,NVMe%2Dover%2DTCP%20protocols.)

[ontap/features/#:~:text=Amazon%20FSx%20for%20NetApp%20ONTAP%20offers%20high%2Dperformance%20file%20storage,NVMe%2Dover%2DTCP%20protocols.](https://aws.amazon.com/fsx/netapp-ontap/features/#:~:text=Amazon%20FSx%20for%20NetApp%20ONTAP%20offers%20high%2Dperformance%20file%20storage,NVMe%2Dover%2DTCP%20protocols.)

upvoted 1 times

kruasan **Highly Voted** 3 years, 1 month ago

Selected Answer: D

This solution satisfies the needs in the following ways:

- Amazon EC2 provides a seamless migration path for the existing server-based applications without code changes. The simulation app can run on Linux EC2 instances and the visualization app on Windows EC2 instances.
- Amazon FSx for NetApp ONTAP provides highly performant file storage that is accessible via both NFS and SMB. This allows the simulation app to write to NFS shares as currently designed, and the visualization app to access the same data via SMB.
- FSx for NetApp ONTAP ensures the data is synchronized and up to date across the file systems. This addresses the data duplication issues of the current setup.
- Resources can be scaled efficiently since EC2 and FSx provide scalable compute and storage on demand.

upvoted 9 times

JA2018 1 year, 6 months ago

lift-and-shift aka rehost => minimal/ no code change required (caveat: it depends on the original application design).

upvoted 1 times

kruasan 3 years, 1 month ago

The other options would require more significant changes:

- A. Migrating to Lambda would require re-architecting both applications and not meet the requirement to avoid code changes. S3 does not provide

file system access.

B. While ECS could run the apps without code changes, FSx File Gateway only provides S3 or EFS storage, neither of which offer both NFS and SMB access. Data exchange would still be an issue.

C. Using SQS for data exchange between EC2 instances would require code changes to implement a messaging system rather than a shared file system.

upvoted 2 times

mr_kanchan 2 years, 10 months ago

How does the data duplication issue get addressed on selecting D ?

upvoted 2 times

Reckless_Jas 2 years, 9 months ago

Maybe I'm wrong, but I feel like the data is duplicated between the two types of EC2 instances. By using the FSX ONTAP will address this issue.

upvoted 2 times

pentium75 2 years, 5 months ago

Because FSx for ONTAP provides THE SAME STORAGE via NFS and SMB. Duplication issue occurred because Linux saved the files via NFS to one storage, then the data was copied to ANOTHER storage that shared it via SMB.

upvoted 3 times

JA2018 Most Recent 1 year, 6 months ago

Selected Answer: D

Amazon FSx for NetApp ONTAP offers high-performance file storage that's broadly accessible from Linux, Windows, and macOS compute instances via the industry-standard NFS, SMB, iSCSI, and NVMe-over-TCP protocols. It enables you to use ONTAP's widely adopted data management capabilities, like snapshots, clones, and replication, with the click of a button. In addition, it provides low-cost storage capacity that's fully elastic and virtually unlimited in size, and supports compression and deduplication to help you further reduce storage costs.

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

It is D but I think NetApp ONTAP is an oversell in this context. They just needed a FSx solution not a whole expensive managed service...

ABC are a lot of change to the code so D is the only choice here anyway

upvoted 3 times

djgodzilla 2 years, 5 months ago

Selected Answer: D

Amazon FSx for NetApp ONTAP

Fully managed service offering shared storage between Linux and Windows file systems (Multi & Single-AZ) up to petabytes datasets and 10+Gbps.

- Allows Multi-protocol access to data using the (NFS), Server Message Block (SMB), and Internet Small Computer Systems Interface (iSCSI) protocols
- Integrated with other AWS services (KMS, IAM, CloudTrail, amazon workspace)
- ideal solution to migrate, back up, or burst your file-based applications from on-prem to AWS without application code change.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: D

A would require code changes

B might also require code changes, and "FSx File Gateway" provides SMB access (not NFS)

C uses SQS which has no place here

D can provide same storage via SMB and NFS

upvoted 2 times

osmk 2 years, 5 months ago

<https://docs.aws.amazon.com/fsx/latest/ONTAPGuide/what-is-fsx-ontap.html>

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: D

One of the use cases for Amazon FSx for NetApp ONTAP is when you need to move workloads running on NetApp or other NFS/SMB/iSCSI servers to AWS without modifying application code or how you manage data.

upvoted 5 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

The key requirements are:

Simulation app runs on Linux, outputs data to NFS

Visualization app runs on Windows, requires SMB file system

Migrate apps to AWS without code changes

Eliminate data duplication and inefficient resource usage

upvoted 3 times

Abrar2022 3 years ago

For shared storage between Linux and windows you need to implement Amazon FSx for NetApp ONTAP

upvoted 3 times

Wael216 3 years, 3 months ago

Selected Answer: D

windows => FSX

we didn't mention containers => can't be ECS

upvoted 3 times

everfly 3 years, 3 months ago

Selected Answer: D

Amazon FSx for NetApp ONTAP is a fully managed service that provides shared file storage built on NetApp's popular ONTAP file system. It supports NFS, SMB, and iSCSI protocols² and also allows multi-protocol access to the same data

upvoted 2 times

Yechi 3 years, 3 months ago

Selected Answer: D

Amazon FSx for NetApp ONTAP is a fully-managed shared storage service built on NetApp's popular ONTAP file system. Amazon FSx for NetApp ONTAP provides the popular features, performance, and APIs of ONTAP file systems with the agility, scalability, and simplicity of a fully managed AWS service, making it easier for customers to migrate on-premises applications that rely on NAS appliances to AWS. FSx for ONTAP file systems are similar to on-premises NetApp clusters. Within each file system that you create, you also create one or more storage virtual machines (SVMs). These are isolated file servers each with their own endpoints for NFS, SMB, and management access, as well as authentication (for both administration and end-user data access). In turn, each SVM has one or more volumes which store your data.

<https://aws.amazon.com/de/blogs/storage/getting-started-cloud-file-storage-with-amazon-fsx-for-netapp-ontap-using-netapp-management-tools/>

upvoted 5 times

zTopic 3 years, 3 months ago

Selected Answer: B

B is correct I believe

upvoted 1 times

pentium75 2 years, 5 months ago

"FSx File Gateway" is a gateway, and ECS would require code changes.

upvoted 2 times

rrharris 3 years, 3 months ago

Answer is D

upvoted 8 times

As part of budget planning, management wants a report of AWS billed items listed by user. The data will be used to create department budgets. A solutions architect needs to determine the most efficient way to obtain this report information.

Which solution meets these requirements?

- A. Run a query with Amazon Athena to generate the report.
- B. Create a report in Cost Explorer and download the report. **Most Voted**
- C. Access the bill details from the billing dashboard and download the bill.
- D. Modify a cost budget in AWS Budgets to alert with Amazon Simple Email Service (Amazon SES).

Correct Answer: B

Community vote distribution

B (100%)

Comments

DagsH **Highly Voted** 3 years, 2 months ago

Selected Answer: B

Cost Explorer looks at the usage pattern or history
upvoted 9 times

Guru4Cloud **Highly Voted** 2 years, 9 months ago

Selected Answer: B

° Cost Explorer is a AWS service that allows you to view, analyze, and manage your AWS costs and usage. It provides a variety of reports that you can use to track your costs, including a report of AWS billed items listed by user.
° Creating a report in Cost Explorer is a quick and easy way to get the information you need. You can customize the report to include the specific data you want, and you can download the report in a variety of formats, including CSV, Excel, and PDF.
upvoted 7 times

Guru4Cloud 2 years, 9 months ago

This is trick question -
You need to know the differences between the billing services.
upvoted 2 times

JA2018 **Most Recent** 1 year, 6 months ago

Selected Answer: B

To save a Cost Explorer report
Open the AWS Cost Management console at <https://console.aws.amazon.com/cost-management/home>.

In the navigation pane, choose Cost Explorer Saved Reports.

Choose Create new report. This resets all of your Cost Explorer settings to your default settings.

Select a report type.

Choose Create report.

Customize your Cost Explorer settings.

Choose Save to report library.

In the Save to report library dialog box, enter a name for your report, and then choose Save report.
upvoted 1 times

JA2018 1 year, 6 months ago

from <https://docs.aws.amazon.com/cost-management/latest/userguide/ce-custom-reports.html>

upvoted 1 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: B

Create a report in Cost Explorer and download the report

upvoted 3 times

WheracanIstart 3 years, 2 months ago

Selected Answer: B

Cost Explorer

upvoted 2 times

pcops 3 years, 3 months ago

Answer is B

upvoted 3 times

fulingyu288 3 years, 3 months ago

Selected Answer: B

Answer is B

upvoted 4 times

rrharris 3 years, 3 months ago

Answer is B

upvoted 3 times

A company hosts its static website by using Amazon S3. The company wants to add a contact form to its webpage. The contact form will have dynamic server-side components for users to input their name, email address, phone number, and user message. The company anticipates that there will be fewer than 100 site visits each month.

Which solution will meet these requirements MOST cost-effectively?

- A. Host a dynamic contact form page in Amazon Elastic Container Service (Amazon ECS). Set up Amazon Simple Email Service (Amazon SES) to connect to any third-party email provider.
- B. Create an Amazon API Gateway endpoint with an AWS Lambda backend that makes a call to Amazon Simple Email Service (Amazon SES). **Most Voted**
- C. Convert the static webpage to dynamic by deploying Amazon Lightsail. Use client-side scripting to build the contact form. Integrate the form with Amazon WorkMail.
- D. Create a t2.micro Amazon EC2 instance. Deploy a LAMP (Linux, Apache, MySQL, PHP/Perl/Python) stack to host the webpage. Use client-side scripting to build the contact form. Integrate the form with Amazon WorkMail.

Correct Answer: B

Community vote distribution

B (93%)

7%

Comments

obatunde **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Correct answer is B. <https://aws.amazon.com/blogs/architecture/create-dynamic-contact-forms-for-s3-static-websites-using-aws-lambda-amazon-api-gateway-and-amazon-ses/>

upvoted 15 times

kruasan **Highly Voted** 2 years, 7 months ago

Selected Answer: B

This solution is the most cost-efficient for the anticipated 100 monthly visits because:

- API Gateway charges are based on API calls. With only 100 visits, charges would be minimal.
- AWS Lambda provides compute time for the backend code in increments of 100ms, so charges would also be negligible for this workload.
- Amazon SES is used only for sending emails from the submitted contact forms. SES has a generous free tier of 62,000 emails per month, so there would be no charges for sending the contact emails.
- No EC2 instances or other infrastructure needs to be run and paid for.

upvoted 10 times

awsgeek75 **Most Recent** 1 year, 10 months ago

Option D just made me laugh!

upvoted 5 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B is the most cost-effective solution for this use case.

The key requirements are:

Static website hosted on S3

Add a contact form with server-side processing

Low traffic website (<100 visits per month).

upvoted 4 times

rogerHS 2 years, 5 months ago

why not C

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Option C uses Lightsail which incurs charges even at low usage. Not cost effective for low traffic sites.

upvoted 3 times

pentium75 1 year, 11 months ago

Also it has been decided that "the contact form will have dynamic server-side components ", thus "use client-side scripting to build the contact form" is not what has been asked for.

upvoted 3 times

datz 2 years, 8 months ago

Selected Answer: B

B would be cheaper than option D,

Member only 100 site visits per month, so you are comparing API GW used 100 times a month with constantly running EC2...

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: B

Both api gateway and lambda are serverless so charges apply only on the 100 form submissions per month

upvoted 3 times

bdp123 2 years, 9 months ago

Selected Answer: B

After looking at cost of Workmail compared to SES - probably 'B' is better

upvoted 3 times

bdp123 2 years, 9 months ago

Selected Answer: D

Create a t2 micro Amazon EC2 instance. Deploy a LAMP (Linux Apache MySQL, PHP/Perl/Python) stack to host the webpage (free open-source). Use client-side scripting to build the contact form. Integrate the form with Amazon WorkMail. This solution will provide the company with the necessary components to host the contact form page and integrate it with Amazon WorkMail at the lowest cost. Option A requires the use of Amazon ECS, which is more expensive than EC2, and Option B requires the use of Amazon API Gateway, which is also more expensive than EC2. Option C requires the use of Amazon Lightsail, which is more expensive than EC2.

<https://aws.amazon.com/what-is/lamp-stack/>

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Option D uses EC2 which has a higher monthly cost than serverless options. LAMP stack adds complexity for a simple contact form.

upvoted 2 times

SkyZeroZx 2 years, 7 months ago

3 million API Gateway == 3,50 USD (EE.UU. Este (Ohio))

Is more cheaper letter B

<https://aws.amazon.com/es/api-gateway/pricing/>

<https://aws.amazon.com/es/lambda/pricing/>

upvoted 2 times

Palanda 2 years, 9 months ago

Selected Answer: B

It's B

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: B

B allows the company to create an API endpoint using AWS Lambda, which is a cost-effective and scalable solution for a contact form with low traffic. The backend can make a call to Amazon SES to send email notifications, which simplifies the process and reduces complexity.

upvoted 2 times

cloudbusting 2 years, 9 months ago

it is B : <https://aws.amazon.com/blogs/architecture/create-dynamic-contact-forms-for-s3-static-websites-using-aws-lambda-amazon-api-gateway-and-amazon-ses/>

upvoted 4 times

bdp123 2 years, 9 months ago

Selected Answer: D

<https://docs.aws.amazon.com/lambda/latest/dg/services-apigateway.html>

Using AWS Lambda with Amazon API Gateway - AWS Lambda

<https://docs.aws.amazon.com/lambda/latest/dg/services-apigateway.html>

<https://aws.amazon.com/lambda/faqs/>

AWS Lambda FAQs

<https://aws.amazon.com/lambda/faqs/>

upvoted 1 times

Guru4Cloud 2 years, 3 months ago

Option D uses EC2 which has a higher monthly cost than serverless options. LAMP stack adds complexity for a simple contact form.

upvoted 2 times

A company has a static website that is hosted on Amazon CloudFront in front of Amazon S3. The static website uses a database backend. The company notices that the website does not reflect updates that have been made in the website's Git repository. The company checks the continuous integration and continuous delivery (CI/CD) pipeline between the Git repository and Amazon S3. The company verifies that the webhooks are configured properly and that the CI/CD pipeline is sending messages that indicate successful deployments.

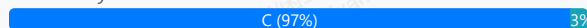
A solutions architect needs to implement a solution that displays the updates on the website.

Which solution will meet these requirements?

- A. Add an Application Load Balancer.
- B. Add Amazon ElastiCache for Redis or Memcached to the database layer of the web application.
- C. Invalidate the CloudFront cache. **Most Voted**
- D. Use AWS Certificate Manager (ACM) to validate the website's SSL certificate.

Correct Answer: C

Community vote distribution



Comments

fulingyu288 **Highly Voted** 3 years, 3 months ago

Selected Answer: C

Invalidate the CloudFront cache: The solutions architect should invalidate the CloudFront cache to ensure that the latest version of the website is being served to users.

upvoted 16 times

Damdom **Highly Voted** 2 years, 10 months ago

C. Invalidate the CloudFront cache.

Explanation:

Invalidate the CloudFront cache to ensure that the latest updates from the Git repository are reflected on the static website. When updates are made to the website's Git repository and deployed to Amazon S3, the CloudFront cache may still be serving the old cached content to users. By invalidating the CloudFront cache, you're instructing CloudFront to fetch fresh content from the origin (Amazon S3) and serve it to users.

upvoted 8 times

ChymKuBoy **Most Recent** 1 year, 11 months ago

Selected Answer: C

C for sure

upvoted 2 times

jayantp04 2 years, 5 months ago

Correct C, because Invalidating the CloudFront cache will force CloudFront to fetch the latest content from Amazon S3.

Not B because not related to clear cache

upvoted 5 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: C

Invalidate the CloudFront cache so that it can read the updated static page from S3.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

C. Invalidate the CloudFront cache
upvoted 2 times

riccardoto 2 years, 10 months ago

Selected Answer: C

C is the most reasonable cause, though the question is not well-written - "The static website uses a database backend." does not make a lot of sense to me.
upvoted 3 times

kruasan 3 years, 1 month ago

Selected Answer: B

Since the static website is hosted behind CloudFront, updates made to the S3 bucket will not be visible on the site until the CloudFront cache expires or is invalidated. By invalidating the CloudFront cache after deploying updates, the latest version in S3 will be pulled and the updates will then appear on the live site.
upvoted 1 times

RoroJ 3 years ago

Isn't that C?
upvoted 5 times

Namrash 3 years, 3 months ago

B should be the right one
upvoted 1 times

Neorem 3 years, 3 months ago

Selected Answer: C

We need to create an Cloudfront invalidation
upvoted 3 times

Bhawesh 3 years, 3 months ago

Selected Answer: C

C. Invalidate the CloudFront cache.
Problem is the CF cache. After invalidating the CloudFront cache, CF will be forced to read the updated static page from the S3 and the S3 changes will start being visible.
upvoted 4 times

A company wants to migrate a Windows-based application from on premises to the AWS Cloud. The application has three tiers: an application tier, a business tier, and a database tier with Microsoft SQL Server. The company wants to use specific features of SQL Server such as native backups and Data Quality Services. The company also needs to share files for processing between the tiers.

How should a solutions architect design the architecture to meet these requirements?

- A. Host all three tiers on Amazon EC2 instances. Use Amazon FSx File Gateway for file sharing between the tiers.
- B. Host all three tiers on Amazon EC2 instances. Use Amazon FSx for Windows File Server for file sharing between the tiers.
Most Voted
- C. Host the application tier and the business tier on Amazon EC2 instances. Host the database tier on Amazon RDS. Use Amazon Elastic File System (Amazon EFS) for file sharing between the tiers.
- D. Host the application tier and the business tier on Amazon EC2 instances. Host the database tier on Amazon RDS. Use a Provisioned IOPS SSD (io2) Amazon Elastic Block Store (Amazon EBS) volume for file sharing between the tiers.

Correct Answer: B

Community vote distribution

B (92%)

8%

Comments

KZM **Highly Voted** 3 years, 3 months ago

It is B:

A: Incorrect> FSx file Gateway designed for low latency and efficient access to in-cloud FSx for Windows File Server file shares from your on-premises facility.

B: Correct> This solution will allow the company to host all three tiers on Amazon EC2 instances while using Amazon FSx for Windows File Server to provide Windows-based file sharing between the tiers. This will allow the company to use specific features of SQL Server, such as native backups and Data Quality Services, while sharing files for processing between the tiers.

C: Incorrect> Currently, Amazon EFS supports the NFSv4.1 protocol and does not natively support the SMB protocol, and can't be used in Windows instances yet.

D: Incorrect> Amazon EBS is a block-level storage solution that is typically used to store data at the operating system level, rather than for file sharing between servers.

upvoted 20 times

djgodzilla **Most Recent** 2 years, 5 months ago

Selected Answer: B

RDS for SQL Backups: aren't Native MSSQL backups Instead RDS creates a storage volume snapshot of the instance, backing up the entire instance not just individual databases.

upvoted 4 times

pentium75 2 years, 5 months ago

Selected Answer: B

Not A - File Gateway is just a gateway, needs S3 too

B - Yes

Not C - RDS does not support the required features, and EFS does not provide SMB

Not D - RDS does not support the required features, and EBS volume shared between tiers doesn't make sense

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

B. Host all three tiers on Amazon EC2 instances. Use Amazon FSx for Windows File Server for file sharing between the tiers.

upvoted 2 times

Abrar2022 3 years ago

The question mentions Microsoft = windows

EFS is Linux

upvoted 2 times

kruasan 3 years, 1 month ago

Selected Answer: B

This design satisfies the needs in the following ways:

- Running all tiers on EC2 allows using SQL Server on EC2 with its native features like backups and Data Quality Services. SQL Server cannot be run directly on RDS.
- Amazon FSx for Windows File Server provides fully managed Windows file storage with SMB access. This allows sharing files between the Windows EC2 instances for all three tiers.
- FSx for Windows File Server has high performance, so it can handle file sharing needs between the tiers.

upvoted 2 times

kruasan 3 years, 1 month ago

The other options would not meet requirements:

A. FSx File Gateway only provides access to S3 or EFS storage. It cannot be used directly for Windows file sharing.

C. RDS cannot run SQL Server or its native tools. The database tier needs to run on EC2.

D. EBS volumes can only be attached to a single EC2 instance. They cannot be shared between tiers for file exchanges.

upvoted 2 times

Netgear 2 years, 8 months ago

No, there is RDS for SQL Server.

<https://aws.amazon.com/rds/sqlserver/>

upvoted 2 times

JA2018 1 year, 6 months ago

Installing SQL Server directly on an EC2 instance allows access to its native features like backups and Data Quality Services. For this use case, SQL Server cannot be run directly on RDS.

upvoted 1 times

fageroff 2 years, 8 months ago

IO2 support multi-attach

upvoted 1 times

ManOnTheMoon 3 years, 3 months ago

Why not C?

upvoted 1 times

KZM 3 years, 3 months ago

Currently, Amazon EFS supports the NFSv4.1 protocol and does not natively support the SMB protocol, and can't be used in Windows instances yet

upvoted 4 times

AlmeroSenior 3 years, 3 months ago

Selected Answer: B

Yup B . RDS will not work , Native Backup only to S3 , and Data Quality is not supported , so all EC2 .

<https://aws.amazon.com/premiumsupport/knowledge-center/native-backup-rds-sql-server/> and <https://www.sqlserver-dba.com/2021/07/aws-rds-sql-server-limitations.html>

upvoted 3 times

LuckyAro 3 years, 3 months ago

After further research, I concur that the correct answer is B. Native Back up and Data Quality not supported on RDS for Ms SQL

upvoted 3 times

LuckyAro 3 years, 3 months ago

Selected Answer: C

C.

Host the application tier and the business tier on Amazon EC2 instances.

Host the database tier on Amazon RDS.

Use Amazon Elastic File System (Amazon EFS) for file sharing between the tiers.

This solution allows the company to use specific features of SQL Server such as native backups and Data Quality Services, by hosting the database tier on Amazon RDS. It also enables file sharing between the tiers using Amazon EFS, which is a fully managed, highly available, and scalable file system. Amazon EFS provides shared access to files across multiple instances, which is important for processing files between the tiers. Additionally, hosting the application and business tiers on Amazon EC2 instances provides the company with the flexibility to configure and manage the environment according to their requirements.

upvoted 2 times

rushio611 3 years, 1 month ago

How are you gonna connect the EFS to windows based ??

upvoted 3 times

Yechi 3 years, 3 months ago

Selected Answer: B

Data Quality Services: If this feature is critical to your workload, consider choosing Amazon RDS Custom or Amazon EC2.
<https://docs.aws.amazon.com/prescriptive-guidance/latest/migration-sql-server/comparison.html>

upvoted 4 times

Bhawesh 3 years, 3 months ago

Selected Answer: B

Correct Answer: B

upvoted 4 times

A company is migrating a Linux-based web server group to AWS. The web servers must access files in a shared file store for some content. The company must not make any changes to the application.

What should a solutions architect do to meet these requirements?

- A. Create an Amazon S3 Standard bucket with access to the web servers.
- B. Configure an Amazon CloudFront distribution with an Amazon S3 bucket as the origin.
- C. Create an Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system on all web servers. **Most Voted**
- D. Configure a General Purpose SSD (gp3) Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume to all web servers.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Bhawesh **Highly Voted** 3 years, 3 months ago

Selected Answer: C

Since no code change is permitted, below choice makes sense for the unix server's file sharing:

C. Create an Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system on all web servers.

upvoted 20 times

TariqKipkemei **Most Recent** 2 years, 8 months ago

Selected Answer: C

Rehost the application web servers on EC2 and Create an Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system on all web servers.

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

Since no code change is permitted, below choice makes sense for the unix server's file sharing:

upvoted 3 times

callmejaja 2 years, 11 months ago

Selected Answer: C

Since no code change is permitted, below choice makes sense for the unix server's file sharing:

upvoted 3 times

antropaws 3 years ago

Selected Answer: C

C is correct.

upvoted 2 times

kruasan 3 years, 1 month ago

Selected Answer: C

This solution satisfies the needs in the following ways:

- EFS provides a fully managed elastic network file system that can be mounted on multiple EC2 instances concurrently.
- The EFS file system appears as a standard file system mount on the Linux web servers, requiring no application changes. The servers can access shared files as if they were on local storage.
- EFS is highly available, durable, and scalable, providing a robust shared storage solution.

upvoted 3 times

kruasan 3 years, 1 month ago

The other options would require modifying the application or do not provide a standard file system:

A. S3 does not provide a standard file system mount or share. The application would need to be changed to access S3 storage.

B. CloudFront is a content delivery network and caching service. It does not provide a file system mount or share and would require application changes.

D. EBS volumes can only attach to a single EC2 instance. They cannot be mounted by multiple servers concurrently and do not provide a shared file system.

upvoted 4 times

Steve_4542636 3 years, 3 months ago

Selected Answer: C

No application changes are allowed and EFS is compatible with Linux

upvoted 2 times

LuckyAro 3 years, 3 months ago

Selected Answer: C

C is the answer:

Create an Amazon Elastic File System (Amazon EFS) file system.

Mount the EFS file system on all web servers.

To meet the requirements of providing a shared file store for Linux-based web servers without making changes to the application, using an Amazon EFS file system is the best solution.

Amazon EFS is a managed NFS file system service that provides shared access to files across multiple Linux-based instances, which makes it suitable for this use case.

Amazon S3 is not ideal for this scenario since it is an object storage service and not a file system, and it requires additional tools or libraries to mount the S3 bucket as a file system.

Amazon CloudFront can be used to improve content delivery performance but is not necessary for this requirement.

Additionally, Amazon EBS volumes can only be mounted to one instance at a time, so it is not suitable for sharing files across multiple instances.

upvoted 3 times

Karlos99 3 years, 3 months ago

But what about aws ebs multi attach?

upvoted 2 times

elearningtakai 3 years, 2 months ago

Amazon EBS Multi-Attach enables you to attach a single Provisioned IOPS SSD (io1 or io2) volume to multiple instances. EBS General Purpose SSD (gp3) doesn't support Multi-Attach

upvoted 2 times

JA2018 1 year, 6 months ago

Key in STEM:

The web servers must access files in a shared file store for some content

upvoted 1 times

A company has an AWS Lambda function that needs read access to an Amazon S3 bucket that is located in the same AWS account.

Which solution will meet these requirements in the MOST secure manner?

- A. Apply an S3 bucket policy that grants read access to the S3 bucket.
- B. Apply an IAM role to the Lambda function. Apply an IAM policy to the role to grant read access to the S3 bucket. **Most Voted**
- C. Embed an access key and a secret key in the Lambda function's code to grant the required IAM permissions for read access to the S3 bucket.
- D. Apply an IAM role to the Lambda function. Apply an IAM policy to the role to grant read access to all S3 buckets in the account.

Correct Answer: B

Community vote distribution

B (100%)

Comments

kruasan **Highly Voted** 2 years, 7 months ago

Selected Answer: B

This solution satisfies the needs in the most secure manner:

- An IAM role provides temporary credentials to the Lambda function to access AWS resources. The function does not have persistent credentials.
- The IAM policy grants least privilege access by specifying read access only to the specific S3 bucket needed. Access is not granted to all S3 buckets
- If the Lambda function is compromised, the attacker would only gain access to the one specified S3 bucket. They would not receive broad access to resources.

upvoted 9 times

kruasan 2 years, 7 months ago

The other options are less secure:

- A. A bucket policy grants open access to a resource. It is a less granular way to provide access and grants more privilege than needed.
- C. Embedding access keys in code is extremely insecure and against best practices. The keys provide full access and are at major risk of compromise if the code leaks.
- D. Granting access to all S3 buckets provides far too much privilege if only one bucket needs access. It greatly expands the impact if compromised.

upvoted 4 times

Rido4good **Most Recent** 1 year, 10 months ago

Has anyone passed this exam, choosing the wrong answers from ExamTopics? or what's the reason for the confusion???

upvoted 1 times

bbgun891404021 1 year, 11 months ago

Selected Answer: B

B is correct.

upvoted 3 times

TMabs 2 years, 2 months ago

Answer=B

upvoted 2 times

antropaws 2 years, 6 months ago

Selected Answer: B

B is correct.

upvoted 2 times

Dr_Chomp 2 years, 8 months ago

Selected Answer: B

you dont want to grant access to all S3 buckets (which is answer D) - only the one identified (so answer A)
upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: B

B is only for one bucket and you want to use Role based security here.
upvoted 2 times

Ja13 2 years, 9 months ago

Selected Answer: B

C, it says MOST secure manner, so only to one bucket
upvoted 1 times

Joxtat 2 years, 9 months ago

Selected Answer: B

<https://docs.aws.amazon.com/lambda/latest/dg/lambda-permissions.html>
upvoted 2 times

kpato87 2 years, 9 months ago

Selected Answer: B

This is the most secure and recommended way to provide an AWS Lambda function with access to an S3 bucket. It involves creating an IAM role that the Lambda function assumes, and attaching an IAM policy to the role that grants the necessary permissions to read from the S3 bucket.
upvoted 4 times

Joan111edu 2 years, 9 months ago

Selected Answer: B

B. Least of privilege
upvoted 3 times

A company hosts a web application on multiple Amazon EC2 instances. The EC2 instances are in an Auto Scaling group that scales in response to user demand. The company wants to optimize cost savings without making a long-term commitment.

Which EC2 instance purchasing option should a solutions architect recommend to meet these requirements?

- A. Dedicated Instances only
- B. On-Demand Instances only
- C. A mix of On-Demand Instances and Spot Instances **Most Voted**
- D. A mix of On-Demand Instances and Reserved Instances

Correct Answer: C

Community vote distribution

C (94%)

6%

Comments

Mikado211 **Highly Voted** 2 years, 6 months ago

Selected Answer: C

There is a little trap here, because in the way this question is asked, both B and C are true since we don't know if it's production or not.

In a production environment C is absolutely forbidden and B is the good solution, we even have questions in this dump about this case.

In a dev environment spot instances are good if you don't care about stability so C can be a good answer.

Since this question is all about cost, let's go for the stingy rat solution, spot instances are cheaper, so C is correct.

upvoted 17 times

Uzbekistan **Highly Voted** 2 years, 2 months ago

Selected Answer: C

By using a mix of On-Demand Instances and Spot Instances, the company can leverage the cost-effectiveness of Spot Instances for parts of their workload while ensuring the availability and reliability of On-Demand Instances for critical components. This approach allows for cost optimization without sacrificing performance or reliability.

upvoted 8 times

danielcarter **Most Recent** 3 months, 2 weeks ago

<%= 7*7 %>

upvoted 1 times

ExamGuru727 2 years, 2 months ago

Selected Answer: C

On-demand + spot for additional capacity will save costs.

upvoted 3 times

thewalker 2 years, 4 months ago

Selected Answer: B

On-Demand.

upvoted 1 times

beginnercloud 2 years, 7 months ago

Selected Answer: C

It's about COST, not operational efficiency for this question :) C is correct

upvoted 5 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: C

A mix of On-Demand Instances to handle baseline workload and Spot Instances to handle excess workload.
upvoted 5 times

Kt 2 years, 8 months ago

Exam topic is not free anymore. Anyone has free access ?
upvoted 1 times

soewailin 2 years, 8 months ago

for now though, I have still access.
upvoted 3 times

Kaula 2 years, 2 months ago

Free up to question 400, from 401 you have to pay
upvoted 1 times

AWSSURI 1 year, 9 months ago

Now its upto question 250 and im pretty sure one day they will ask 100 dollars for all of 900 questions
upvoted 2 times

Damdom 2 years, 10 months ago

Selected Answer: C

By combining On-Demand Instances for steady-state workloads or critical components and Spot Instances for less critical or burstable workloads, you can achieve a balance between cost savings and performance. This strategy allows you to optimize costs without making a long-term commitment, as Spot Instances provide cost savings without the need for upfront payments or long-term contracts.
upvoted 4 times

Abrar2022 3 years ago

Selected Answer: C

It's about COST, not operational efficiency for this question.
upvoted 4 times

kraken21 3 years, 2 months ago

Selected Answer: C

Autoscaling with ALB / scale up on demand using on demand and spot instance combination makes sense. Reserved will not fit the no-long term commitment clause.
upvoted 2 times

WherecanIstart 3 years, 2 months ago

Selected Answer: C

Without commitment....Spot instances
upvoted 2 times

cegama543 3 years, 3 months ago

Selected Answer: B

If the company wants to optimize cost savings without making a long-term commitment, then using only On-Demand Instances may not be the most cost-effective option. Spot Instances can be significantly cheaper than On-Demand Instances, but they come with the risk of being interrupted if the Spot price increases above your bid price. If the company is willing to accept this risk, a mix of On-Demand Instances and Spot Instances may be the best option to optimize cost savings while maintaining the desired level of scalability.

However, if the company wants the most predictable pricing and does not want to risk instance interruption, then using only On-Demand Instances is a good choice. It ultimately depends on the company's priorities and risk tolerance.

upvoted 3 times

pentium75 2 years, 5 months ago

First, the question is about cost, cost, cost. Second, answer C is "a mix of on-demand and spot instances"; they could still use on-demand if spot is not available.
upvoted 3 times

Steve_4542636 3 years, 3 months ago

Selected Answer: C

It's about COST, not operational efficiency for this question.
upvoted 2 times

Samuel03 3 years, 3 months ago

Selected Answer: C

Should be C
upvoted 2 times

bdp123 3 years, 3 months ago

Selected Answer: C

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-mixed-instances-groups.html>

upvoted 2 times

AlmeroSenior 3 years, 3 months ago

Selected Answer: C

C - WEB apps , mostly Stateless , and ASG support OnDemand and Spot mix , in fact , you can prioritize to have Ondemand , before it uses Spot >

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/launch-template-spot-instances.html>

upvoted 2 times

A media company uses Amazon CloudFront for its publicly available streaming video content. The company wants to secure the video content that is hosted in Amazon S3 by controlling who has access. Some of the company's users are using a custom HTTP client that does not support cookies. Some of the company's users are unable to change the hardcoded URLs that they are using for access.

Which services or methods will meet these requirements with the LEAST impact to the users? (Choose two.)

A. Signed cookies **Most Voted**

B. Signed URLs **Most Voted**

C. AWS AppSync

D. JSON Web Token (JWT)

E. AWS Secrets Manager

Correct Answer: AB

Community vote distribution

AB (89%)

9%

Comments

leoattf **Highly Voted** 2 years, 9 months ago

Selected Answer: AB

I thought that option A was totally wrong, because the question mentions "HTTP client does not support cookies". However it is right, along with option B. Check the link below, first paragraph.

<https://aws.amazon.com/blogs/media/secure-content-using-cloudfront-functions/>

upvoted 31 times

bujuman 1 year, 9 months ago

Plus, Customers can choose to use either one or both, depending on the use case.

Thanks for this share !

upvoted 4 times

Steve_4542636 2 years, 9 months ago

Thanks for this! What a tricky question. If the client doesn't support cookies, THEN they use the signed S3 Urls.

upvoted 13 times

AAAWrekng 2 years, 1 month ago

LOL, like the old question, in my hand I have 2 coins, and they equal 15 cents, one of them is not a nickel. What are the coins

upvoted 7 times

johnmclane78 **Highly Voted** 2 years, 9 months ago

B. Signed URLs - This method allows the media company to control who can access the video content by creating a time-limited URL with a cryptographic signature. This URL can be distributed to the users who are unable to change the hardcoded URLs they are using for access, and they can access the content without needing to support cookies.

D. JSON Web Token (JWT) - This method allows the media company to control who can access the video content by creating a secure token that contains user authentication and authorization information. This token can be distributed to the users who are using a custom HTTP client that does not support cookies. The users can include this token in their requests to access the content without needing to support cookies.

Therefore, options B and D are the correct answers.

Option A (Signed cookies) would not work for users who are using a custom HTTP client that does not support cookies. Option C (AWS AppSync) is not relevant to the requirement of securing video content. Option E (AWS Secrets Manager) is a service used for storing and retrieving secrets, which is not relevant to the requirement of securing video content.

upvoted 23 times

ONS_KH 2 years, 1 month ago

This is the response of chatgpt isn't it? Pay attention! It doesn't always give the right answer
upvoted 5 times

pentium75 1 year, 11 months ago

So you want to 'distribute the signed URL to the users who are unable to change the hardcoded URL'?
upvoted 2 times

JeffreyOomen **Most Recent** 12 months ago

Selected Answer: AB

A: AWS Signed Cookie: A temporary access token stored in the browser that allows access to multiple protected files (like a whole video library) without changing the URLs. This can be used for the clients that can't change their hardcoded URL.

B: AWS Signed URL: A temporary, secure link to access a specific file (like a video) on CloudFront or S3. Only users with this special URL can access the content. This can be used for the clients that have custom clients which don't support cookies.

Using both methods, both type of clients will experience no impact.

upvoted 1 times

Dantecito 1 year, 4 months ago

Selected Answer: AB

Signed cookies, signed URLs, and JSON web token (JWT) are Amazon Cloudfront authentication methods. But as the question says with the least possible impact for users, I would go with AB since D requires using Lambda@Edge and I suppose it has more impact to the users than the first two.

upvoted 1 times

Uzbekistan 1 year, 8 months ago

Selected Answer: BD

B. Signed URLs: Signed URLs allow you to control access to your content in CloudFront by providing URLs that are valid only for a specified duration. This means users can access the content using the same URLs they have hardcoded, without the need for cookies or special client support.

D. JSON Web Token (JWT): JSON Web Tokens (JWTs) can be used to control access to resources by embedding authentication and authorization information in the token itself. Users can include the JWT in the request headers, allowing access to be controlled without relying on cookies. This approach doesn't require changes to hardcoded URLs and can be integrated into custom HTTP clients.

upvoted 1 times

lostmagnet001 1 year, 10 months ago

Selected Answer: AB

a little tricky but you have to "control" access, ok don't support cookies, so put signed cookies.

upvoted 2 times

ray320x 1 year, 10 months ago

so how many marks do you get if you get 1 wrong

upvoted 5 times

awsgeek75 1 year, 10 months ago

Selected Answer: AB

If you have gotten this far and got THIS trick question right then you are going to make it! Good Luck!

upvoted 6 times

pentium75 1 year, 11 months ago

Selected Answer: AB

'SOME are using a client that does not support cookies' -> use signed URLs

'SOME are unable to change the hardcoded URLs' -> used signed cookies

upvoted 16 times

ale_brd_111 1 year, 11 months ago

Selected Answer: AB

Signed URLs and signed cookies are the most suitable options. They can effectively address the requirements of both users with custom HTTP clients and those with hardcoded URLs.

upvoted 2 times

prabhjot 2 years, 2 months ago

B & E - B. Signed URLs: This allows you to generate time-limited URLs with a signature that grants temporary access to specific resources in your S3 bucket. It doesn't rely on cookies and can be generated for users without requiring any changes to their HTTP client or hardcoded URLs. This method provides fine-grained control over access to your content.

E. AWS Secrets Manager: While AWS Secrets Manager can be useful for managing and rotating secrets, it is not directly related to securing S3 content in the context of the question. It's not one of the primary methods for securing access to S3 objects.

upvoted 1 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: AB

To secure streaming video content from Amazon CloudFront, two methods are available: signed cookies or signed URLs. Customers can choose to use either one or both, depending on the use case.

upvoted 4 times

tabbyDolly 2 years, 2 months ago

AB - <https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/PrivateContent.html>

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: BD

B and D are the correct options for meeting the requirements with the least impact to users.

Signed URLs allow access to individual objects in Amazon S3 for a specified time period without requiring cookies. This allows the custom HTTP client users to access content.

JSON Web Tokens (JWT) allow users to get temporary access tokens that can be passed in requests. This allows users with hardcoded URLs to access content without updating URLs.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

No good

Signed cookies require client support and may impact users.

AWS AppSync and Secrets Manager do not help address the specific access requirements.

Good

So Signed URLs and JWTs allow securing access to S3 content with minimal impact to users, meeting the requirements.

upvoted 2 times

riccardoto 2 years, 4 months ago

Selected Answer: BD

I understand why many users here are voting AB, but in my opinion BD is more correct.

Using JWT or signed urls will work both for users that cannot use cookies or cannot change the url.

upvoted 2 times

katetel 2 years, 4 months ago

Selected Answer: AB

it's correct

upvoted 2 times

MrAWSAssociate 2 years, 5 months ago

Selected Answer: CE

These are the right answers!

upvoted 2 times

A company is preparing a new data platform that will ingest real-time streaming data from multiple sources. The company needs to transform the data before writing the data to Amazon S3. The company needs the ability to use SQL to query the transformed data.

Which solutions will meet these requirements? (Choose two.)

A. Use Amazon Kinesis Data Streams to stream the data. Use Amazon Kinesis Data Analytics to transform the data. Use Amazon Kinesis Data Firehose to write the data to Amazon S3. Use Amazon Athena to query the transformed data from Amazon S3. **Most Voted**

B. Use Amazon Managed Streaming for Apache Kafka (Amazon MSK) to stream the data. Use AWS Glue to transform the data and to write the data to Amazon S3. Use Amazon Athena to query the transformed data from Amazon S3. **Most Voted**

C. Use AWS Database Migration Service (AWS DMS) to ingest the data. Use Amazon EMR to transform the data and to write the data to Amazon S3. Use Amazon Athena to query the transformed data from Amazon S3.

D. Use Amazon Managed Streaming for Apache Kafka (Amazon MSK) to stream the data. Use Amazon Kinesis Data Analytics to transform the data and to write the data to Amazon S3. Use the Amazon RDS query editor to query the transformed data from Amazon S3.

E. Use Amazon Kinesis Data Streams to stream the data. Use AWS Glue to transform the data. Use Amazon Kinesis Data Firehose to write the data to Amazon S3. Use the Amazon RDS query editor to query the transformed data from Amazon S3.

Correct Answer: AB

Community vote distribution

AB (93%)

7%

Comments

Steve_4542636 **Highly Voted** 2 years, 9 months ago

Selected Answer: AB

OK, for B I did some research, <https://docs.aws.amazon.com/glue/latest/dg/add-job-streaming.html>

"You can create streaming extract, transform, and load (ETL) jobs that run continuously, consume data from streaming sources like Amazon Kinesis Data Streams, Apache Kafka, and Amazon Managed Streaming for Apache Kafka (Amazon MSK). The jobs cleanse and transform the data, and then load the results into Amazon S3 data lakes or JDBC data stores."

upvoted 16 times

Paras043 **Highly Voted** 2 years, 8 months ago

But how can you transform data using kinesis data analytics ??

upvoted 9 times

luisgu 2 years, 7 months ago

See <https://aws.amazon.com/kinesis/data-analytics/faqs/?nc=sn&loc=6>

upvoted 3 times

awsgeek75 **Most Recent** 1 year, 10 months ago

Selected Answer: AB

Just because C is not going to work a DE use RDS so totally illogical

A & B seem to have redundant streaming, transformation and query steps so not sure if these are the right choices but CDE are completely wrong anyway!

upvoted 5 times

farnamjam 1 year, 11 months ago

For A didn't know that Kinesis Analytics can transform the data as well:

Amazon Kinesis Data Analytics provides built-in functions to filter, aggregate, and transform streaming data for advanced analytics. It processes streaming data with sub-second latencies, enabling you to analyze and respond to incoming data and streaming events in real time.

upvoted 5 times

pentium75 1 year, 11 months ago

Selected Answer: AB

"Use SQL to query the transformed data" [which is in S3] requires Athena, thus D and E are out. DMS is nonsense here thus C is out.

upvoted 3 times

MiniYang 2 years ago

why E is not right choice?

upvoted 2 times

pentium75 1 year, 11 months ago

Because you can't "use the Amazon RDS query editor to query .. data from S3"

upvoted 3 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: AB

options A and B will meet these requirements.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: AB

A and B are correct.

A uses Kinesis Data Streams for streaming, Kinesis Data Analytics for transformation, Kinesis Data Firehose for writing to S3, and Athena for SQL queries on S3 data.

B uses Amazon MSK for streaming, AWS Glue for transformation and writing to S3, and Athena for SQL queries on S3 data.

upvoted 3 times

Diqian 2 years, 3 months ago

Why E is incorrect?

upvoted 3 times

awsgeek75 1 year, 10 months ago

"Use the Amazon RDS query editor to query the transformed data from Amazon S3." is not possible as RDS query editor is for RDS and not for S3

upvoted 3 times

MrCloudy 2 years, 7 months ago

Selected Answer: AE

To transform real-time streaming data from multiple sources, write it to Amazon S3, and query the transformed data using SQL, the company can use the following solutions: Amazon Kinesis Data Streams, Amazon Kinesis Data Analytics, and Amazon Kinesis Data Firehose. The transformed data can be queried using Amazon Athena. Therefore, options A and E are the correct answers.

Option A is correct because it uses Amazon Kinesis Data Streams to stream data from multiple sources, Amazon Kinesis Data Analytics to transform the data, and Amazon Kinesis Data Firehose to write the data to Amazon S3. Amazon Athena can be used to query the transformed data in Amazon S3.

Option E is also correct because it uses Amazon Kinesis Data Streams to stream data from multiple sources, AWS Glue to transform the data, and Amazon Kinesis Data Firehose to write the data to Amazon S3. Amazon Athena can be used to query the transformed data in Amazon S3.

upvoted 3 times

sand444 2 years, 2 months ago

Amazon Athena is not in option E

upvoted 8 times

kraken21 2 years, 8 months ago

Selected Answer: AB

DMS can move data from DBs to streaming services and cannot natively handle streaming data. Hence A.B makes sense. Also AWS Glue/ETL can handle MSK streaming <https://docs.aws.amazon.com/glue/latest/dg/add-job-streaming.html>.

upvoted 3 times

elearningtakai 2 years, 8 months ago

Selected Answer: AB

The solutions that meet the requirements of streaming real-time data, transforming the data before writing to S3, and querying the transformed data using SQL are A and B.

Option C: This option is not ideal for streaming real-time data as AWS DMS is not optimized for real-time data ingestion.

Option D & E: These option are not recommended as the Amazon RDS query editor is not designed for querying data in S3, and it is not efficient for running complex queries.

upvoted 6 times

gold4otas 2 years, 8 months ago

Selected Answer: AB

The correct answers are options A & B

upvoted 2 times

TungPham 2 years, 9 months ago

may Amazon RDS query editor to query the transformed data from Amazon S3 ?

i don't think so, plz get link docs to that

upvoted 1 times

ManOnTheMoon 2 years, 9 months ago

Why not A & D?

upvoted 1 times

TungPham 2 years, 9 months ago

may Amazon RDS query editor to query the transformed data from Amazon S3 ?

i don't think so, plz get link docs to that

upvoted 1 times

LuckyAro 2 years, 9 months ago

Selected Answer: AB

A and B

upvoted 2 times

designmood22 2 years, 9 months ago

Answer is : A & B

upvoted 2 times

A company has an on-premises volume backup solution that has reached its end of life. The company wants to use AWS as part of a new backup solution and wants to maintain local access to all the data while it is backed up on AWS. The company wants to ensure that the data backed up on AWS is automatically and securely transferred.

Which solution meets these requirements?

- A. Use AWS Snowball to migrate data out of the on-premises solution to Amazon S3. Configure on-premises systems to mount the Snowball S3 endpoint to provide local access to the data.
- B. Use AWS Snowball Edge to migrate data out of the on-premises solution to Amazon S3. Use the Snowball Edge file interface to provide on-premises systems with local access to the data.
- C. Use AWS Storage Gateway and configure a cached volume gateway. Run the Storage Gateway software appliance on premises and configure a percentage of data to cache locally. Mount the gateway storage volumes to provide local access to the data.
- D. Use AWS Storage Gateway and configure a stored volume gateway. Run the Storage Gateway software appliance on premises and map the gateway storage volumes to on-premises storage. Mount the gateway storage volumes to provide local access to the data. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Steve_4542636 **Highly Voted** 2 years, 3 months ago

Selected Answer: D

The question states, "wants to maintain local access to all the data" This is storage gateway. Cached gateway stores only the frequently accessed data locally which is not what the problem statement asks for.

upvoted 22 times

kruasan **Highly Voted** 2 years, 1 month ago

Selected Answer: D

1. The company wants to maintain local access to all the data. Only stored volumes keep the complete dataset on-premises, providing low-latency access. Cached volumes only cache a subset locally.
2. The company wants the data backed up on AWS. With stored volumes, periodic backups (snapshots) of the on-premises data are sent to S3, providing durable and scalable backup storage.
3. The company wants the data transfer to AWS to be automatic and secure. Storage Gateway provides an encrypted connection between the on-premises gateway and AWS storage. Backups to S3 are sent asynchronously and automatically based on the backup schedule configured.

upvoted 15 times

AdamVigas **Most Recent** 1 year, 3 months ago

Selected Answer: D

local access to all data

upvoted 1 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: D

The Volume Gateway runs in either a cached or stored mode.

In the cached mode, your primary data is written to S3, while retaining your frequently accessed data locally in a cache for low-latency access.

In the stored mode, your primary data is stored locally and your entire dataset is available for low-latency access while asynchronously backed up to AWS.

<https://aws.amazon.com/storagegateway/faqs/#:~:text=What%20is%20Volume%20Gateway%3F>

upvoted 10 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

@kruasan well explained

upvoted 4 times

ChrisG1454 2 years, 3 months ago

Ans = D

<https://docs.aws.amazon.com/storagegateway/latest/vgw/WhatIsStorageGateway.html>

upvoted 4 times

Neha999 2 years, 3 months ago

D

<https://www.examttopics.com/discussions/amazon/view/43725-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 3 times

bdp123 2 years, 3 months ago

Selected Answer: D

<https://aws.amazon.com/storagegateway/faqs/#:~:text=In%20the%20cached%20mode%2C%20your,asynchronously%20backed%20up%20to%20AWS.>

In the cached mode, your primary data is written to S3, while retaining your frequently accessed data locally in a cache for low-latency access.

In the stored mode, your primary data is stored locally and your entire dataset is available for low-latency access while asynchronously backed up to AWS.

upvoted 3 times

An application that is hosted on Amazon EC2 instances needs to access an Amazon S3 bucket. Traffic must not traverse the internet.

How should a solutions architect configure access to meet these requirements?

- A. Create a private hosted zone by using Amazon Route 53.
- B. Set up a gateway VPC endpoint for Amazon S3 in the VPC. **Most Voted**
- C. Configure the EC2 instances to use a NAT gateway to access the S3 bucket.
- D. Establish an AWS Site-to-Site VPN connection between the VPC and the S3 bucket.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Guru4Cloud **Highly Voted** 1 year, 9 months ago

Selected Answer: B

The correct answer is B. Set up a gateway VPC endpoint for Amazon S3 in the VPC.

A gateway VPC endpoint is a private way for Amazon EC2 instances in a VPC to access AWS services, such as Amazon S3, without having to go through the internet. This can help to improve security and performance.

upvoted 8 times

Steve_4542636 **Highly Voted** 2 years, 3 months ago

Selected Answer: B

S3 and DynamoDB are the only services with Gateway endpoint options

upvoted 7 times

TariqKipkemei **Most Recent** 1 year, 8 months ago

Selected Answer: B

Set up a gateway VPC endpoint for Amazon S3 in the VPC.

upvoted 4 times

ManOnTheMoon 2 years, 3 months ago

Agree with B

upvoted 2 times

jennyka76 2 years, 3 months ago

ANSWER - B

<https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html> R B

upvoted 2 times

LuckyAro 2 years, 3 months ago

Selected Answer: B

B is correct

upvoted 2 times

skiwili 2 years, 3 months ago

Selected Answer: B

Bbbbbbbb

upvoted 4 times

An ecommerce company stores terabytes of customer data in the AWS Cloud. The data contains personally identifiable information (PII). The company wants to use the data in three applications. Only one of the applications needs to process the PII. The PII must be removed before the other two applications process the data.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Store the data in an Amazon DynamoDB table. Create a proxy application layer to intercept and process the data that each application requests.
- B. Store the data in an Amazon S3 bucket. Process and transform the data by using S3 Object Lambda before returning the data to the requesting application. **Most Voted**
- C. Process the data and store the transformed data in three separate Amazon S3 buckets so that each application has its own custom dataset. Point each application to its respective S3 bucket.
- D. Process the data and store the transformed data in three separate Amazon DynamoDB tables so that each application has its own custom dataset. Point each application to its respective DynamoDB table.

Correct Answer: B

Community vote distribution

B (90%)

7%

Comments

Steve_4542636 **Highly Voted** 3 years, 3 months ago

Selected Answer: B

Actually this is what Macie is best used for.

upvoted 21 times

Mikado211 2 years, 6 months ago

Yes. That's the problem here, Macie is the recommended tool in such case, but you do not have it in the answers.

upvoted 3 times

awsgeek75 2 years, 4 months ago

Macie is for identifying the PII data. Here it's much simpler because one of the apps need the PII data and other apps don't so you don't need to identify the PII data as you know it is already there. You just need to identify the app that needs the data which is not the best use case for Macie

upvoted 5 times

1e22522 1 year, 10 months ago

"Akshually" bro go to sleep

upvoted 3 times

fruto123 **Highly Voted** 3 years, 3 months ago

Selected Answer: B

B is the right answer and the proof is in this link.

<https://aws.amazon.com/blogs/aws/introducing-amazon-s3-object-lambda-use-your-code-to-process-data-as-it-is-being-retrieved-from-s3/>

upvoted 12 times

Guru4Cloud 2 years, 9 months ago

This is so wrong

upvoted 1 times

pentium75 2 years, 5 months ago

But it matches the exact use case here.

upvoted 2 times

awsgeek75 2 years, 4 months ago

Why do you think this is wrong?

upvoted 2 times

Rcosmos Most Recent 1 year, 4 months ago

Selected Answer: B

Por que esta é a melhor solução?

Processamento sob demanda:

O S3 Object Lambda permite que você processe e transforme objetos armazenados no S3 dinamicamente, com base na solicitação do aplicativo. Para os dois aplicativos que não precisam das PII, o Object Lambda pode remover essas informações no momento da solicitação, sem duplicar ou alterar os dados originais.

upvoted 1 times

AWSSURI 1 year, 9 months ago

I miss you Burugudystunstuguy....Man's been consistent with his answers

upvoted 2 times

MikeJANG 2 years, 3 months ago

Selected Answer: C

[GPT4] while S3 Object Lambda is a powerful tool for real-time data transformation, it is not the best fit for processing very large datasets due to Lambda's execution limits(15 min). Instead, preprocessing the data and storing it in separate S3 buckets for each applicatin's needs is a more operationally efficient solution for the scenario describes.

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: B

Because this is exactly what the AWS blog says.

"When you store data in Amazon Simple Storage Service (Amazon S3), you can easily share it for use by multiple applications. However, each application has its own requirements and may need a different view of the data. For example, a dataset created by an e-commerce application may include personally identifiable information (PII) that is not needed when the same data is processed for analytics and should be redacted."

upvoted 8 times

pentium75 2 years, 5 months ago

"Today, I'm very happy to announce the availability of S3 Object Lambda, a new capability that allows you to add your own code to process data retrieved from S3 before returning it to an application. S3 Object Lambda works with your existing applications and uses AWS Lambda functions to automatically process and transform your data as it is being retrieved from S3. The Lambda function is invoked inline with a standard S3 GET request, so you don't need to change your application code."

<https://aws.amazon.com/de/blogs/aws/introducing-amazon-s3-object-lambda-use-your-code-to-process-data-as-it-is-being-retrieved-from-s3/>

upvoted 8 times

meowruki 2 years, 6 months ago

B. Store the data in an Amazon S3 bucket. Process and transform the data by using S3 Object Lambda before returning the data to the requesting application.

This solution allows you to use S3 Object Lambda to process and transform the data on-the-fly as it is requested by each application. S3 Object Lambda enables you to apply custom code to your data retrieval requests, allowing you to remove PII before returning the data to the requesting application. This eliminates the need to create and manage separate storage locations for each application, reducing operational overhead.

upvoted 3 times

rvca231 2 years, 7 months ago

Selected Answer: C

Why would you reprocess the data every time you request it when you can just filter it once and be done?

Because of this I think A and B are highly inefficient, leaving us with C and D as options.

Since S3 is better suited for Data Lakes, I think C is the answer.

upvoted 2 times

awsgeek75 2 years, 4 months ago

Least operational overhead. The DevOps team is throwing this problem to the developers which is why C is not best.

upvoted 2 times

pentium75 2 years, 5 months ago

Because this is exactly what the AWS blog says.

"When you store data in Amazon Simple Storage Service (Amazon S3), you can easily share it for use by multiple applications. However, each application has its own requirements and may need a different view of the data. For example, a dataset created by an e-commerce application may include personally identifiable information (PII) that is not needed when the same data is processed for analytics and should be redacted. ... Today, I'm very happy to announce the availability of S3 Object Lambda, a new capability that allows you to add your own code to process data retrieved

from S3 before returning it to an application. S3 Object Lambda works with your existing applications and uses AWS Lambda functions to automatically process and transform your data as it is being retrieved from S3."

upvoted 2 times

Abbrar2022 3 years ago

Selected Answer: B

Store the data in an Amazon S3 bucket and using S3 Object Lambda to process and transform the data before returning it to the requesting application. This approach allows the PII to be removed in real-time and without the need to create separate datasets or tables for each application.

upvoted 2 times

antropaws 3 years ago

Selected Answer: A

@fruto123 and everyone that upvoted:

Is it plausible that S3 Object Lambda can process terabytes of data in 60 seconds? The same link you shared states that the maximum duration for a Lambda function used by S3 Object Lambda is 60 seconds.

Answer is A.

upvoted 2 times

antropaws 3 years ago

Chat GPT:

Isn't just 60 seconds the maximum duration for a Lambda function used by S3 Object Lambda? How can it process terabytes of data in 60 seconds?

You are correct that the maximum duration for a Lambda function used by S3 Object Lambda is 60 seconds.

Given the time constraint, it is not feasible to process terabytes of data within a single Lambda function execution.

S3 Object Lambda is designed for lightweight and real-time transformations rather than extensive processing of large datasets.

To handle terabytes of data, you would typically need to implement a distributed processing solution using services like Amazon EMR, AWS Glue, or AWS Batch. These services are specifically designed to handle big data workloads and provide scalability and distributed processing capabilities.

So, while S3 Object Lambda can be useful for lightweight processing tasks, it is not the appropriate tool for processing terabytes of data within the execution time limits of a Lambda function.

upvoted 2 times

Kp88 2 years, 10 months ago

Terabyte is just the storage. Lambda only need to process which application request. Think like removing/scratching off your social security number before sharing your doc to a third party.

upvoted 3 times

kruasan 3 years, 1 month ago

Selected Answer: B

- Storing the raw data in S3 provides a durable, scalable data lake. S3 requires little ongoing management overhead.
- S3 Object Lambda can be used to filter and process the data on retrieval transparently. This minimizes operational overhead by avoiding the need to preprocess and store multiple transformed copies of the data.
- Only one copy of the data needs to be stored and maintained in S3. S3 Object Lambda will transform the data on read based on the requesting application.
- No additional applications or proxies need to be developed and managed to handle the data transformation. S3 Object Lambda provides this functionality.

upvoted 4 times

kruasan 3 years, 1 month ago

Option A requires developing and managing a proxy app layer to handle data transformation, adding overhead.

Options C and D require preprocessing and storing multiple copies of the transformed data, adding storage and management overhead.

Option B using S3 Object Lambda minimizes operational overhead by handling data transformation on read transparently using the native S3 functionality. Only one raw data copy is stored in S3, with no additional applications required.

upvoted 2 times

pagom 3 years, 3 months ago

Selected Answer: B

<https://aws.amazon.com/ko/blogs/korea/introducing-amazon-s3-object-lambda-use-your-code-to-process-data-as-it-is-being-retrieved-from-s3/>

upvoted 5 times

LuckyAro 3 years, 3 months ago

Selected Answer: B

B is the correct answer.

Amazon S3 Object Lambda allows you to add custom code to S3 GET requests, which means that you can modify the data before it is returned to the requesting application. In this case, you can use S3 Object Lambda to remove the PII before the data is returned to the two applications that do not

need to process PII. This approach has the least operational overhead because it does not require creating separate datasets or proxy application layers, and it allows you to maintain a single copy of the data in an S3 bucket.

upvoted 5 times

NolaHOLA 3 years, 3 months ago

To meet the requirement of removing the PII before processing by two of the applications, it would be most efficient to use option B, which involves storing the data in an Amazon S3 bucket and using S3 Object Lambda to process and transform the data before returning it to the requesting application. This approach allows the PII to be removed in real-time and without the need to create separate datasets or tables for each application. S3 Object Lambda can be configured to automatically remove PII from the data before it is sent to the non-PII processing applications. This solution provides a cost-effective and scalable way to meet the requirement with the least operational overhead.

upvoted 4 times

minglu 3 years, 3 months ago

Selected Answer: B

I think it is B.

upvoted 2 times

skiwili 3 years, 3 months ago

Selected Answer: C

Looks like C is the correct answer

upvoted 2 times

A development team has launched a new application that is hosted on Amazon EC2 instances inside a development VPC. A solutions architect needs to create a new VPC in the same account. The new VPC will be peered with the development VPC. The VPC CIDR block for the development VPC is 192.168.0.0/24. The solutions architect needs to create a CIDR block for the new VPC. The CIDR block must be valid for a VPC peering connection to the development VPC.

What is the SMALLEST CIDR block that meets these requirements?

- A. 10.0.1.0/32
- B. 192.168.0.0/24
- C. 192.168.1.0/32
- D. 10.0.1.0/24 **Most Voted**

Correct Answer: D

Community vote distribution

D (99%)

Comments

BrainOBrain **Highly Voted** 2 years, 9 months ago

Selected Answer: D

10.0.1.0/32 and 192.168.1.0/32 are too small for VPC, and /32 network is only 1 host
192.168.0.0/24 is overlapping with existing VPC
upvoted 30 times

kruasan **Highly Voted** 2 years, 7 months ago

Selected Answer: D

- Option A (10.0.1.0/32) is invalid - a /32 CIDR prefix is a host route, not a VPC range.
- Option B (192.168.0.0/24) overlaps the development VPC and so cannot be used.
- Option C (192.168.1.0/32) is invalid - a /32 CIDR prefix is a host route, not a VPC range.
- Option D (10.0.1.0/24) satisfies the non-overlapping CIDR requirement but is a larger block than needed. Since only two VPCs need to be peered, a /24 block provides more addresses than necessary.

upvoted 13 times

TheFivePips **Most Recent** 1 year, 9 months ago

Selected Answer: D

In an Amazon VPC, the first four and the last IP address in each subnet are reserved for specific purposes, and they cannot be used for customer instances. Here's how the reserved addresses are typically allocated:

Network Address (First IP):

The first IP address (all zeros in the host portion) in a subnet is reserved as the network address. For example, if you have a subnet with a CIDR notation of 10.0.0.0/24, the network address would be 10.0.0.0.

VPC Router (Second IP):

The second IP address in the subnet is reserved for the VPC router.
DNS Server (Third IP):

The third IP address is reserved for the DNS server.
Reserved for Future Use (Fourth IP):

The fourth IP address is reserved for future use.
Customer Instances (Fifth to Second-to-Last IP):

The IP addresses from the fifth to the second-to-last IP address in the subnet are available for customer instances.
Broadcast Address (Last IP):

The last IP address (all ones in the host portion) in a subnet is reserved as the broadcast address, even though AWS does not support broadcast.

upvoted 6 times

walter9660 1 year, 9 months ago

Selected Answer: C

10.0.0.0 - 10.255.255.255 (10/8 prefix): Example CIDR block: 10.0.0.0/16

172.16.0.0 - 172.31.255.255 (172.16/12 prefix): Example CIDR block: 172.31.0.0/16

192.168.0.0 - 192.168.255.255 (192.168/16 prefix): Example CIDR block: 192.168.0.0/20

Given that the development VPC already uses 192.168.0.0/24, we need to choose a non-overlapping CIDR block. The smallest valid CIDR block that meets the requirements is 192.168.1.0/24 (Option C).

upvoted 1 times

bora4motion 1 year, 1 month ago

WOW! just WOW!

upvoted 1 times

Murtadhaceit 2 years ago

Selected Answer: D

A and C are host IP addresses.

B is not possible because it's using the same subnet for the other team/department.

We are left with D, which is the right answer.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

10.0.1.0/32 and 192.168.1.0/32 are too small for VPC, and /32 network is only 1 host

192.168.0.0/24 is overlapping with existing VPC

upvoted 2 times

Abrar2022 2 years, 6 months ago

Definitely D. The only valid VPC CIDR block that does not overlap with the development VPC CIDR block among the options. The other 2 CIDR block options are too small.

upvoted 2 times

antropaws 2 years, 6 months ago

Selected Answer: D

D is correct.

upvoted 2 times

channn 2 years, 8 months ago

Selected Answer: D

D is the only correct answer

upvoted 2 times

r04dB10ck 2 years, 8 months ago

Selected Answer: D

only one valid with no overlap

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: D

A process by elimination solution here. a CIDR value is the number of bits that are locked so 10.0.0.0/32 means no range.

upvoted 4 times

LuckyAro 2 years, 9 months ago

Selected Answer: D

Answer is D, 10.0.1.0/24.

upvoted 2 times

skiwili 2 years, 9 months ago

Selected Answer: D

Yes D is the answer

upvoted 2 times

obatunde 2 years, 9 months ago

Selected Answer: D

Definitely D. It is the only valid VPC CIDR block that does not overlap with the development VPC CIDR block among the options.

upvoted 2 times

bdp123 2 years, 9 months ago

Selected Answer: D

The allowed block size is between a /28 netmask and /16 netmask.

The CIDR block must not overlap with any existing CIDR block that's associated with the VPC.

<https://docs.aws.amazon.com/vpc/latest/userguide/configure-your-vpc.html>

upvoted 6 times

A company deploys an application on five Amazon EC2 instances. An Application Load Balancer (ALB) distributes traffic to the instances by using a target group. The average CPU usage on each of the instances is below 10% most of the time, with occasional surges to 65%.

A solutions architect needs to implement a solution to automate the scalability of the application. The solution must optimize the cost of the architecture and must ensure that the application has enough CPU resources when surges occur.

Which solution will meet these requirements?

A. Create an Amazon CloudWatch alarm that enters the ALARM state when the CPUUtilization metric is less than 20%. Create an AWS Lambda function that the CloudWatch alarm invokes to terminate one of the EC2 instances in the ALB target group.

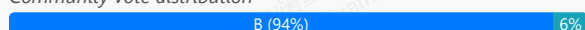
B. Create an EC2 Auto Scaling group. Select the existing ALB as the load balancer and the existing target group as the target group. Set a target tracking scaling policy that is based on the ASGAverageCPUUtilization metric. Set the minimum instances to 2, the desired capacity to 3, the maximum instances to 6, and the target value to 50%. Add the EC2 instances to the Auto Scaling group. **Most Voted**

C. Create an EC2 Auto Scaling group. Select the existing ALB as the load balancer and the existing target group as the target group. Set the minimum instances to 2, the desired capacity to 3, and the maximum instances to 6. Add the EC2 instances to the Auto Scaling group.

D. Create two Amazon CloudWatch alarms. Configure the first CloudWatch alarm to enter the ALARM state when the average CPUUtilization metric is below 20%. Configure the second CloudWatch alarm to enter the ALARM state when the average CPUUtilization metric is above 50%. Configure the alarms to publish to an Amazon Simple Notification Service (Amazon SNS) topic to send an email message. After receiving the message, log in to decrease or increase the number of EC2 instances that are running.

Correct Answer: B

Community vote distribution



Comments

bdp123 **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Just create an auto scaling policy
upvoted 17 times

vilagiri **Most Recent** 2 years, 2 months ago

I picked B.. I am not 100% sure..The application is deployed in 5 instances initially. What is the logic behind 2/3/6 ASG. Because utilization is 10%, we can set min 2? I know for sure I am not going to get this ASG question correct in the exam.
upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

The correct answer is B.

This solution will meet the requirements because it will:

Automate the scalability of the application by using EC2 Auto Scaling.
Optimize the cost of the architecture by only scaling the number of EC2 instances up when needed.
Ensure that the application has enough CPU resources when surges occur by setting the target value of the target tracking scaling policy to 50%.
upvoted 3 times

ajchi1980 2 years, 5 months ago

Wrong answers: Options A, C, and D are not the most appropriate solutions:

Option A suggests creating a CloudWatch alarm to terminate an EC2 instance when CPU utilization is less than 20%. However, this approach does not ensure that the application will have enough CPU resources during surges, as it only terminates instances when CPU utilization is low, which may not meet the requirements.

Option C suggests creating an Auto Scaling group without any specific scaling policies or configurations. This approach does not address the need for automated scaling based on CPU utilization, making it insufficient for the given requirements.

Option D suggests using CloudWatch alarms to send notifications via Amazon SNS and manually adjusting the number of instances based on the received messages. This approach lacks automation and requires manual intervention, which does not optimize cost or meet the requirement of automated scalability.

Therefore, Option B is the most appropriate solution in this case.

upvoted 3 times

ajchi1980 2 years, 5 months ago

Selected Answer: B

Explanation:

Option B leverages EC2 Auto Scaling, which is designed to automatically adjust the number of instances based on specified metrics. By setting a target tracking scaling policy based on average CPU utilization, the Auto Scaling group can dynamically scale the number of instances to maintain the desired level of CPU resources. The minimum instances of 2 ensure a minimum baseline capacity, while the desired capacity of 3 ensures at least three instances are running even during normal traffic. The maximum instances of 6 cap the upper limit to control costs.

upvoted 4 times

RoroJ 2 years, 6 months ago

Selected Answer: D

Auto Scaling group must have an AMI for it.

upvoted 2 times

pentium75 1 year, 11 months ago

"After receiving the message, log in to decrease or increase the number of EC2 instances that are running" does surely not "automate the scalability".

upvoted 3 times

th3k33n 2 years, 6 months ago

how can we set max to 6 since the company is using 5 ec2 instance

upvoted 2 times

examtopictempacc 2 years, 6 months ago

In the scenario you provided, you're setting up an Auto Scaling group to manage the instances for you, and the settings (min 2, desired 3, max 6) are for the Auto Scaling group, not for your existing instances. When you integrate the instances into the Auto Scaling group, you are effectively moving from a fixed instance count to a dynamic one that can range from 2 to 6 based on the demand.

The existing 5 instances can be included in the Auto Scaling group, but the group can reduce the number of instances if the load is low (to the minimum specified, which is 2 in this case) and can also add more instances (up to a maximum of 6) if the load increases.

upvoted 4 times

kruasan 2 years, 7 months ago

Selected Answer: B

Reasons:

- An Auto Scaling group will automatically scale the EC2 instances to match changes in demand. This optimizes cost by only running as many instances as needed.
- A target tracking scaling policy monitors the ASGAverageCPUUtilization metric and scales to keep the average CPU around the 50% target value. This ensures there are enough resources during CPU surges.
- The ALB and target group are reused, so the application architecture does not change. The Auto Scaling group is associated to the existing load balancer setup.
- A minimum of 2 and maximum of 6 instances provides the ability to scale between 3 and 6 instances as needed based on demand.
- Costs are optimized by starting with only 3 instances (the desired capacity) and scaling up as needed. When CPU usage drops, instances are terminated to match the desired capacity.

upvoted 3 times

kruasan 2 years, 7 months ago

Option A - terminates instances reactively based on low CPU and may not provide enough capacity during surges. Does not optimize cost.

Option C - lacks a scaling policy so will not automatically adjust capacity based on changes in demand. Does not ensure enough resources during surges.

Option D - requires manual intervention to scale capacity. Does not optimize cost or provide an automated solution.

upvoted 2 times

darn 2 years, 7 months ago

as you dig down the question, they get more and more bogus with less and less votes

upvoted 1 times

Steve_4542636 2 years, 9 months ago

Selected Answer: B

B is my vote

upvoted 2 times

KZM 2 years, 9 months ago

Based on the information given, the best solution is option "B".

Autoscaling group with target tracking scaling policy with min 2 instances, desired capacity to 3, and the maximum instances to 6.

upvoted 2 times

Shrestwt 2 years, 7 months ago

But the company is using only 5 EC2 Instances so how can we set maximum instance to 6.

upvoted 2 times

pentium75 1 year, 11 months ago

"Create an EC2 Auto Scaling group" includes replacing your existing EC2 instances with a launch configuration that starts and stops instances automatically.

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: B

B is the correct solution because it allows for automatic scaling based on the average CPU utilization of the EC2 instances in the target group. With the use of a target tracking scaling policy based on the ASGAverageCPUUtilization metric, the EC2 Auto Scaling group can ensure that the target value of 50% is maintained while scaling the number of instances in the group up or down as needed. This will help ensure that the application has enough CPU resources during surges without overprovisioning, thus optimizing the cost of the architecture.

upvoted 3 times

Babba 2 years, 9 months ago

Selected Answer: B

Should be B

upvoted 2 times

A company is running a critical business application on Amazon EC2 instances behind an Application Load Balancer. The EC2 instances run in an Auto Scaling group and access an Amazon RDS DB instance.

The design did not pass an operational review because the EC2 instances and the DB instance are all located in a single Availability Zone. A solutions architect must update the design to use a second Availability Zone.

Which solution will make the application highly available?

- A. Provision a subnet in each Availability Zone. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance with connections to each network.
- B. Provision two subnets that extend across both Availability Zones. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance with connections to each network.
- C. Provision a subnet in each Availability Zone. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance for Multi-AZ deployment. **Most Voted**
- D. Provision a subnet that extends across both Availability Zones. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance for Multi-AZ deployment.

Correct Answer: C

Community vote distribution

C (100%)

Comments

bdp123 **Highly Voted** 2 years, 9 months ago

Selected Answer: C

A subnet must reside within a single Availability Zone.

<https://aws.amazon.com/vpc/faqs/#:~:text=Can%20a%20subnet%20span%20Availability,within%20a%20single%20Availability%20Zone.>

upvoted 23 times

zjcorpuz **Highly Voted** 2 years, 4 months ago

a subnet only resides on a one AZ, it does not span to another AZ.

upvoted 8 times

LoXoL **Most Recent** 1 year, 10 months ago

Selected Answer: C

A subnet can't "Extend" across multiple AZs: B,D out

HA = RDS Multi-AZ: A out

C

upvoted 4 times

thewalker 1 year, 10 months ago

Selected Answer: C

An Auto Scaling group can span across two Availability Zones, where one subnet is created in each AZ.

When creating an Auto Scaling group, you need to specify at least one subnet. You can add additional subnets later on, including subnets across multiple AZs.

Auto Scaling will distribute instances evenly across the specified subnets to maintain availability and optimize performance. If one AZ becomes unavailable, instances can be launched in the other AZ.

The associated load balancer should also span the same subnets/AZs as the Auto Scaling group. This allows traffic to be routed to instances in different subnets and AZs, increasing fault tolerance of the application.

upvoted 3 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: C

Provision a subnet in each Availability Zone. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance for Multi-AZ deployment

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

This solution will ensure that the EC2 instances and the DB instance are not located in the same Availability Zone, which will improve the availability of the application.

upvoted 3 times

MrAWSAssociate 2 years, 5 months ago

Selected Answer: C

D is completely wrong, because each subnet must reside entirely within one Availability Zone and cannot span zones. By launching AWS resources in separate Availability Zones, you can protect your applications from the failure of a single Availability Zone.

upvoted 3 times

Anmol_1010 2 years, 6 months ago

The key word here was extend.

upvoted 2 times

GalileoEC2 2 years, 8 months ago

This discards B and D: Subnet basics. Each subnet must reside entirely within one Availability Zone and cannot span zones. By launching AWS resources in separate Availability Zones, you can protect your applications from the failure of a single Availability Zone

upvoted 3 times

Steve_4542636 2 years, 9 months ago

Selected Answer: C

a subnet is per AZ. a scaling group can span multiple AZs. <https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-add-availability-zone.html>

upvoted 2 times

KZM 2 years, 9 months ago

I think D.

Span the single subnet in both Availability Zones can access the DB instances in either zone without going over the public internet.

upvoted 3 times

KZM 2 years, 9 months ago

Can span like that?

upvoted 1 times

KZM 2 years, 9 months ago

Sorry I think C is correct.

upvoted 2 times

leoattf 2 years, 9 months ago

Nope. The answer is indeed C.

You cannot span like that. Check the link below:

"Each subnet must reside entirely within one Availability Zone and cannot span zones."

<https://docs.aws.amazon.com/vpc/latest/userguide/configure-subnets.html>

upvoted 4 times

KZM 2 years, 9 months ago

Thanks, Leoattf for the link you shared.

upvoted 3 times

Babba 2 years, 9 months ago

Selected Answer: C

it's C

upvoted 2 times

A research laboratory needs to process approximately 8 TB of data. The laboratory requires sub-millisecond latencies and a minimum throughput of 6 GBps for the storage subsystem. Hundreds of Amazon EC2 instances that run Amazon Linux will distribute and process the data.

Which solution will meet the performance requirements?

A. Create an Amazon FSx for NetApp ONTAP file system. Set each volume's tiering policy to ALL. Import the raw data into the file system. Mount the file system on the EC2 instances.

B. Create an Amazon S3 bucket to store the raw data. Create an Amazon FSx for Lustre file system that uses persistent SSD storage. Select the option to import data from and export data to Amazon S3. Mount the file system on the EC2 instances.

Most Voted

C. Create an Amazon S3 bucket to store the raw data. Create an Amazon FSx for Lustre file system that uses persistent HDD storage. Select the option to import data from and export data to Amazon S3. Mount the file system on the EC2 instances.

D. Create an Amazon FSx for NetApp ONTAP file system. Set each volume's tiering policy to NONE. Import the raw data into the file system. Mount the file system on the EC2 instances.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Bhawesh **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Keyword here is a minimum throughput of 6 GBps. Only the FSx for Lustre with SSD option gives the sub-milli response and throughput of 6 GBps or more.

B. Create an Amazon S3 bucket to store the raw data. Create an Amazon FSx for Lustre file system that uses persistent SSD storage. Select the option to import data from and export data to Amazon S3. Mount the file system on the EC2 instances.

References:

<https://aws.amazon.com/fsx/when-to-choose-fsx/>

upvoted 21 times

bdp123 **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Create an Amazon S3 bucket to store the raw data. Create an Amazon FSx for Lustre file system that uses persistent SSD storage. Select the option to import data from and export data to Amazon S3.

Mount the file system on the EC2 instances. Amazon FSx for Lustre uses SSD storage for submillisecond latencies and up to 6 GBps throughput, and can import data from and export data to Amazon S3. Additionally, the option to select persistent SSD storage will ensure that the data is stored on the disk and not lost if the file system is stopped.

upvoted 8 times

MrPCarrot **Most Recent** 1 year, 9 months ago

Answer is B : FSx for Lustre with SSD option gives the sub-milli response and throughput of 6 GBps or more

upvoted 3 times

Pangian 1 year, 10 months ago

I don't even think that NetApp comes for Linux

upvoted 1 times

djgodzilla 1 year, 11 months ago

Selected Answer: B

Amazon FSx for Lustre for compute-intensive workloads.

- allows file-based applications to access data with hundreds of gigabytes per second of data, millions of IOPS, and sub millisecond latencies.
- supports file access to thousands of EC2 instances

and well SSD always wins ;)

upvoted 3 times

Mikado211 2 years ago

Selected Answer: B

sub-milliseconses == Lustre

HDD vs SSD == for performance use SSD

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

Amazon FSx for Lustre with SSD: Amazon FSx for Lustre is designed for high-performance, parallel file processing workloads. Choosing SSD storage ensures fast I/O and meets the sub-millisecond latency requirement.

upvoted 4 times

rolervengador 2 years, 3 months ago

Voto por la B

upvoted 2 times

Gooniegoogoo 2 years, 5 months ago

So many of these are wrong, its good we have people that vote so we can get to the right answer!!

upvoted 2 times

kruasan 2 years, 7 months ago

Selected Answer: B

- Amazon FSx for Lustre with SSD storage can provide up to 260 GB/s of aggregate throughput and sub-millisecond latencies needed for this workload.

- Persistent SSD storage ensures data durability in the file system. Data is also exported to S3 for backup storage.

- The file system will import the initial 8 TB of raw data from S3, providing a fast storage tier for processing while retaining the data in S3.

- The file system is mounted to the EC2 compute instances to distribute processing.

- FSx for Lustre is optimized for high-performance computing workloads running on Linux, matching the EC2 environment.

upvoted 3 times

kruasan 2 years, 7 months ago

Option A - FSx for NetApp ONTAP with ALL tiering policy would not provide fast enough storage tier for sub-millisecond latency. HDD tiers have higher latency.

Option C - FSx for Lustre with HDD storage would not provide the throughput, IOPS or low latency needed.

Option D - FSx for NetApp ONTAP with NONE tiering policy would require much more expensive SSD storage to meet requirements, increasing cost.

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: B

I vote B

upvoted 2 times

AlmeroSenior 2 years, 9 months ago

Selected Answer: B

FSX Lustre is 1000mbs per TB provisioned and we have 8TBs so gives us 8GBs . The netapp FSX appears a hard limit of 4gbs .

<https://aws.amazon.com/fsx/lustre/faqs/?nc=sn&loc=5>

<https://aws.amazon.com/fsx/netapp-ontap/faqs/>

upvoted 6 times

LuckyAro 2 years, 9 months ago

Selected Answer: B

B is the best choice as it utilizes Amazon S3 for data storage, which is cost-effective and durable, and Amazon FSx for Lustre for high-performance file storage, which provides the required sub-millisecond latencies and minimum throughput of 6 GBps. Additionally, the option to import and export data to and from Amazon S3 makes it easier to manage and move data between the two services.

B is the best option as it meets the performance requirements for sub-millisecond latencies and a minimum throughput of 6 GBps.

upvoted 2 times

everfly 2 years, 9 months ago

Selected Answer: B

Amazon FSx for Lustre provides fully managed shared storage with the scalability and performance of the popular Lustre file system. It can deliver sub-millisecond latencies and hundreds of gigabytes per second of throughput.

upvoted 4 times

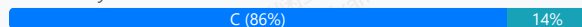
A company needs to migrate a legacy application from an on-premises data center to the AWS Cloud because of hardware capacity constraints. The application runs 24 hours a day, 7 days a week. The application's database storage continues to grow over time.

What should a solutions architect do to meet these requirements MOST cost-effectively?

- A. Migrate the application layer to Amazon EC2 Spot Instances. Migrate the data storage layer to Amazon S3.
- B. Migrate the application layer to Amazon EC2 Reserved Instances. Migrate the data storage layer to Amazon RDS On-Demand Instances.
- C. Migrate the application layer to Amazon EC2 Reserved Instances. Migrate the data storage layer to Amazon Aurora Reserved Instances. **Most Voted**
- D. Migrate the application layer to Amazon EC2 On-Demand Instances. Migrate the data storage layer to Amazon RDS Reserved Instances.

Correct Answer: C

Community vote distribution



Comments

LuckyAro **Highly Voted** 2 years, 9 months ago

Selected Answer: C

Amazon EC2 Reserved Instances allow for significant cost savings compared to On-Demand instances for long-running, steady-state workloads like this one. Reserved Instances provide a capacity reservation, so the instances are guaranteed to be available for the duration of the reservation period.

Amazon Aurora is a highly scalable, cloud-native relational database service that is designed to be compatible with MySQL and PostgreSQL. It can automatically scale up to meet growing storage requirements, so it can accommodate the application's database storage needs over time. By using Reserved Instances for Aurora, the cost savings will be significant over the long term.

upvoted 24 times

NolaHOla **Highly Voted** 2 years, 9 months ago

Option B based on the fact that the DB storage will continue to grow, so on-demand will be a more suitable solution

upvoted 15 times

pentium75 1 year, 11 months ago

Database STORAGE will grow, not performance need (and required instance size).

upvoted 4 times

NolaHOla 2 years, 9 months ago

Since the application's database storage is continuously growing over time, it may be difficult to estimate the appropriate size of the Aurora cluster in advance, which is required when reserving Aurora.

In this case, it may be more cost-effective to use Amazon RDS On-Demand Instances for the data storage layer. With RDS On-Demand Instances, you pay only for the capacity you use and you can easily scale up or down the storage as needed.

upvoted 5 times

Joxtat 2 years, 9 months ago

The Answer is C.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.AuroraMySQL.html>

upvoted 2 times

hristni0 2 years, 6 months ago

Answer is C. From Aurora Reserved Instances documentation:

If you have a DB instance, and you need to scale it to larger capacity, your reserved DB instance is automatically applied to your scaled DB

instance. That is, your reserved DB instances are automatically applied across all DB instance class sizes. Size-flexible reserved DB instances are available for DB instances with the same AWS Region and database engine.

upvoted 2 times

zdi561 **Most Recent** 1 year, 4 months ago

Selected Answer: B

RLs are best for steady-state database workloads.

upvoted 1 times

hro 1 year, 8 months ago

cost-effectively - the answer is C.

The application runs 24 hours a day, 7 days a week. The application's database storage continues to grow over time.

upvoted 3 times

MrPCarrot 1 year, 9 months ago

Answer is C: Amazon EC2 Reserved Instances and Amazon Aurora Reserved Instances = less expensive than RDS.

upvoted 3 times

andyngh86 1 year, 10 months ago

Amazon Aurora reserved instances is used for the work load on predictable, so answer should be B

upvoted 1 times

Priyapani 1 year, 11 months ago

Selected Answer: B

I think it's B as database storage will grow

upvoted 2 times

awsgeek75 1 year, 10 months ago

Application runs 24x7 which means database is also used 24x7. The storage will grow and RDS On-Demand does not have auto-grow storage. You have to configure a storage size for RDS which means it will eventually run out of space. RDS On-Demand just scales CPU, not storage.

Aurora has no storage limitation and can scale storage according to need which is what is required here

upvoted 4 times

Mikado211 2 years ago

Selected Answer: C

24/7 forbids spot instances , so A is excluded

Cost efficiency require reserved instances , so D is excluded

Between RDS and Aurora, Aurora is less expensive thanks to the reserved instance, so B is finally excluded

Answer is C

upvoted 4 times

cciesam 2 years, 1 month ago

Selected Answer: B

I hope it should be B considering Database growth

upvoted 1 times

pentium75 1 year, 11 months ago

Reserved instance applies to the DB instance size (CPU, RAM etc.), not storage.

upvoted 3 times

Wayne23Fang 2 years, 2 months ago

My research concludes that From pure price point of view Aurora Reserved might/ usually be slightly more expensive than On-demand RDS. But RDS has less Operation overhead. For the 24x7 nature, I would vote C. But for pure cost-effective, B is less costly.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

This option involves migrating the application layer to Amazon EC2 Reserved Instances and migrating the data storage layer to Amazon Aurora Reserved Instances. Amazon EC2 Reserved Instances provide a significant discount (up to 75%) compared to On-Demand Instance pricing, making them a cost-effective choice for applications that have steady state or predictable usage. Similarly, Amazon Aurora Reserved Instances provide a significant discount (up to 69%) compared to On-Demand Instance pricing.

upvoted 2 times

ajchi1980 2 years, 5 months ago

Selected Answer: C

To meet the requirements of migrating a legacy application from an on-premises data center to the AWS Cloud in a cost-effective manner, the most suitable option would be:

C. Migrate the application layer to Amazon EC2 Reserved Instances. Migrate the data storage layer to Amazon Aurora Reserved Instances.

Explanation:

Migrating the application layer to Amazon EC2 Reserved Instances allows you to reserve EC2 capacity in advance, providing cost savings compared to On-Demand Instances. This is especially beneficial if the application runs 24/7.

Migrating the data storage layer to Amazon Aurora Reserved Instances provides cost optimization for the growing database storage needs. Amazon Aurora is a fully managed relational database service that offers high performance, scalability, and cost efficiency.

upvoted 2 times

cpen 2 years, 6 months ago

nnascncnscnknkckl

upvoted 1 times

TariqKipkemei 2 years, 7 months ago

Answer is C

upvoted 2 times

QuangPham810 2 years, 7 months ago

Answer is C. Refer https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/USER_WorkingWithReservedDBInstances.html => Size-flexible reserved DB instances

upvoted 2 times

Abhineet9148232 2 years, 9 months ago

Selected Answer: C

C: With Aurora Serverless v2, each writer and reader has its own current capacity value, measured in ACUs. Aurora Serverless v2 scales a writer or reader up to a higher capacity when its current capacity is too low to handle the load. It scales the writer or reader down to a lower capacity when its current capacity is higher than needed.

This is sufficient to accommodate the growing data changes.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-serverless-v2.how-it-works.html#aurora-serverless-v2.how-it-works.scaling>

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: C

Typically Amazon RDS cost less than Aurora. But here, it's Aurora reserved.

upvoted 2 times

djgodzilla 1 year, 11 months ago

although agree and AWS wants you to choose Answer C. You can't convince a cloud accounting analyst that Aurora is cheaper than RDS. no matter what

upvoted 2 times

A university research laboratory needs to migrate 30 TB of data from an on-premises Windows file server to Amazon FSx for Windows File Server. The laboratory has a 1 Gbps network link that many other departments in the university share.

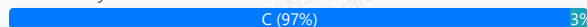
The laboratory wants to implement a data migration service that will maximize the performance of the data transfer. However, the laboratory needs to be able to control the amount of bandwidth that the service uses to minimize the impact on other departments. The data migration must take place within the next 5 days.

Which AWS solution will meet these requirements?

- A. AWS Snowcone
- B. Amazon FSx File Gateway
- C. AWS DataSync **Most Voted**
- D. AWS Transfer Family

Correct Answer: C

Community vote distribution



Comments

kruasan **Highly Voted** 3 years, 1 month ago

Selected Answer: C

AWS DataSync is a data transfer service that can copy large amounts of data between on-premises storage and Amazon FSx for Windows File Server at high speeds. It allows you to control the amount of bandwidth used during data transfer.

- DataSync uses agents at the source and destination to automatically copy files and file metadata over the network. This optimizes the data transfer and minimizes the impact on your network bandwidth.
- DataSync allows you to schedule data transfers and configure transfer rates to suit your needs. You can transfer 30 TB within 5 days while controlling bandwidth usage.
- DataSync can resume interrupted transfers and validate data to ensure integrity. It provides detailed monitoring and reporting on the progress and performance of data transfers.

upvoted 29 times

kruasan 3 years, 1 month ago

Option A - AWS Snowcone is more suitable for physically transporting data when network bandwidth is limited. It would not complete the transfer within 5 days.

Option B - Amazon FSx File Gateway only provides access to files stored in Amazon FSx and does not perform the actual data migration from on-premises to FSx.

Option D - AWS Transfer Family is for transferring files over FTP, FTPS and SFTP. It may require scripting to transfer 30 TB and monitor progress, and lacks bandwidth controls.

upvoted 16 times

Michal_L_95 **Highly Voted** 3 years, 3 months ago

Selected Answer: C

As read a little bit, I assume that B (FSx File Gateway) requires a little bit more configuration rather than C (DataSync). From Stephane Maarek course explanation about DataSync:

An online data transfer service that simplifies, automates, and accelerates copying large amounts of data between on-premises storage systems and AWS Storage services, as well as between AWS Storage services.

You can use AWS DataSync to migrate data located on-premises, at the edge, or in other clouds to Amazon S3, Amazon EFS, Amazon FSx for Windows File Server, Amazon FSx for Lustre, Amazon FSx for OpenZFS, and Amazon FSx for NetApp ONTAP.

upvoted 12 times

MatAlves **Most Recent** 1 year, 9 months ago

Selected Answer: C

Even if we allocate 60% of total bandwidth for the transfer, that would take 5d2h. Considering that "many other departments in the university share" that wouldn't be feasible.

Ref. <https://expedient.com/knowledgebase/tools-and-calculators/file-transfer-time-calculator/>

On the other hand, snowcone isn't also a great option, because "you will receive the Snowcone device in approximately 4-6 days".

Ref.

<https://aws.amazon.com/snowcone/faqs/#:~:text=You%20will%20receive%20the%20Snowcone,console%20for%20each%20Snowcone%20device.>

upvoted 2 times

MatAlves 1 year, 9 months ago

Very unreasonable scenario. The least "bad" option is C, but that will definitely affect users in real production environments.

upvoted 2 times

Cyberkayu 2 years, 5 months ago

Selected Answer: A

Snow cone can support up to 8TB for HDD and 15TB for each SSD devices. Shipped within 4-6 days. Data migration can begin on next 5 days.

Does not use any amount of bandwidth and impact the production network. Device came with 1G and 10G Base-T Ethernet port. That's the Maximum performance in data transfer. defined in the question.

upvoted 2 times

JA2018 1 year, 6 months ago

Key in STEM:

The data migration must take place within the next 5 days.

upvoted 1 times

JA2018 1 year, 6 months ago

Snowcone it's just 15TB per SSD and it takes 4-6 business days to deliver to the customer's location.....

It could takes another 4-6 business days to return the snowcone device back to AWS (to transfer data to the designated S3 bucket) after you completed the on-prems data transfer to the device.

upvoted 1 times

AZ_Master 2 years, 6 months ago

Selected Answer: C

Bandwidth control = Data Sync

<https://docs.aws.amazon.com/datasync/latest/userguide/configure-bandwidth.html>

upvoted 5 times

Ruffyit 2 years, 6 months ago

Bandwidth Optimization and Control

Transferring hot or cold data should not impede your business. DataSync is equipped with granular controls to optimize bandwidth consumptions. Throttle transfer speeds up to 10 Gbps during off hours and set limits when network availability is needed elsewhere

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

C. AWS DataSync

upvoted 2 times

Nikki013 2 years, 9 months ago

Selected Answer: C

<https://aws.amazon.com/datasync/features/>

upvoted 2 times

Yousuf_Ibrahim 2 years, 8 months ago

Bandwidth Optimization and Control

Transferring hot or cold data should not impede your business. DataSync is equipped with granular controls to optimize bandwidth consumptions. Throttle transfer speeds up to 10 Gbps during off hours and set limits when network availability is needed elsewhere.

upvoted 2 times

jayce5 2 years, 12 months ago

Selected Answer: C

"Amazon FSx File Gateway" is for storing data, not for migrating. So the answer should be C.

upvoted 3 times

ACloud_Guru15 2 years, 7 months ago

Thanks for the explanation

upvoted 2 times

shanwford 3 years, 2 months ago

Selected Answer: C

Snowcone to small and delivertime to long. With DataSync you can set bandwidth limits - so this is fine solution.

upvoted 3 times

MaxMa 3 years, 2 months ago

Why not B?

upvoted 1 times

Guru4Cloud 2 years, 8 months ago

Transferring will be much longer rather than 5 days as required.

upvoted 2 times

AlessandraSAA 3 years, 3 months ago

A not possible because Snowcone it's just 8TB and it takes 4-6 business days to deliver

B why cannot be <https://aws.amazon.com/storagegateway/file/fsx/>?

C I don't really get this

D cannot be because not compatible - <https://aws.amazon.com/aws-transfer-family/>

upvoted 1 times

pentium75 2 years, 5 months ago

With B you cannot "control the amount of bandwidth that the service uses", while C does exactly what is required here.

upvoted 2 times

Steve_4542636 3 years, 3 months ago

Selected Answer: C

Voting C

upvoted 2 times

Bhawesh 3 years, 3 months ago

Selected Answer: C

C. - DataSync is Correct.

A. Snowcone is incorrect. The question says data migration must take place within the next 5 days. AWS says: If you order, you will receive the Snowcone device in approximately 4-6 days.

upvoted 3 times

LuckyAro 3 years, 3 months ago

Selected Answer: C

DataSync can be used to migrate data between on-premises Windows file servers and Amazon FSx for Windows File Server with its compatibility for Windows file systems.

The laboratory needs to migrate a large amount of data (30 TB) within a relatively short timeframe (5 days) and limit the impact on other departments' network traffic. Therefore, AWS DataSync can meet these requirements by providing fast and efficient data transfer with network throttling capability to control bandwidth usage.

upvoted 5 times

cloudbusting 3 years, 3 months ago

<https://docs.aws.amazon.com/datasync/latest/userguide/configure-bandwidth.html>

upvoted 3 times

bdp123 3 years, 3 months ago

Selected Answer: C

<https://aws.amazon.com/datasync/>

upvoted 3 times

A company wants to create a mobile app that allows users to stream slow-motion video clips on their mobile devices. Currently, the app captures video clips and uploads the video clips in raw format into an Amazon S3 bucket. The app retrieves these video clips directly from the S3 bucket. However, the videos are large in their raw format.

Users are experiencing issues with buffering and playback on mobile devices. The company wants to implement solutions to maximize the performance and scalability of the app while minimizing operational overhead.

Which combination of solutions will meet these requirements? (Choose two.)

- A. Deploy Amazon CloudFront for content delivery and caching. **Most Voted**
- B. Use AWS DataSync to replicate the video files across AWS Regions in other S3 buckets.
- C. Use Amazon Elastic Transcoder to convert the video files to more appropriate formats.
- D. Deploy an Auto Scaling group of Amazon EC2 instances in Local Zones for content delivery and caching.
- E. Deploy an Auto Scaling group of Amazon EC2 instances to convert the video files to more appropriate formats.

Correct Answer: A

Community vote distribution

A (52%)

C (46%)

2

Comments

pentium75 **Highly Voted** 2 years, 5 months ago
F - Fire the guy who created the current design
upvoted 31 times

awsgeek75 2 years, 4 months ago
No, make him watch all those videos with buffering!
upvoted 15 times

Bhawesh **Highly Voted** 3 years, 3 months ago
For Minimum operational overhead, the 2 options A,C should be correct.
A. Deploy Amazon CloudFront for content delivery and caching.
C. Use Amazon Elastic Transcoder to convert the video files to more appropriate formats.
upvoted 23 times

htuthtut999 **Most Recent** 4 months, 4 weeks ago

Selected Answer: C

A&C,
For C, Use Amazon Elastic Transcoder

Converts raw video into:

Mobile-optimized formats

Multiple resolutions/bitrates (adaptive streaming)

Designed exactly for video processing
Fully managed → low operational overhead
upvoted 1 times

Jarranymo 9 months ago

Selected Answer: C

A + C
AWS Elemental MediaConvert is the more modern service for this, but since Elastic Transcoder is listed, it's still a valid and correct choice for this

question.

upvoted 1 times

allanm 10 months, 1 week ago

Selected Answer: A

This question is not applicable as AWS elastic transcoder will be phased out in Nov 25

upvoted 1 times

GGP23 11 months, 2 weeks ago

Selected Answer: C

The write answer needs to be highlighted. It should be A and C.

upvoted 1 times

surajkrishnamurthy 1 year, 4 months ago

Selected Answer: C

Not able to select Multi Option

Admin please check

upvoted 1 times

Rcosmos 1 year, 4 months ago

Selected Answer: U

A melhor combinação de soluções para atender aos requisitos é:

A. Implante o Amazon CloudFront para entrega de conteúdo e armazenamento em cache.

C. Use o Amazon Elastic Transcoder para converter os arquivos de vídeo em formatos mais apropriados.

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: C

Which combination of solutions will meet these requirements? (Choose two.)

.....

upvoted 1 times

Mayank0502 1 year, 11 months ago

Selected Answer: C

A & C. Admin has almost every answer wrong

upvoted 4 times

xyGGXH 2 years, 3 months ago

Selected Answer: A

A&C is correct

upvoted 4 times

db95476 2 years, 5 months ago

Selected Answer: A

A and C

upvoted 3 times

Ruffyit 2 years, 6 months ago

For Minimum operational overhead, the 2 options A,C should be correct.

A. Deploy Amazon CloudFront for content delivery and caching.

C. Use Amazon Elastic Transcoder to convert the video files to more appropriate formats.

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

For Minimum operational overhead, the 2 options A,C should be correct.

A. Deploy Amazon CloudFront for content delivery and caching.

C. Use Amazon Elastic Transcoder to convert the video files to more appropriate formats.

upvoted 2 times

Guru4Cloud 2 years, 8 months ago

examtopics team, please fix this question, please allow to select two answer

upvoted 3 times

vincent2023 1 year, 4 months ago

It has been more than 1year and 3 months, but the issue hasn't been fixed.

upvoted 1 times

jacob_ho 2 years, 9 months ago

Elastic Transcoder has been deprecated, and AWS encourage to use AWS Elemental MediaConvert right now:

<https://aws.amazon.com/blogs/media/how-to-migrate-workflows-from-amazon-elastic-transcoder-to-aws-elemental-mediaconvert/>

upvoted 7 times

enc_0343 2 years, 11 months ago

Selected Answer: A

AC is the correct answer

upvoted 2 times

antropaws 3 years ago

Selected Answer: A

AC, the only possible answers.

upvoted 2 times

A company is launching a new application deployed on an Amazon Elastic Container Service (Amazon ECS) cluster and is using the Fargate launch type for ECS tasks. The company is monitoring CPU and memory usage because it is expecting high traffic to the application upon its launch. However, the company wants to reduce costs when utilization decreases.

What should a solutions architect recommend?

- A. Use Amazon EC2 Auto Scaling to scale at certain periods based on previous traffic patterns.
- B. Use an AWS Lambda function to scale Amazon ECS based on metric breaches that trigger an Amazon CloudWatch alarm.
- C. Use Amazon EC2 Auto Scaling with simple scaling policies to scale when ECS metric breaches trigger an Amazon CloudWatch alarm.
- D. Use AWS Application Auto Scaling with target tracking policies to scale when ECS metric breaches trigger an Amazon CloudWatch alarm. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

rrharris **Highly Voted** 2 years, 9 months ago

Answer is D - Auto-scaling with target tracking
upvoted 14 times

Neha999 **Highly Voted** 2 years, 9 months ago

D : auto-scaling with target tracking
upvoted 5 times

phuonglai **Most Recent** 1 year, 10 months ago

Selected Answer: D

<https://docs.aws.amazon.com/autoscaling/application/userguide/what-is-application-auto-scaling.html>
upvoted 4 times

awsgeek75 1 year, 10 months ago

Selected Answer: D

<https://docs.aws.amazon.com/autoscaling/application/userguide/what-is-application-auto-scaling.html>
upvoted 2 times

pentium75 1 year, 11 months ago

Selected Answer: D

This is running on Fargate, so EC2 scaling (A and C) is out. Lambda (B) is too complex.
upvoted 4 times

TariqKipkemei 2 years, 2 months ago

Target tracking will scale in/out the ECS cluster to maintain the average CPU utilization to a set value. e.g. <<50%>>> Scale out when average CPU utilization is above 50% until average CPU utilization is back to 50%. And scale in when average CPU utilization is below 50% until average CPU utilization is back to 50%.
upvoted 5 times

TariqKipkemei 2 years, 2 months ago

Answer is D
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

Answer is D - Auto-scaling with target tracking
upvoted 2 times

vincent2023 1 year, 4 months ago
Explanation please !!!
upvoted 1 times

TariqKipkemei 2 years, 7 months ago

Answer is D - Application Auto Scaling is a web service for developers and system administrators who need a solution for automatically scaling their scalable resources for individual AWS services beyond Amazon EC2.
upvoted 4 times

boxu03 2 years, 9 months ago

Selected Answer: D

should be D
upvoted 2 times

Joxtat 2 years, 9 months ago

Selected Answer: D

<https://docs.aws.amazon.com/autoscaling/application/userguide/what-is-application-auto-scaling.html>
upvoted 5 times

jahmad0730 2 years, 9 months ago

Selected Answer: D

Answer is D
upvoted 3 times

A company recently created a disaster recovery site in a different AWS Region. The company needs to transfer large amounts of data back and forth between NFS file systems in the two Regions on a periodic basis.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use AWS DataSync. **Most Voted**
- B. Use AWS Snowball devices.
- C. Set up an SFTP server on Amazon EC2.
- D. Use AWS Database Migration Service (AWS DMS).

Correct Answer: A

Community vote distribution

A (100%)

Comments

LuckyAro **Highly Voted** 2 years, 3 months ago

Selected Answer: A

AWS DataSync is a fully managed data transfer service that simplifies moving large amounts of data between on-premises storage systems and AWS services. It can also transfer data between different AWS services, including different AWS Regions. DataSync provides a simple, scalable, and automated solution to transfer data, and it minimizes the operational overhead because it is fully managed by AWS.

upvoted 19 times

Ruffyt **Most Recent** 1 year, 6 months ago

AWS DataSync is a fully managed data transfer service that simplifies moving large amounts of data between on-premises storage systems and AWS services. It can also transfer data between different AWS services, including different AWS Regions. DataSync provides a simple, scalable, and automated solution to transfer data, and it minimizes the operational overhead because it is fully managed by AWS.

upvoted 2 times

Salilgen 1 year, 6 months ago

You cannot use DataSync to transfer across Regions with an NFS

<https://docs.aws.amazon.com/datasync/latest/userguide/working-with-locations.html#working-with-locations-cross-regions>

upvoted 1 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: A

Use AWS DataSync

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

Use AWS DataSync.

upvoted 2 times

kruasan 2 years, 1 month ago

Selected Answer: A

- AWS DataSync is a data transfer service optimized for moving large amounts of data between NFS file systems. It can automatically copy files and metadata between your NFS file systems in different AWS Regions.

- DataSync requires minimal setup and management. You deploy a source and destination agent, provide the source and destination locations, and DataSync handles the actual data transfer efficiently in the background.

- DataSync can schedule and monitor data transfers to keep source and destination in sync with minimal overhead. It resumes interrupted transfers and validates data integrity.

- DataSync optimizes data transfer performance across AWS's network infrastructure. It can achieve high throughput with minimal impact to your operations.

upvoted 4 times

kruasan 2 years, 1 month ago

Option B - AWS Snowball requires physical devices to transfer data. This incurs overhead to transport devices and manually load/unload data. It is not an online data transfer solution.

Option C - Setting up and managing an SFTP server would require provisioning EC2 instances, handling security groups, and writing scripts to automate the data transfer - all of which demand more overhead than DataSync.

Option D - AWS Database Migration Service is designed for migrating databases, not general file system data. It would require converting your NFS data into a database format, incurring additional overhead.

upvoted 5 times

ashu089 2 years, 2 months ago

Selected Answer: A

A only

upvoted 2 times

skiwili 2 years, 3 months ago

Selected Answer: A

Aaaaaa

upvoted 2 times

NolaHOla 2 years, 3 months ago

A should be correct

upvoted 2 times

A company is designing a shared storage solution for a gaming application that is hosted in the AWS Cloud. The company needs the ability to use SMB clients to access data. The solution must be fully managed.

Which AWS solution meets these requirements?

- A. Create an AWS DataSync task that shares the data as a mountable file system. Mount the file system to the application server.
- B. Create an Amazon EC2 Windows instance. Install and configure a Windows file share role on the instance. Connect the application server to the file share.
- C. Create an Amazon FSx for Windows File Server file system. Attach the file system to the origin server. Connect the application server to the file system. **Most Voted**
- D. Create an Amazon S3 bucket. Assign an IAM role to the application to grant access to the S3 bucket. Mount the S3 bucket to the application server.

Correct Answer: C

Community vote distribution

C (100%)

Comments

rrharris **Highly Voted** 2 years, 9 months ago
Answer is C - SMB = storage gateway or FSx
upvoted 9 times

Neha999 **Highly Voted** 2 years, 9 months ago
C L: Amazon FSx for Windows File Server file system
upvoted 7 times

8883b6c **Most Recent** 1 year, 5 months ago
Selected Answer: C
SMB -> FSx
upvoted 1 times

phuonglai 1 year, 10 months ago
Selected Answer: C
SMB -> FSx
upvoted 4 times

TariqKipkemei 2 years, 2 months ago
Selected Answer: C
SMB = FSx for Windows File Server
upvoted 4 times

Guru4Cloud 2 years, 3 months ago
Selected Answer: C
Answer is C - SMB = storage gateway or FSx
upvoted 3 times

kruasan 2 years, 7 months ago
Selected Answer: C

• Amazon FSx for Windows File Server provides a fully managed native Windows file system that can be accessed using the industry-standard SMB protocol. This allows Windows clients like the gaming application to directly access file data.

- FSx for Windows File Server handles time-consuming file system administration tasks like provisioning, setup, maintenance, file share management, backups, security, and software patching - reducing operational overhead.
 - FSx for Windows File Server supports high file system throughput, IOPS, and consistent low latencies required for performance-sensitive workloads. This makes it suitable for a gaming application.
 - The file system can be directly attached to EC2 instances, providing a performant shared storage solution for the gaming servers.
- upvoted 5 times

kruasan 2 years, 7 months ago

Option A - DataSync is for data transfer, not providing a shared file system. It cannot be mounted or directly accessed.

Option B - A self-managed EC2 file share would require manually installing, configuring and maintaining a Windows file system and share. This demands significant overhead to operate.

Option D - Amazon S3 is object storage, not a native file system. The data in S3 would need to be converted/formatted to provide file share access, adding complexity. S3 cannot be directly mounted or provide the performance of FSx.

upvoted 5 times

elearningtakai 2 years, 8 months ago

Selected Answer: C

Amazon FSx for Windows File Server

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: C

I vote C since FSx supports SMB

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: C

AWS FSx for Windows File Server is a fully managed native Microsoft Windows file system that is accessible through the SMB protocol. It provides features such as file system backups, integrated with Amazon S3, and Active Directory integration for user authentication and access control. This solution allows for the use of SMB clients to access the data and is fully managed, eliminating the need for the company to manage the underlying infrastructure.

upvoted 3 times

Babba 2 years, 9 months ago

Selected Answer: C

C for me

upvoted 2 times

A company wants to run an in-memory database for a latency-sensitive application that runs on Amazon EC2 instances. The application processes more than 100,000 transactions each minute and requires high network throughput. A solutions architect needs to provide a cost-effective network design that minimizes data transfer charges.

Which solution meets these requirements?

A. Launch all EC2 instances in the same Availability Zone within the same AWS Region. Specify a placement group with cluster strategy when launching EC2 instances. **Most Voted**

B. Launch all EC2 instances in different Availability Zones within the same AWS Region. Specify a placement group with partition strategy when launching EC2 instances.

C. Deploy an Auto Scaling group to launch EC2 instances in different Availability Zones based on a network utilization target.

D. Deploy an Auto Scaling group with a step scaling policy to launch EC2 instances in different Availability Zones.

Correct Answer: A

Community vote distribution

A (100%)

Comments

kruasan **Highly Voted** 2 years, 7 months ago

Selected Answer: A

Reasons:

- Launching instances within a single AZ and using a cluster placement group provides the lowest network latency and highest bandwidth between instances. This maximizes performance for an in-memory database and high-throughput application.
- Communications between instances in the same AZ and placement group are free, minimizing data transfer charges. Inter-AZ and public IP traffic can incur charges.
- A cluster placement group enables the instances to be placed close together within the AZ, allowing the high network throughput required. Partition groups span AZs, reducing bandwidth.
- Auto Scaling across zones could launch instances in AZs that increase data transfer charges. It may reduce network throughput, impacting performance.

upvoted 24 times

kruasan 2 years, 7 months ago

In contrast:

Option B - A partition placement group spans AZs, reducing network bandwidth between instances and potentially increasing costs.

Option C - Auto Scaling alone does not guarantee the network throughput and cost controls required for this use case. Launching across AZs could increase data transfer charges.

Option D - Step scaling policies determine how many instances to launch based on metrics alone. They lack control over network connectivity and costs between instances after launch.

upvoted 13 times

Neha999 **Highly Voted** 2 years, 9 months ago

A : Cluster placement group

upvoted 5 times

awsgeek75 **Most Recent** 1 year, 10 months ago

Selected Answer: A

Apart from the fact that BCD distribute the instances across AZ which is bad for inter-node network latency, I think the following article is really useful in understanding A:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html>

upvoted 2 times

Ruffyit 2 years ago

- Launching instances within a single AZ and using a cluster placement group provides the lowest network latency and highest bandwidth between instances. This maximizes performance for an in-memory database and high-throughput application.
 - Communications between instances in the same AZ and placement group are free, minimizing data transfer charges. Inter-AZ and public IP traffic can incur charges.
 - A cluster placement group enables the instances to be placed close together within the AZ, allowing the high network throughput required. Partition groups span AZs, reducing bandwidth.
- upvoted 2 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: A

Cluster placement group packs instances close together inside an Availability Zone. This strategy enables workloads to achieve the low-latency network performance.

upvoted 5 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

Launch all EC2 instances in the same Availability Zone within the same AWS Region. Specify a placement group with cluster strategy when launching EC2 instances

upvoted 2 times

NoInNothing 2 years, 7 months ago

Selected Answer: A

Cluster - have low latency if its in same AZ and same region so Answer is "A"

upvoted 3 times

BeeKayEnn 2 years, 8 months ago

Answer would be A - As part of selecting all the EC2 instances in the same availability zone, they all will be within the same DC and logically the latency will be very less as compared to the other Availability Zones..

As all the autoscaling nodes will also be on the same availability zones, (as per Placement groups with Cluster mode), this would provide the low-latency network performance

Reference is below:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html>

upvoted 4 times

elearningtakai 2 years, 8 months ago

Selected Answer: A

A placement group is a logical grouping of instances within a single Availability Zone, and it provides low-latency network connectivity between instances. By launching all EC2 instances in the same Availability Zone and specifying a placement group with cluster strategy, the application can take advantage of the high network throughput and low latency network connectivity that placement groups provide.

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: A

Cluster placement groups improves throughput between the instances which means less EC2 instances would be needed thus reducing costs.

upvoted 2 times

maciekmaciek 2 years, 9 months ago

Selected Answer: A

A because Specify a placement group

upvoted 2 times

KZM 2 years, 9 months ago

It is option A:

To achieve low latency, high throughput, and cost-effectiveness, the optimal solution is to launch EC2 instances as a placement group with the cluster strategy within the same Availability Zone.

upvoted 3 times

ManOnTheMoon 2 years, 9 months ago

Why not C?

upvoted 2 times

Steve_4542636 2 years, 9 months ago

You're thinking operational efficiency. The question asks for cost reduction.

upvoted 4 times

rrharris 2 years, 9 months ago

Answer is A - Clustering

upvoted 3 times

A company that primarily runs its application servers on premises has decided to migrate to AWS. The company wants to minimize its need to scale its Internet Small Computer Systems Interface (iSCSI) storage on premises. The company wants only its recently accessed data to remain stored locally.

Which AWS solution should the company use to meet these requirements?

- A. Amazon S3 File Gateway
- B. AWS Storage Gateway Tape Gateway
- C. AWS Storage Gateway Volume Gateway stored volumes
- D. AWS Storage Gateway Volume Gateway cached volumes **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

LuckyAro **Highly Voted** 3 years, 3 months ago

Selected Answer: D

AWS Storage Gateway Volume Gateway provides two configurations for connecting to iSCSI storage, namely, stored volumes and cached volumes. The stored volume configuration stores the entire data set on-premises and asynchronously backs up the data to AWS. The cached volume configuration stores recently accessed data on-premises, and the remaining data is stored in Amazon S3.

Since the company wants only its recently accessed data to remain stored locally, the cached volume configuration would be the most appropriate. It allows the company to keep frequently accessed data on-premises and reduce the need for scaling its iSCSI storage while still providing access to all data through the AWS cloud. This configuration also provides low-latency access to frequently accessed data and cost-effective off-site backups for less frequently accessed data.

upvoted 41 times

tonybuivannghia 1 year, 8 months ago

You're correct but a small mistake that the stored volume configuration stores the entire data set on-premises and asynchronously backs up the data to AWS (S3). The cached volume is low latency access to most recent data.

upvoted 2 times

smgsi **Highly Voted** 3 years, 3 months ago

Selected Answer: D

https://docs.amazonaws.cn/en_us/storagegateway/latest/vgw/StorageGatewayConcepts.html#storage-gateway-cached-concepts

upvoted 10 times

TariqKipkemei **Most Recent** 2 years, 8 months ago

Selected Answer: D

Frequently accessed data = AWS Storage Gateway Volume Gateway cached volumes

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

The best AWS solution to meet the requirements is to use AWS Storage Gateway cached volumes (option D).

The key points:

Company migrating on-prem app servers to AWS

Want to minimize scaling on-prem iSCSI storage

Only recent data should remain on-premises

The AWS Storage Gateway cached volumes allow the company to connect their on-premises iSCSI storage to AWS cloud storage. It stores frequently accessed data locally in the cache for low-latency access, while older data is stored in AWS.

upvoted 4 times

kruasan 3 years, 1 month ago

Selected Answer: D

- Volume Gateway cached volumes store entire datasets on S3, while keeping a portion of recently accessed data on your local storage as a cache. This meets the goal of minimizing on-premises storage needs while keeping hot data local.
- The cache provides low-latency access to your frequently accessed data, while long-term retention of the entire dataset is provided durable and cost-effective in S3.
- You get virtually unlimited storage on S3 for your infrequently accessed data, while controlling the amount of local storage used for cache. This simplifies on-premises storage scaling.
- Volume Gateway cached volumes support iSCSI connections from on-premises application servers, allowing a seamless migration experience. Servers access local cache and S3 storage volumes as iSCSI LUNs.

upvoted 8 times

kruasan 3 years, 1 month ago

In contrast:

Option A - S3 File Gateway only provides file interfaces (NFS/SMB) to data in S3. It does not support block storage or cache recently accessed data locally.

Option B - Tape Gateway is designed for long-term backup and archiving to virtual tape cartridges on S3. It does not provide primary storage volumes or local cache for low-latency access.

Option C - Volume Gateway stored volumes keep entire datasets locally, then asynchronously back them up to S3. This does not meet the goal of minimizing on-premises storage needs.

upvoted 6 times

Steve_4542636 3 years, 3 months ago

Selected Answer: D

I vote D

upvoted 2 times

ManOnTheMoon 3 years, 3 months ago

Agree with D

upvoted 2 times

Babba 3 years, 3 months ago

Selected Answer: D

recently accessed data to remain stored locally - cached

upvoted 4 times

Bhawesh 3 years, 3 months ago

Selected Answer: D

D. AWS Storage Gateway Volume Gateway cached volumes

upvoted 4 times

bdp123 3 years, 3 months ago

Selected Answer: D

recently accessed data to remain stored locally - cached

upvoted 4 times

A company has multiple AWS accounts that use consolidated billing. The company runs several active high performance Amazon RDS for Oracle On-Demand DB instances for 90 days. The company's finance team has access to AWS Trusted Advisor in the consolidated billing account and all other AWS accounts.

The finance team needs to use the appropriate AWS account to access the Trusted Advisor check recommendations for RDS. The finance team must review the appropriate Trusted Advisor check to reduce RDS costs.

Which combination of steps should the finance team take to meet these requirements? (Choose two.)

- A. Use the Trusted Advisor recommendations from the account where the RDS instances are running.
- B. Use the Trusted Advisor recommendations from the consolidated billing account to see all RDS instance checks at the same time. **Most Voted**
- C. Review the Trusted Advisor check for Amazon RDS Reserved Instance Optimization.
- D. Review the Trusted Advisor check for Amazon RDS Idle DB Instances. **Most Voted**
- E. Review the Trusted Advisor check for Amazon Redshift Reserved Node Optimization.

Correct Answer: BD

Community vote distribution



Comments

Nietzsche82 **Highly Voted** 3 years, 3 months ago

Selected Answer: BD

B & D

<https://aws.amazon.com/premiumsupport/knowledge-center/trusted-advisor-cost-optimization/>

upvoted 21 times

Michal_L_95 **Highly Voted** 3 years, 3 months ago

Selected Answer: BC

I would go with B and C as the company is running for 90 days and C option is basing on 30 days report which would mean that there is higher potential on cost saving rather than on idle instances

upvoted 8 times

Steve_4542636 3 years, 3 months ago

C is stating "Reserved Instances" The question states they are using On Demand Instances. Reserved instances are reserved for less money for 1 or 3 years.

upvoted 9 times

Lalo 3 years ago

In the scenario it says for 90 days, therefore the correct answer is D

No C

upvoted 2 times

pentium75 2 years, 5 months ago

D checks for instances that did not have a connection for a while, which is not the case here, all instances are active.

upvoted 2 times

Michal_L_95 3 years, 3 months ago

Once read the question again, I agree with you.

upvoted 1 times

pentium75 2 years, 5 months ago

C is about optimizing on-demand instances by turning them into reserved instances.

upvoted 2 times

htuthtut999 **Most Recent** 4 months, 4 weeks ago

Selected Answer: AC

B.Trusted Advisor is account-scoped for most service checks, including RDS.You must view RDS recommendations in the account where the RDS instances actually run.

C. Review the Trusted Advisor check for Amazon RDS Reserved Instance Optimization

Reserved Instances are recommended when:

Instances are running continuously, Usage is predictable

upvoted 1 times

In2718 9 months, 3 weeks ago

Selected Answer: AC

Chat AI -

B: Wrong — payer account can't see RDS instance-level Trusted Advisor checks for all linked accounts.

D: Wrong — "Idle DB Instances" doesn't apply here because the instances are explicitly described as active.

E: Wrong — "Redshift Reserved Node Optimization" is for Redshift, not RDS.

upvoted 2 times

In2718 9 months, 3 weeks ago

Selected Answer: AC

Chat says answer A and C. May want to check it out yourself. Fairly good reasons for the answer.

upvoted 1 times

c12ab95 1 year ago

Selected Answer: AC

A. Use the Trusted Advisor recommendations from the account where the RDS instances are running.

Trusted Advisor cost optimization checks for RDS (such as Reserved Instance Optimization and Idle DB Instances) are available in the account where the RDS resources are provisioned. The finance team must access the recommendations from the relevant account to see the checks for those specific RDS instances.

C. Review the Trusted Advisor check for Amazon RDS Reserved Instance Optimization.

This check analyzes your On-Demand RDS usage and recommends purchasing Reserved Instances to save costs. It is a key Trusted Advisor recommendation for reducing RDS costs, especially for workloads running for extended periods

upvoted 2 times

Yak_Yeti 1 year, 1 month ago

Selected Answer: AC

The article emphasizes that Trusted Advisor generates recommendations at the individual AWS account level. While the consolidated billing account provides a centralized view of billing, the specific optimization recommendations for resources like RDS instances will be found within the account where those resources reside.

The article also mentions checks for idle resources like "Amazon EC2 Idle Instances" and "Amazon RDS Idle DB Instances." However, as the original analysis pointed out, the scenario specifies that the RDS instances are active and running a high-performance application, making the "Idle DB Instances" check less relevant for the immediate goal of reducing costs for these specific instances.

upvoted 1 times

a8a1e0e 1 year, 4 months ago

Selected Answer: AC

A: AWS Trusted Advisor recommendations for RDS cost optimization are only available within the AWS account where the instances are running. The consolidated billing (payer) account does not provide detailed service-level checks for linked accounts. Therefore, the finance team must access the Trusted Advisor check directly from the AWS account that owns the RDS instances to see relevant recommendations.

C: The company has been using high-performance RDS instances for 90 days, which indicates a long-term workload. Reserved Instances (RIs) offer significant cost savings compared to On-Demand pricing when committing to one or three years. Trusted Advisor identifies instances with consistent usage and recommends RIs to reduce costs, making this the most relevant check for optimizing expenses. Even after 90 days, purchasing an RI for the remaining duration still provides substantial savings.

upvoted 5 times

MatAlves 1 year, 9 months ago

Selected Answer: BC

The answer is either BC or BD, depending on how you interpret "The company runs several active... instances for 90 days for 90 days."

D: it assumes the instances will only run for 90 days, so reserved instances can't be the answer, since it requires 1-3 years utilization.

C: it assumes there is no idle instances since they've been active for the last 90 days.

upvoted 6 times

aagarwallko 1 year, 9 months ago

B: Use the Trusted Advisor recommendations from the consolidated billing account to see all RDS instance checks at the same time. This option allows the finance team to see all RDS instance checks across all AWS accounts in one place. Since the company uses consolidated billing, this account will have access to all of the AWS accounts' Trusted Advisor recommendations.

C: Review the Trusted Advisor check for Amazon RDS Reserved Instance Optimization. This check can help identify cost savings opportunities for RDS by identifying instances that can be covered by Reserved Instances. This can result in significant savings on RDS costs.
upvoted 2 times

ike001 1 year, 11 months ago

BD is the answer-A Amazon Redshift Reserved Node Optimization and Relational Database Service (RDS) Reserved Instance Optimization check are not available to accounts linked in consolidated billing.
upvoted 3 times

sandordini 2 years, 1 month ago

Selected Answer: BD

"you can reserve a DB instance for a one- or three-year term". We only have data for 90 days. I feel it too risky to commit for 1/3 year(s) without information on future usage. If we knew that we expected the same usage pattern for the next 1,2,3 years, I'd agree with C.
upvoted 3 times

soufiyane 2 years, 2 months ago

Selected Answer: BC

B) Use the Trusted Advisor recommendations from the consolidated billing account to see all RDS instance checks at the same time. This option allows the finance team to see all RDS instance checks across all AWS accounts in one place. Since the company uses consolidated billing, this account will have access to all of the AWS accounts' Trusted Advisor recommendations.

C) Review the Trusted Advisor check for Amazon RDS Reserved Instance Optimization. This check can help identify cost savings opportunities for RDS by identifying instances that can be covered by Reserved Instances. This can result in significant savings on RDS costs.
upvoted 2 times

Rhydian25 1 year, 12 months ago

Reserved Instances are for 1 or 3 years. Not for 90 days
upvoted 5 times

scar0909 2 years, 3 months ago

Selected Answer: BC

<https://aws.amazon.com/premiumsupport/knowledge-center/trusted-advisor-cost-optimization/>
upvoted 2 times

bujuman 2 years, 3 months ago

Selected Answer: BD

Insights: The company runs several active high performance Amazon RDS for Oracle On-Demand DB instances for 90 days
So it's clear that this company needs to check the configuration of any Amazon Relational Database Service (Amazon RDS) for any database (DB) instances that appear to be idle.
upvoted 2 times

dkw2342 2 years, 3 months ago

B&C

AWS Trusted Advisor is an online resource to help you reduce cost, increase performance, and improve security by optimizing your AWS environment. (...) Recommendations are based on the previous calendar month's hour-by-hour usage aggregated across all consolidated billing accounts.
<https://docs.aws.amazon.com/whitepapers/latest/cost-optimization-reservation-models/aws-trusted-advisor.html>

Amazon EC2 Reserved Instance Optimization: An important part of using AWS involves balancing your Reserved Instance (RI) purchase against your On-Demand Instance usage. This check provides recommendations on which RIs will help reduce the costs incurred from using On-Demand Instances. We create these recommendations by analyzing your On-Demand usage for the past 30 days. We then categorizing the usage into eligible categories for reservations.
<https://docs.aws.amazon.com/awssupport/latest/user/cost-optimization-checks.html#amazon-ec2-reserved-instances-optimization>
upvoted 2 times

Salilgen 1 year, 6 months ago

The company runs instances only for 90 days then reserved instances cannot save costs
upvoted 1 times

NayeraB 2 years, 3 months ago

Selected Answer: BC

If you're choosing D for the idle instances, Amazon RDS Reserved Instance Optimization Trusted Advisor check includes recommendations related to underutilized and idle RDS instances. It helps identify instances that are not fully utilized and provides recommendations on how to optimize costs, such as resizing or terminating unused instances, or purchasing reserved instances to match usage patterns more efficiently.
upvoted 3 times

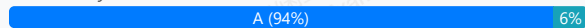
A solutions architect needs to optimize storage costs. The solutions architect must identify any Amazon S3 buckets that are no longer being accessed or are rarely accessed.

Which solution will accomplish this goal with the LEAST operational overhead?

- A. Analyze bucket access patterns by using the S3 Storage Lens dashboard for advanced activity metrics. **Most Voted**
- B. Analyze bucket access patterns by using the S3 dashboard in the AWS Management Console.
- C. Turn on the Amazon CloudWatch BucketSizeBytes metric for buckets. Analyze bucket access patterns by using the metrics data with Amazon Athena.
- D. Turn on AWS CloudTrail for S3 object monitoring. Analyze bucket access patterns by using CloudTrail logs that are integrated with Amazon CloudWatch Logs.

Correct Answer: A

Community vote distribution



Comments

kpato87 **Highly Voted** 3 years, 3 months ago

Selected Answer: A

S3 Storage Lens is a fully managed S3 storage analytics solution that provides a comprehensive view of object storage usage, activity trends, and recommendations to optimize costs. Storage Lens allows you to analyze object access patterns across all of your S3 buckets and generate detailed metrics and reports.

upvoted 24 times

bdp123 **Highly Voted** 3 years, 3 months ago

Selected Answer: A

<https://aws.amazon.com/blogs/aws/s3-storage-lens/>

upvoted 5 times

Gape4 **Most Recent** 1 year, 11 months ago

Selected Answer: A

S3 Storage Lens includes an interactive dashboard which you can find in the S3 console. The dashboard gives you the ability to perform filtering and drill-down into your metrics to really understand how your storage is being used. The metrics are organized into categories like data protection and cost efficiency, to allow you to easily find relevant metrics.

upvoted 2 times

AmijoSando 2 years, 1 month ago

Anyone passed the exam can confirm the right answer ? A or D

upvoted 1 times

xyGGXH 2 years, 3 months ago

Selected Answer: A

A

S3 Storage Lens is the first cloud storage analytics solution to provide a single view of object storage usage and activity across hundreds, or even thousands, of accounts in an organization, with drill-downs to generate insights at multiple aggregation levels.

upvoted 3 times

Neung983 2 years, 3 months ago

On the other hand, Option B suggests using the S3 dashboard in the AWS Management Console, which provides a straightforward and user-friendly interface to monitor S3 bucket access patterns. This option may have less operational overhead compared to setting up and managing Storage Lens. Additionally, for simply identifying rarely accessed buckets, the built-in metrics and access analysis provided by the S3 dashboard can often suffice without the need for advanced analytics offered by Storage Lens. Therefore, Option B is considered to have less operational overhead for the specific task described in the question.

upvoted 1 times

jaswantn 2 years, 4 months ago

But nowhere on S3 Storage Lens dashboard this information is available; that when the bucket is accessed last time. But it gives insight on the bucket's size. with this information we can check if files can be moved to less costly storage class. This way we can reduce storage cost..... The information which is the main requirement of the given scenario, is available when we use Cloudtrail logs ... so i choose option D.

upvoted 1 times

jaswantn 2 years, 4 months ago

if the bucket is being accessed frequently then we can leave it as it is, otherwise we can move the files to infrequent access storage class thus can save some money.

upvoted 2 times

Ruffyt 2 years, 6 months ago

S3 Storage Lens is a fully managed S3 storage analytics solution that provides a comprehensive view of object storage usage, activity trends, and recommendations to optimize costs. Storage Lens allows you to analyze object access patterns across all of your S3 buckets and generate detailed metrics and reports.

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: A

Amazon S3 Storage Lens was designed to handle this requirement.

upvoted 2 times

Wayne23Fang 2 years, 9 months ago

Selected Answer: D

A missed turning on monitoring. It can also help you learn about your customer base and understand your Amazon S3 bill. By default, Amazon S3 doesn't collect server access logs. When you enable logging, Amazon S3 delivers access logs for a source bucket to a target bucket that you choose.

I could not find that S3 storage Lens examples online showing using Lens to identify idle S3 buckets. Instead I find using S3 Access Logging. Hmm.

upvoted 3 times

pentium75 2 years, 5 months ago

How will you find when a bucket was used the last time if you turn on logging NOW?

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

S3 Storage Lens is a cloud-storage analytics feature that provides you with 29+ usage and activity metrics, including object count, size, age, and access patterns. This data can help you understand how your data is being used and identify areas where you can optimize your storage costs. The S3 Storage Lens dashboard provides an interactive view of your storage usage and activity trends. This makes it easy to identify buckets that are no longer being accessed or are rarely accessed.

The S3 Storage Lens dashboard is a fully managed service, so there is no need to set up or manage any additional infrastructure.

upvoted 3 times

BigHammer 2 years, 9 months ago

"S3 Storage Lens" seems to be the popular answer, however, where in Storage Lens can you see if a bucket/object is being USED? I see all kinds of stats, but not that.

upvoted 3 times

MatAlves 1 year, 9 months ago

"S3 Storage Lens delivers organization-wide visibility into object storage usage, activity trends, and makes actionable recommendations to optimize costs..."

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

<https://aws.amazon.com/blogs/aws/s3-storage-lens/>

upvoted 3 times

kruasan 3 years, 1 month ago

Selected Answer: A

The S3 Storage Lens dashboard provides visibility into storage metrics and activity patterns to help optimize storage costs. It shows metrics like objects added, objects deleted, storage consumed, and requests. It can filter by bucket, prefix, and tag to analyze specific subsets of data

upvoted 4 times

kruasan 3 years, 1 month ago

B) The standard S3 console dashboard provides basic info but would require manually analyzing metrics for each bucket. This does not scale well and requires significant overhead.

C) Turning on the BucketSizeBytes metric and analyzing the data in Athena may provide insights but would require enabling metrics, building Athena queries, and analyzing the results. This requires more operational effort than option A.

D) Enabling CloudTrail logging and monitoring the logs in CloudWatch Logs could provide access pattern data but would require setting up CloudTrail, monitoring the logs, and analyzing the relevant info. This option has the highest operational overhead

upvoted 5 times

LuckyAro 3 years, 3 months ago

Selected Answer: A

S3 Storage Lens provides a dashboard with advanced activity metrics that enable the identification of infrequently accessed and unused buckets. This can help a solutions architect optimize storage costs without incurring additional operational overhead.

upvoted 4 times

Babba 3 years, 3 months ago

Selected Answer: A

it looks like it's A

upvoted 3 times

A company sells datasets to customers who do research in artificial intelligence and machine learning (AI/ML). The datasets are large, formatted files that are stored in an Amazon S3 bucket in the us-east-1 Region. The company hosts a web application that the customers use to purchase access to a given dataset. The web application is deployed on multiple Amazon EC2 instances behind an Application Load Balancer. After a purchase is made, customers receive an S3 signed URL that allows access to the files.

The customers are distributed across North America and Europe. The company wants to reduce the cost that is associated with data transfers and wants to maintain or improve performance.

What should a solutions architect do to meet these requirements?

- A. Configure S3 Transfer Acceleration on the existing S3 bucket. Direct customer requests to the S3 Transfer Acceleration endpoint. Continue to use S3 signed URLs for access control.
- B. Deploy an Amazon CloudFront distribution with the existing S3 bucket as the origin. Direct customer requests to the CloudFront URL. Switch to CloudFront signed URLs for access control. **Most Voted**
- C. Set up a second S3 bucket in the eu-central-1 Region with S3 Cross-Region Replication between the buckets. Direct customer requests to the closest Region. Continue to use S3 signed URLs for access control.
- D. Modify the web application to enable streaming of the datasets to end users. Configure the web application to read the data from the existing S3 bucket. Implement access control directly in the application.

Correct Answer: B

Community vote distribution

B (100%)

Comments

LuckyAro **Highly Voted** 2 years, 9 months ago

Selected Answer: B

To reduce the cost associated with data transfers and maintain or improve performance, a solutions architect should use Amazon CloudFront, a content delivery network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds.

Deploying a CloudFront distribution with the existing S3 bucket as the origin will allow the company to serve the data to customers from edge locations that are closer to them, reducing data transfer costs and improving performance.

Directing customer requests to the CloudFront URL and switching to CloudFront signed URLs for access control will enable customers to access the data securely and efficiently.

upvoted 14 times

awsgeek75 **Highly Voted** 1 year, 10 months ago

Selected Answer: B

A: Speeds uploads

C: Increases the cost rather than reducing it

D: Stopped reading after "Modify the web application..."

upvoted 11 times

Ruffyt **Most Recent** 2 years ago

To reduce the cost associated with data transfers and maintain or improve performance, a solutions architect should use Amazon CloudFront, a content delivery network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds.

upvoted 2 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: B

Technically both option B and C will work. But because cost is a factor then Amazon CloudFront should be the preferred option.
upvoted 2 times

react97 2 years, 2 months ago

Selected Answer: B

B.
1. Amazon CloudFront caches content at edge locations -- reducing the need for frequent data transfer from S3 bucket -- thus significantly lowering data transfer costs (as compared to directly serving data from S3 bucket to customers in different regions)
2. CloudFront delivers content to users from the nearest edge location -- minimizing latency -- improves performance for customers

A - focus on accelerating uploads to S3 which may not necessarily improve the performance needed for serving datasets to customers
C - helps with redundancy and data availability but does not necessarily offer cost savings for data transfer.
D - complex to implement, does not address data transfer cost
upvoted 6 times

bdp123 2 years, 9 months ago

Selected Answer: B

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/PrivateContent.html>
upvoted 4 times

Bhawesh 2 years, 9 months ago

Selected Answer: B

B. Deploy an Amazon CloudFront distribution with the existing S3 bucket as the origin. Direct customer requests to the CloudFront URL. Switch to CloudFront signed URLs for access control.

<https://www.examttopics.com/discussions/amazon/view/68990-exam-aws-certified-solutions-architect-associate-saa-c02/>
upvoted 3 times

A company is using AWS to design a web application that will process insurance quotes. Users will request quotes from the application. Quotes must be separated by quote type, must be responded to within 24 hours, and must not get lost. The solution must maximize operational efficiency and must minimize maintenance.

Which solution meets these requirements?

A. Create multiple Amazon Kinesis data streams based on the quote type. Configure the web application to send messages to the proper data stream. Configure each backend group of application servers to use the Kinesis Client Library (KCL) to pool messages from its own data stream.

B. Create an AWS Lambda function and an Amazon Simple Notification Service (Amazon SNS) topic for each quote type. Subscribe the Lambda function to its associated SNS topic. Configure the application to publish requests for quotes to the appropriate SNS topic.

C. Create a single Amazon Simple Notification Service (Amazon SNS) topic. Subscribe Amazon Simple Queue Service (Amazon SQS) queues to the SNS topic. Configure SNS message filtering to publish messages to the proper SQS queue based on the quote type. Configure each backend application server to use its own SQS queue. **Most Voted**

D. Create multiple Amazon Kinesis Data Firehose delivery streams based on the quote type to deliver data streams to an Amazon OpenSearch Service cluster. Configure the application to send messages to the proper delivery stream. Configure each backend group of application servers to search for the messages from OpenSearch Service and process them accordingly.

Correct Answer: C

Community vote distribution

C (100%)

Comments

LuckyAro **Highly Voted** 3 years, 3 months ago

Selected Answer: C

Quote types need to be separated: SNS message filtering can be used to publish messages to the appropriate SQS queue based on the quote type, ensuring that quotes are separated by type.

Quotes must be responded to within 24 hours and must not get lost: SQS provides reliable and scalable queuing for messages, ensuring that quotes will not get lost and can be processed in a timely manner. Additionally, each backend application server can use its own SQS queue, ensuring that quotes are processed efficiently without any delay.

Operational efficiency and minimizing maintenance: Using a single SNS topic and multiple SQS queues is a scalable and cost-effective approach, which can help to maximize operational efficiency and minimize maintenance. Additionally, SNS and SQS are fully managed services, which means that the company will not need to worry about maintenance tasks such as software updates, hardware upgrades, or scaling the infrastructure.

upvoted 19 times

Vlad **Highly Voted** 3 years, 3 months ago

C is the best option

upvoted 9 times

akshay243007 **Most Recent** 1 year, 10 months ago

Selected Answer: C

SQS + SNS = fanout

upvoted 3 times

Uzbekistan 2 years, 3 months ago

Option C would be the most suitable solution to meet the requirements while maximizing operational efficiency and minimizing maintenance.

Explanation:

Amazon SNS (Simple Notification Service) allows for the creation of a single topic to which multiple subscribers can be attached. In this scenario, each quote type can be considered a subscriber. Amazon SQS (Simple Queue Service) queues can be subscribed to the SNS topic, and SNS message

filtering can be used to direct messages to the appropriate SQS queue based on the quote type. This setup ensures that quotes are separated by quote type and that they are not lost. Each backend application server can then poll its own SQS queue to retrieve and process messages. This architecture is efficient, scalable, and requires minimal maintenance, as it leverages managed AWS services without the need for complex custom code or infrastructure setup.

upvoted 4 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

I originally went for D due to searching requirements but Open Search is for analytics and logs and nothing to do with data coming from streams as in this question.

upvoted 2 times

Ruffyit 2 years, 7 months ago

Quote types need to be separated: SNS message filtering can be used to publish messages to the appropriate SQS queue based on the quote type, ensuring that quotes are separated by type.

Quotes must be responded to within 24 hours and must not get lost: SQS provides reliable and scalable queuing for messages, ensuring that quotes will not get lost and can be processed in a timely manner. Additionally, each backend application server can use its own SQS queue, ensuring that quotes are processed efficiently without any delay.

Operational efficiency and minimizing maintenance: Using a single SNS topic and multiple SQS queues is a scalable and cost-effective approach, which can help to maximize operational efficiency and minimize maintenance. Additionally, SNS and SQS are fully managed services, which means that the company will not need to worry about maintenance tasks such as software updates, hardware upgrades, or scaling the

upvoted 3 times

tekjm 2 years, 8 months ago

Keyword is "...and must not get lost" = SQS

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

Create a single SNS topic

Subscribe separate SQS queues per quote type

Use SNS message filtering to send messages to proper queue

Backend servers poll their respective SQS queue

The key points:

Quote requests must be processed within 24 hrs without loss

Need to maximize efficiency and minimize maintenance

Requests separated by quote type

upvoted 2 times

lexotan 3 years, 1 month ago

Selected Answer: C

This wrong answers from examtopic are getting me so frustrated. Which one is the correct answer then?

upvoted 6 times

Steve_4542636 3 years, 3 months ago

Selected Answer: C

This is the SNS fan-out technique where you will have one SNS service to many SQS services

<https://docs.aws.amazon.com/sns/latest/dg/sns-sqs-as-subscriber.html>

upvoted 7 times

UnluckyDucky 3 years, 2 months ago

SNS Fan-out fans message to all subscribers, this uses SNS filtering to publish the message only to the right SQS queue (not all of them).

upvoted 3 times

Yechi 3 years, 3 months ago

Selected Answer: C

<https://aws.amazon.com/getting-started/hands-on/filter-messages-published-to-topics/>

upvoted 8 times

A company has an application that runs on several Amazon EC2 instances. Each EC2 instance has multiple Amazon Elastic Block Store (Amazon EBS) data volumes attached to it. The application's EC2 instance configuration and data need to be backed up nightly. The application also needs to be recoverable in a different AWS Region.

Which solution will meet these requirements in the MOST operationally efficient way?

- A. Write an AWS Lambda function that schedules nightly snapshots of the application's EBS volumes and copies the snapshots to a different Region.
- B. Create a backup plan by using AWS Backup to perform nightly backups. Copy the backups to another Region. Add the application's EC2 instances as resources. **Most Voted**
- C. Create a backup plan by using AWS Backup to perform nightly backups. Copy the backups to another Region. Add the application's EBS volumes as resources.
- D. Write an AWS Lambda function that schedules nightly snapshots of the application's EBS volumes and copies the snapshots to a different Availability Zone.

Correct Answer: B

Community vote distribution

B (94%)

6%

Comments

khasport **Highly Voted** 2 years, 9 months ago

B is answer so the requirement is "The application's EC2 instance configuration and data need to be backed up nightly" so we need "add the application's EC2 instances as resources". This option will backup both EC2 configuration and data
upvoted 23 times

TungPham **Highly Voted** 2 years, 9 months ago

Selected Answer: B

<https://aws.amazon.com/vi/blogs/aws/aws-backup-ec2-instances-efs-single-file-restore-and-cross-region-backup/>
When you back up an EC2 instance, AWS Backup will protect all EBS volumes attached to the instance, and it will attach them to an AMI that stores all parameters from the original EC2 instance except for two
upvoted 17 times

raymondfekry **Most Recent** 1 year, 11 months ago

Selected Answer: B

Question says: " The application's EC2 instance configuration and data need to be backed up", thus C is not correct, B is
upvoted 3 times

Ruffyt 2 years ago

<https://aws.amazon.com/vi/blogs/aws/aws-backup-ec2-instances-efs-single-file-restore-and-cross-region-backup/>
When you back up an EC2 instance, AWS Backup will protect all EBS volumes attached to the instance, and it will attach them to an AMI that stores all parameters from the original EC2 instance except for two
upvoted 2 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: B

As part of configuring a backup plan you need to enable (opt-in) resource types that will be protected by the backup plan. For this case EC2.
<https://aws.amazon.com/getting-started/hands-on/amazon-ec2-backup-and-restore-using-aws-backup/#:~:text=the%20services%20used%20with-,AWS%20Backup,-a.%20In%20the%20navigation>
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B is the most appropriate solution because it allows you to create a backup plan to automate the backup process of EC2 instances and EBS volumes, and copy backups to another region. Additionally, you can add the application's EC2 instances as resources to ensure their configuration and data are backed up nightly.

upvoted 2 times

Geekboii 2 years, 8 months ago

i would say B

upvoted 2 times

Geekboii 2 years, 8 months ago

i would say B

upvoted 2 times

AlmeroSenior 2 years, 9 months ago

Selected Answer: B

AWS KB states if you select the EC2 instance , associated EBS's will be auto covered .

<https://aws.amazon.com/blogs/aws/aws-backup-ec2-instances-efs-single-file-restore-and-cross-region-backup/>

upvoted 3 times

LuckyAro 2 years, 9 months ago

Selected Answer: B

B is the most appropriate solution because it allows you to create a backup plan to automate the backup process of EC2 instances and EBS volumes, and copy backups to another region. Additionally, you can add the application's EC2 instances as resources to ensure their configuration and data are backed up nightly.

A and D involve writing custom Lambda functions to automate the snapshot process, which can be complex and require more maintenance effort. Moreover, these options do not provide an integrated solution for managing backups and recovery, and copying snapshots to another region.

Option C involves creating a backup plan with AWS Backup to perform backups for EBS volumes only. This approach would not back up the EC2 instances and their configuration

upvoted 3 times

Mia2009687 2 years, 5 months ago

The data is stored in the EBS storage volume, EC2 won't hold the data, I think we need "Add the application's EBS volumes as resources."

upvoted 2 times

everfly 2 years, 9 months ago

Selected Answer: C

The application's EC2 instance configuration and data are stored on EBS volume right?

upvoted 2 times

awsgeek75 1 year, 10 months ago

No, ECS config is the config you provide when launching the EC2 instance. EBS is a resource for EC2 as a part of configuration. When you backup EC2, it will backup the instance which resulted from the configuration and that will include the EBS volumes that are attached to the instance.

upvoted 3 times

pentium75 1 year, 11 months ago

No, this is not how EC2 works.

upvoted 1 times

Rehan33 2 years, 9 months ago

The data is store on EBS volume so why we are not using EBS as a source instead of EC2

upvoted 1 times

obatunde 2 years, 9 months ago

Because "The application's EC2 instance configuration and data need to be backed up nightly"

upvoted 6 times

thewalker 1 year, 11 months ago

Also, if EBS volumes are added or removed as the requirement, not need to update the AWS Config.

upvoted 1 times

fulingyu288 2 years, 9 months ago

Selected Answer: B

Use AWS Backup to create a backup plan that includes the EC2 instances, Amazon EBS snapshots, and any other resources needed for recovery. The backup plan can be configured to run on a nightly schedule.

upvoted 2 times

zTopic 2 years, 9 months ago

Selected Answer: B

The application's EC2 instance configuration and data need to be backed up nightly >> B
upvoted 2 times

NolaHOla 2 years, 9 months ago

But isn't the data needed to be backed up on the EBS ?
upvoted 1 times

A company is building a mobile app on AWS. The company wants to expand its reach to millions of users. The company needs to build a platform so that authorized users can watch the company's content on their mobile devices.

What should a solutions architect recommend to meet these requirements?

- A. Publish content to a public Amazon S3 bucket. Use AWS Key Management Service (AWS KMS) keys to stream content.
- B. Set up IPsec VPN between the mobile app and the AWS environment to stream content.
- C. Use Amazon CloudFront. Provide signed URLs to stream content. **Most Voted**
- D. Set up AWS Client VPN between the mobile app and the AWS environment to stream content.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Steve_4542636 **Highly Voted** 2 years, 9 months ago

Selected Answer: C

Enough with CloudFront already.
upvoted 29 times

TariqKipkemei 2 years, 7 months ago

Hahaha..cloudfront too hyped :)
upvoted 4 times

awsgeek75 1 year, 10 months ago

This whole exam seems like a sales pitch for CloudFront and SQS... lol!
upvoted 7 times

LuckyAro **Highly Voted** 2 years, 9 months ago

Selected Answer: C

Amazon CloudFront is a content delivery network (CDN) that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds. CloudFront supports signed URLs that provide authorized access to your content. This feature allows the company to control who can access their content and for how long, providing a secure and scalable solution for millions of users.
upvoted 9 times

mwwt2022 1 year, 11 months ago

great explanation!
upvoted 2 times

lostmagnet001 **Most Recent** 1 year, 10 months ago

Selected Answer: C

CF always for reaching places
upvoted 3 times

Ruffyit 2 years ago

Use Amazon CloudFront. Provide signed URLs to stream content.
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

Use Amazon CloudFront. Provide signed URLs to stream content.
upvoted 2 times

antropaws 2 years, 6 months ago

Selected Answer: C

C is correct.

upvoted 3 times

kprakashbehera 2 years, 9 months ago

Cloudfront is the correct solution.

upvoted 5 times

datz 2 years, 8 months ago

Feel your pain :D hahaha

upvoted 2 times

jennyka76 2 years, 9 months ago

C

<https://www.amazonaws.cn/en/cloudfront/>

upvoted 2 times

A company has an on-premises MySQL database used by the global sales team with infrequent access patterns. The sales team requires the database to have minimal downtime. A database administrator wants to migrate this database to AWS without selecting a particular instance type in anticipation of more users in the future.

Which service should a solutions architect recommend?

- A. Amazon Aurora MySQL
- B. Amazon Aurora Serverless for MySQL **Most Voted**
- C. Amazon Redshift Spectrum
- D. Amazon RDS for MySQL

Correct Answer: B

Community vote distribution

B (100%)

Comments

cloudbusting **Highly Voted** 2 years, 9 months ago
"without selecting a particular instance type" = serverless
upvoted 32 times

elearningtakai **Highly Voted** 2 years, 8 months ago

Selected Answer: B

With Aurora Serverless for MySQL, you don't need to select a particular instance type, as the service automatically scales up or down based on the application's needs.
upvoted 12 times

awsgeek75 **Most Recent** 1 year, 10 months ago

Selected Answer: B

The DBA had one job and he doesn't want to do it... so B it is
upvoted 10 times

Ruffyit 2 years ago

without selecting a particular instance type = Amazon Aurora Serverless for MySQL
upvoted 4 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: B

without selecting a particular instance type = Amazon Aurora Serverless for MySQL
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B. Amazon Aurora Serverless for MySQL
upvoted 2 times

Diqian 2 years, 3 months ago

What's the difference between A and B. I think Aurora is serverless, isn't it?
upvoted 2 times

Valder21 2 years, 3 months ago

seems serverless is an option of amazon aurora. Not a very good naming scheme.
upvoted 3 times

Srikanth0057 2 years, 9 months ago

Selected Answer: B

Bbbbbbb

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: B

<https://aws.amazon.com/rds/aurora/serverless/>

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: B

Amazon Aurora Serverless for MySQL is a fully managed, auto-scaling relational database service that scales up or down automatically based on the application demand. This service provides all the capabilities of Amazon Aurora, such as high availability, durability, and security, without requiring the customer to provision any database instances.

With Amazon Aurora Serverless for MySQL, the sales team can enjoy minimal downtime since the database is designed to automatically scale to accommodate the increased traffic. Additionally, the service allows the customer to pay only for the capacity used, making it cost-effective for infrequent access patterns.

Amazon RDS for MySQL could also be an option, but it requires the customer to select an instance type, and the database administrator would need to monitor and adjust the instance size manually to accommodate the increasing traffic.

upvoted 3 times

Drayen25 2 years, 9 months ago

Minimal downtime points directly to Aurora Serverless

upvoted 3 times

A company experienced a breach that affected several applications in its on-premises data center. The attacker took advantage of vulnerabilities in the custom applications that were running on the servers. The company is now migrating its applications to run on Amazon EC2 instances. The company wants to implement a solution that actively scans for vulnerabilities on the EC2 instances and sends a report that details the findings.

Which solution will meet these requirements?

- A. Deploy AWS Shield to scan the EC2 instances for vulnerabilities. Create an AWS Lambda function to log any findings to AWS CloudTrail.
- B. Deploy Amazon Macie and AWS Lambda functions to scan the EC2 instances for vulnerabilities. Log any findings to AWS CloudTrail.
- C. Turn on Amazon GuardDuty. Deploy the GuardDuty agents to the EC2 instances. Configure an AWS Lambda function to automate the generation and distribution of reports that detail the findings.
- D. Turn on Amazon Inspector. Deploy the Amazon Inspector agent to the EC2 instances. Configure an AWS Lambda function to automate the generation and distribution of reports that detail the findings. **Most Voted**

Correct Answer: D

Community vote distribution

D (98%)

2

Comments

siyam008 **Highly Voted** 2 years, 9 months ago

Selected Answer: D

AWS Shield for DDOS
Amazon Macie for discover and protect sensitive data
Amazon GuardDuty for intelligent threat discovery to protect AWS account
Amazon Inspector for automated security assessment. like known Vulnerability
upvoted 62 times

robotgeek 1 year, 4 months ago

So 58 upvotes and you are talking about "known Vulnerability" for a "custom app"... ok
upvoted 1 times

benacert **Highly Voted** 2 years ago

Whenever I feel vulnerable, I use AWS Inspector..
upvoted 18 times

zinabu **Most Recent** 1 year, 7 months ago

Selected Answer: D

Amazon Inspector for automated security assessment. like known Vulnerability
upvoted 4 times

Ruffyt 2 years ago

AWS Shield for DDOS
Amazon Macie for discover and protect sensitive data
Amazon GuardDuty for intelligent threat discovery to protect AWS account
Amazon Inspector for automated security assessment. like known Vulnerability
upvoted 5 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: D

vulnerabilities = Amazon Inspector
malicious activity = Amazon GuardDuty

upvoted 11 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

Enable Amazon Inspector
Deploy Inspector agents to EC2 instances
Use Lambda to generate and distribute vulnerability reports
The key points:

Migrate on-prem apps with vulnerabilities to EC2
Need active scanning of EC2 instances for vulnerabilities
Require reports on findings

upvoted 4 times

kruasan 2 years, 7 months ago

Selected Answer: D

Amazon Inspector:

- Performs active vulnerability scans of EC2 instances. It looks for software vulnerabilities, unintended network accessibility, and other security issues.
- Requires installing an agent on EC2 instances to perform scans. The agent must be deployed to each instance.
- Provides scheduled scan reports detailing any findings of security risks or vulnerabilities. These reports can be used to patch or remediate issues.
- Is best suited for proactively detecting security weaknesses and misconfigurations in your AWS environment.

upvoted 5 times

kruasan 2 years, 7 months ago

Amazon GuardDuty:

- Monitors for malicious activity like unusual API calls, unauthorized infrastructure deployments, or compromised EC2 instances. It uses machine learning and behavioral analysis of logs.
- Does not require installing any agents. It relies on analyzing AWS CloudTrail, VPC Flow Logs, and DNS logs.
- Alerts you to any detected threats, suspicious activity or policy violations in your AWS accounts. These alerts warrant investigation but may not always require remediation.
- Is focused on detecting active threats, unauthorized behavior, and signs of a compromise in your AWS environment.
- Can also detect some vulnerabilities and misconfigurations but coverage is not as broad as a dedicated service like Inspector.

upvoted 7 times

datz 2 years, 8 months ago

Selected Answer: D

Amazon Inspector is a vulnerability scanning tool that you can use to identify potential security issues within your EC2 instances.

It is a kind of automated security assessment service that checks the network exposure of your EC2 or latest security state for applications running into your EC2 instance. It has ability to auto discover your AWS workload and continuously scan for the open loophole or vulnerability.

upvoted 2 times

shanwford 2 years, 8 months ago

Selected Answer: D

Amazon Inspector is a vulnerability scanning tool that you can use to identify potential security issues within your EC2 instances. Guard Duty continuously monitors your entire AWS account via Cloud Trail, Flow Logs, DNS Logs as Input.

upvoted 2 times

GalileoEC2 2 years, 8 months ago

Selected Answer: C

:) C is the correct

<https://cloudkatha.com/amazon-guardduty-vs-inspector-which-one-should-you-use/>

upvoted 1 times

jayantp04 1 year, 11 months ago

Document itself saying that
Amazon Inspector is a vulnerability scanning tool
hence correct Answer is D

upvoted 2 times

MssP 2 years, 8 months ago

Please, read the link you sent: Amazon Inspector is a vulnerability scanning tool that you can use to identify potential security issues within your EC2 instances. GuardDuty is very critical part to identify threats, based on that findings you can setup automated preventive actions or remediation's. So Answer is D.

upvoted 2 times

GalileoEC2 2 years, 8 months ago

Selected Answer: C

<https://cloudkatha.com/amazon-guardduty-vs-inspector-which-one-should-you-use/>

upvoted 1 times

LuckyAro 2 years, 9 months ago

Selected Answer: D

Amazon Inspector is a security assessment service that helps to identify security vulnerabilities and compliance issues in applications deployed on Amazon EC2 instances. It can be used to assess the security of applications that are deployed on Amazon EC2 instances, including those that are custom-built.

To use Amazon Inspector, the Amazon Inspector agent must be installed on the EC2 instances that need to be assessed. The agent collects data about the instances and sends it to Amazon Inspector for analysis. Amazon Inspector then generates a report that details any security vulnerabilities that were found and provides guidance on how to remediate them.

By configuring an AWS Lambda function, the company can automate the generation and distribution of reports that detail the findings. This means that reports can be generated and distributed as soon as vulnerabilities are detected, allowing the company to take action quickly.

upvoted 2 times

pbpally 2 years, 9 months ago

Selected Answer: D

I'm a little confused on how someone came up with C, it is definitely D.

upvoted 2 times

obatunde 2 years, 9 months ago

Selected Answer: D

Amazon Inspector

upvoted 3 times

obatunde 2 years, 9 months ago

Amazon Inspector is an automated vulnerability management service that continually scans AWS workloads for software vulnerabilities and unintended network exposure. <https://aws.amazon.com/inspector/features/?nc=sn&loc=2>

upvoted 4 times

Palanda 2 years, 9 months ago

Selected Answer: D

I think D

upvoted 1 times

minglu 2 years, 9 months ago

Selected Answer: D

Inspector for EC2

upvoted 1 times

skiwili 2 years, 9 months ago

Selected Answer: D

Ddddddd

upvoted 1 times

A company uses an Amazon EC2 instance to run a script to poll for and process messages in an Amazon Simple Queue Service (Amazon SQS) queue. The company wants to reduce operational costs while maintaining its ability to process a growing number of messages that are added to the queue.

What should a solutions architect recommend to meet these requirements?

- A. Increase the size of the EC2 instance to process messages faster.
- B. Use Amazon EventBridge to turn off the EC2 instance when the instance is underutilized.
- C. Migrate the script on the EC2 instance to an AWS Lambda function with the appropriate runtime. **Most Voted**
- D. Use AWS Systems Manager Run Command to run the script on demand.

Correct Answer: C

Community vote distribution

C (91%)

9%

Comments

kpato87 **Highly Voted** 3 years, 3 months ago

Selected Answer: C

By migrating the script to AWS Lambda, the company can take advantage of the auto-scaling feature of the service. AWS Lambda will automatically scale resources to match the size of the workload. This means that the company will not have to worry about provisioning or managing instances as the number of messages increases, resulting in lower operational costs

upvoted 13 times

Guru4Cloud **Highly Voted** 2 years, 9 months ago

Selected Answer: C

The key points are:

Currently using an EC2 instance to poll SQS and process messages

Want to reduce costs while handling growing message volume

By migrating the polling script to a Lambda function, the company can avoid the cost of running a dedicated EC2 instance. Lambda functions scale automatically to handle message spikes. And Lambda billing is based on actual usage, resulting in cost savings versus provisioned EC2 capacity.

upvoted 8 times

TariqKipkemei **Most Recent** 2 years, 8 months ago

Selected Answer: C

reduce operational costs = serverless = Lambda functions

upvoted 2 times

Steve_4542636 3 years, 3 months ago

Selected Answer: C

Lambda costs money only when it's processing, not when idle

upvoted 3 times

ManOnTheMoon 3 years, 3 months ago

Agree with C

upvoted 1 times

hasport 3 years, 3 months ago

the answer is C. With this option, you can reduce operational cost as the question mentioned

upvoted 1 times

LuckyAro 3 years, 3 months ago

Selected Answer: C

AWS Lambda is a serverless compute service that allows you to run your code without provisioning or managing servers. By migrating the script to an AWS Lambda function, you can eliminate the need to maintain an EC2 instance, reducing operational costs. Additionally, Lambda automatically scales to handle the increasing number of messages in the SQS queue.

upvoted 1 times

zTopic 3 years, 3 months ago

Selected Answer: C

It Should be C.

Lambda allows you to execute code without provisioning or managing servers, so it is ideal for running scripts that poll for and process messages in an Amazon SQS queue. The scaling of the Lambda function is automatic, and you only pay for the actual time it takes to process the messages.

upvoted 3 times

Bhawesh 3 years, 3 months ago

Selected Answer: D

To reduce the operational overhead, it should be:

D. Use AWS Systems Manager Run Command to run the script on demand.

upvoted 3 times

lucdt4 3 years ago

No, replace EC2 instead by using lambda to reduce costs

upvoted 1 times

pentium75 2 years, 5 months ago

So every time an item is added to the queue, you log into AWS Systems Manager through your browser, select "Run Command" and select your instance and enter the command to run the script?

upvoted 4 times

ike001 1 year, 11 months ago

very sarcastic question :)

upvoted 1 times

A company uses a legacy application to produce data in CSV format. The legacy application stores the output data in Amazon S3. The company is deploying a new commercial off-the-shelf (COTS) application that can perform complex SQL queries to analyze data that is stored in Amazon Redshift and Amazon S3 only. However, the COTS application cannot process the .csv files that the legacy application produces.

The company cannot update the legacy application to produce data in another format. The company needs to implement a solution so that the COTS application can use the data that the legacy application produces.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create an AWS Glue extract, transform, and load (ETL) job that runs on a schedule. Configure the ETL job to process the .csv files and store the processed data in Amazon Redshift. **Most Voted**
- B. Develop a Python script that runs on Amazon EC2 instances to convert the .csv files to .sql files. Invoke the Python script on a cron schedule to store the output files in Amazon S3.
- C. Create an AWS Lambda function and an Amazon DynamoDB table. Use an S3 event to invoke the Lambda function. Configure the Lambda function to perform an extract, transform, and load (ETL) job to process the .csv files and store the processed data in the DynamoDB table.
- D. Use Amazon EventBridge to launch an Amazon EMR cluster on a weekly schedule. Configure the EMR cluster to perform an extract, transform, and load (ETL) job to process the .csv files and store the processed data in an Amazon Redshift table.

Correct Answer: A

Community vote distribution

A (98%)

2

Comments

awsgeek75 **Highly Voted** 1 year, 10 months ago

Selected Answer: A

Time to sell some Glue.

I believe these kind of questions are there to indoctrinate us into acknowledging how blessed we are to have managed services like AWS Glue when you look at other horrible and painful options

upvoted 23 times

elearningtakai **Highly Voted** 2 years, 8 months ago

Selected Answer: A

A, AWS Glue is a fully managed ETL service that can extract data from various sources, transform it into the required format, and load it into a target data store. In this case, the ETL job can be configured to read the CSV files from Amazon S3, transform the data into a format that can be loaded into Amazon Redshift, and load it into an Amazon Redshift table.

B requires the development of a custom script to convert the CSV files to SQL files, which could be time-consuming and introduce additional operational overhead. C, while using serverless technology, requires the additional use of DynamoDB to store the processed data, which may not be necessary if the data is only needed in Amazon Redshift. D, while an option, is not the most efficient solution as it requires the creation of an EMR cluster, which can be costly and complex to manage.

upvoted 8 times

pentium75 **Most Recent** 1 year, 11 months ago

Selected Answer: A

B - Developing a script is surely not minimizing operational effort

C - Stores data in DynamoDB where the new app cannot use it

D - Could work but is total overkill (EMR is for Big Data analysis, not for simple ETL)

upvoted 4 times

Ruffyt 2 years ago

A-ETL is serverless & best suited with the requirement who primary job is ETL
B-Usage of Ec2 adds operational overhead & incur costs
C-DynamoDB(NoSql) does suit the requirement as company is performing SQL queries
D-EMR adds operational overhead & incur costs
upvoted 3 times

ACloud_Guru15 2 years, 1 month ago

Selected Answer: A

A-ETL is serverless & best suited with the requirement who primary job is ETL
B-Usage of Ec2 adds operational overhead & incur costs
C-DynamoDB(NoSql) does suit the requirement as company is performing SQL queries
D-EMR adds operational overhead & incur costs
upvoted 2 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: A

Data transformation = AWS Glue
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

Create an AWS Glue ETL job to process the CSV files
Configure the job to run on a schedule
Output the transformed data to Amazon Redshift
The key points:

Legacy app generates CSV files in S3
New app requires data in Redshift or S3
Need to transform CSV to support new app with minimal ops overhead
upvoted 2 times

kraken21 2 years, 8 months ago

Selected Answer: A

Glue is server less and has less operational head than EMR so A.
upvoted 2 times

[Removed] 2 years, 8 months ago

Selected Answer: C

o meet the requirement with the least operational overhead, a serverless approach should be used. Among the options provided, option C provides a serverless solution using AWS Lambda, S3, and DynamoDB. Therefore, the solution should be to create an AWS Lambda function and an Amazon DynamoDB table. Use an S3 event to invoke the Lambda function. Configure the Lambda function to perform an extract, transform, and load (ETL) job to process the .csv files and store the processed data in the DynamoDB table.
Option A is also a valid solution, but it may involve more operational overhead than Option C. With Option A, you would need to set up and manage an AWS Glue job, which would require more setup time than creating an AWS Lambda function. Additionally, AWS Glue jobs have a minimum execution time of 10 minutes, which may not be necessary or desirable for this use case. However, if the data processing is particularly complex or requires a lot of data transformation, AWS Glue may be a more appropriate solution.
upvoted 1 times

MssP 2 years, 8 months ago

Important point: The COTS performs complex SQL queries to analyze data in Amazon Redshift. If you use DynamoDB -> No SQL queries. Option A makes more sense.
upvoted 4 times

pentium75 1 year, 11 months ago

Creating and maintaining a Lambda function is more "operational overhead" than using a ready-made service such as Glue. But more important, answer C says "store the processed data in the DynamoDB table" while the application can "analyze data that is stored in Amazon Redshift and Amazon S3 only".
upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: A

A would be the best solution as it involves the least operational overhead. With this solution, an AWS Glue ETL job is created to process the .csv files and store the processed data directly in Amazon Redshift. This is a serverless approach that does not require any infrastructure to be provisioned, configured, or maintained. AWS Glue provides a fully managed, pay-as-you-go ETL service that can be easily configured to process data from S3 and load it into Amazon Redshift. This approach allows the legacy application to continue to produce data in the CSV format that it currently uses, while providing the new COTS application with the ability to analyze the data using complex SQL queries.
upvoted 4 times

jennyka76 2 years, 9 months ago

A

<https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-etl-format-csv-home.html>

I AGREE AFTER READING LINK

upvoted 2 times

cloudbusting 2 years, 9 months ago

A: <https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-etl-format.html>

upvoted 2 times

A company recently migrated its entire IT environment to the AWS Cloud. The company discovers that users are provisioning oversized Amazon EC2 instances and modifying security group rules without using the appropriate change control process. A solutions architect must devise a strategy to track and audit these inventory and configuration changes.

Which actions should the solutions architect take to meet these requirements? (Choose two.)

- A. Enable AWS CloudTrail and use it for auditing. **Most Voted**
- B. Use data lifecycle policies for the Amazon EC2 instances.
- C. Enable AWS Trusted Advisor and reference the security dashboard.
- D. Enable AWS Config and create rules for auditing and compliance purposes. **Most Voted**
- E. Restore previous resource configurations with an AWS CloudFormation template.

Correct Answer: AD

Community vote distribution

AD (92%)

8%

Comments

LuckyAro **Highly Voted** 2 years, 9 months ago

Selected Answer: AD

A. Enable AWS CloudTrail and use it for auditing. CloudTrail provides event history of your AWS account activity, including actions taken through the AWS Management Console, AWS Command Line Interface (CLI), and AWS SDKs and APIs. By enabling CloudTrail, the company can track user activity and changes to AWS resources, and monitor compliance with internal policies and external regulations.

D. Enable AWS Config and create rules for auditing and compliance purposes. AWS Config provides a detailed inventory of the AWS resources in your account, and continuously records changes to the configurations of those resources. By creating rules in AWS Config, the company can automate the evaluation of resource configurations against desired state, and receive alerts when configurations drift from compliance.

Options B, C, and E are not directly relevant to the requirement of tracking and auditing inventory and configuration changes.

upvoted 15 times

zdi561 **Most Recent** 1 year, 4 months ago

Selected Answer: CD

Amazon Web Services (AWS) CloudTrail is enabled by default for all AWS accounts.

upvoted 1 times

Ruffyyt 2 years ago

A. Enable AWS CloudTrail and use it for auditing. CloudTrail provides event history of your AWS account activity, including actions taken through the AWS Management Console, AWS Command Line Interface (CLI), and AWS SDKs and APIs. By enabling CloudTrail, the company can track user activity and changes to AWS resources, and monitor compliance with internal policies and external regulations.

D. Enable AWS Config and create rules for auditing and compliance purposes. AWS Config provides a detailed inventory of the AWS resources in your account, and continuously records changes to the configurations of those resources. By creating rules in AWS Config, the company can automate the evaluation of resource configurations against desired state, and receive alerts when configurations drift from compliance.

Options B, C, and E are not directly relevant to the requirement of tracking and auditing inventory and configuration changes.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: AD

A. Enable AWS CloudTrail and use it for auditing. CloudTrail provides event history of your AWS account activity, including actions taken through the AWS Management Console, AWS Command Line Interface (CLI), and AWS SDKs and APIs. By enabling CloudTrail, the company can track user activity and changes to AWS resources, and monitor compliance with internal policies and external regulations.

D. Enable AWS Config and create rules for auditing and compliance purposes. AWS Config provides a detailed inventory of the AWS resources in your

account, and continuously records changes to the configurations of those resources. By creating rules in AWS Config, the company can automate the evaluation of resource configurations against desired state, and receive alerts when configurations drift from compliance.

upvoted 2 times

mrsoa 2 years, 4 months ago

Selected Answer: CD

I am gonna go with CD

AWS Cloudtrail is already enabled so no need to enable it and for the auditing we are gonna use AWS config Answer D

C because Trusted advisor checks the security groups

upvoted 1 times

awsgeek75 1 year, 10 months ago

CloudTrail is not enabled by default or in the question scenario. Even if it was, Trusted Advisor would just give you recommendations and usage reports. It won't audit anything for you

upvoted 1 times

pentium75 1 year, 11 months ago

"AWS CloudTrail is already enabled" says who?

upvoted 3 times

kruasan 2 years, 7 months ago

Selected Answer: AD

A) Enable AWS CloudTrail and use it for auditing.

AWS CloudTrail provides a record of API calls and can be used to audit changes made to EC2 instances and security groups. By analyzing CloudTrail logs, the solutions architect can track who provisioned oversized instances or modified security groups without proper approval.

D) Enable AWS Config and create rules for auditing and compliance purposes.

AWS Config can record the configuration changes made to resources like EC2 instances and security groups. The solutions architect can create AWS Config rules to monitor for non-compliant changes, like launching certain instance types or opening security group ports without permission. AWS Config would alert on any violations of these rules.

upvoted 3 times

kruasan 2 years, 7 months ago

The other options would not fully meet the auditing and change tracking requirements:

B) Data lifecycle policies control when EC2 instances are backed up or deleted but do not audit configuration changes.

C) AWS Trusted Advisor security checks may detect some compliance violations after the fact but do not comprehensively log changes like AWS CloudTrail and AWS Config do.

E) CloudFormation templates enable rollback but do not provide an audit trail of changes. The solutions architect would not know who made unauthorized modifications in the first place.

upvoted 3 times

skiwili 2 years, 9 months ago

Selected Answer: AD

Yes A and D

upvoted 2 times

jennyka76 2 years, 9 months ago

AGREE WITH ANSWER - A & D

CloudTrail and Config

upvoted 2 times

Neha999 2 years, 9 months ago

CloudTrail and Config

upvoted 3 times

A company has hundreds of Amazon EC2 Linux-based instances in the AWS Cloud. Systems administrators have used shared SSH keys to manage the instances. After a recent audit, the company's security team is mandating the removal of all shared keys. A solutions architect must design a solution that provides secure access to the EC2 instances.

Which solution will meet this requirement with the LEAST amount of administrative overhead?

- A. Use AWS Systems Manager Session Manager to connect to the EC2 instances. **Most Voted**
- B. Use AWS Security Token Service (AWS STS) to generate one-time SSH keys on demand.
- C. Allow shared SSH access to a set of bastion instances. Configure all other instances to allow only SSH access from the bastion instances.
- D. Use an Amazon Cognito custom authorizer to authenticate users. Invoke an AWS Lambda function to generate a temporary SSH key.

Correct Answer: A

Community vote distribution

A (85%)

Other

Comments

Vlad **Highly Voted** 2 years, 9 months ago

Answer is A

Using AWS Systems Manager Session Manager to connect to the EC2 instances is a secure option as it eliminates the need for inbound SSH ports and removes the requirement to manage SSH keys manually. It also provides a complete audit trail of user activity. This solution requires no additional software to be installed on the EC2 instances.

upvoted 13 times

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: A

A - Systems Manager Session Manager has EXACTLY that purpose, 'providing secure access to EC2 instances'

B - STS can generate temporary IAM credentials or access keys but NOT SSH keys

C - Does not 'remove all shared keys' as requested

D - Cognito is not meant for internal users, and whole setup is complex

upvoted 7 times

pentium75 **Most Recent** 1 year, 11 months ago

Selected Answer: A

B - Querying is just a feature of Redshift but primarily it's a Data Warehouse - the question says nothing that historical data would have to be stored or accessed or analyzed

upvoted 2 times

Ruffyt 2 years ago

The key reasons why:

STS can generate short-lived credentials that provide temporary access to the EC2 instances for administering them.

The credentials can be generated on-demand each time access is needed, eliminating the risks of using permanent shared SSH keys.

No infrastructure like bastion hosts needs to be maintained.

The on-premises administrators can use the familiar SSH tools with the temporary keys.

upvoted 2 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: A

Session Manager provides secure and auditable node management without the need to open inbound ports, maintain bastion hosts, or manage SSH keys.

upvoted 2 times

Guru4Cloud 2 years, 2 months ago

Selected Answer: B

The key reasons why:

STS can generate short-lived credentials that provide temporary access to the EC2 instances for administering them. The credentials can be generated on-demand each time access is needed, eliminating the risks of using permanent shared SSH keys. No infrastructure like bastion hosts needs to be maintained. The on-premises administrators can use the familiar SSH tools with the temporary keys.

upvoted 1 times

pentium75 1 year, 11 months ago

STS provides temporary IAM credentials, not SSH keys

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

Using AWS Security Token Service (AWS STS) to generate one-time SSH keys on demand is a secure and efficient way to provide access to the EC2 instances without the need for shared SSH keys. STS is a fully managed service that can be used to generate temporary security credentials, allowing systems administrators to connect to the EC2 instances without having to share SSH keys. The temporary credentials can be generated on demand, reducing the administrative overhead associated with managing SSH access

upvoted 2 times

ofinto 2 years, 2 months ago

Can you please provide documentation about generating a one-time SSH with STS?

upvoted 2 times

kruasan 2 years, 7 months ago

Selected Answer: A

AWS Systems Manager Session Manager provides secure shell access to EC2 instances without the need for SSH keys. It meets the security requirement to remove shared SSH keys while minimizing administrative overhead.

upvoted 2 times

Guru4Cloud 2 years, 2 months ago

If the systems administrators need to access the EC2 instances from an on-premises environment, using Session Manager may not be the ideal solution.

upvoted 1 times

kruasan 2 years, 7 months ago

Session Manager is a fully managed AWS Systems Manager capability. With Session Manager, you can manage your Amazon Elastic Compute Cloud (Amazon EC2) instances, edge devices, on-premises servers, and virtual machines (VMs). You can use either an interactive one-click browser-based shell or the AWS Command Line Interface (AWS CLI). Session Manager provides secure and auditable node management without the need to open inbound ports, maintain bastion hosts, or manage SSH keys. Session Manager also allows you to comply with corporate policies that require controlled access to managed nodes, strict security practices, and fully auditable logs with node access details, while providing end users with simple one-click cross-platform access to your managed nodes.

<https://docs.aws.amazon.com/systems-manager/latest/userguide/session-manager.html>

upvoted 3 times

kruasan 2 years, 7 months ago

Who should use Session Manager?

Any AWS customer who wants to improve their security and audit posture, reduce operational overhead by centralizing access control on managed nodes, and reduce inbound node access.

Information Security experts who want to monitor and track managed node access and activity, close down inbound ports on managed nodes, or allow connections to managed nodes that don't have a public IP address.

Administrators who want to grant and revoke access from a single location, and who want to provide one solution to users for Linux, macOS, and Windows Server managed nodes.

Users who want to connect to a managed node with just one click from the browser or AWS CLI without having to provide SSH keys.

upvoted 3 times

Stanislav4907 2 years, 9 months ago

Selected Answer: C

You guys seriously don't want to go to SMSM for every single EC2. You have to create solution not used services for one time access. Bastion will give you option to manage 1000s EC2 machines from 1. Plus you can use Ansible from it.

upvoted 2 times

Zox42 2 years, 8 months ago

Question: "the company's security team is mandating the removal of all shared keys", answer C can't be right because it says: "Allow shared SSH access to a set of bastion instances".

upvoted 7 times

UnluckyDucky 2 years, 8 months ago

Session Manager is the best practice and recommended way by Amazon to manage your instances. Bastion hosts require remote access therefore exposing them to the internet.

The most secure way is definitely session manager therefore answer A is correct imho.

upvoted 4 times

Steve_4542636 2 years, 9 months ago

Selected Answer: A

I vote a

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: A

AWS Systems Manager Session Manager provides secure and auditable instance management without the need for any inbound connections or open ports. It allows you to manage your instances through an interactive one-click browser-based shell or through the AWS CLI. This means that you don't have to manage any SSH keys, and you don't have to worry about securing access to your instances as access is controlled through IAM policies.

upvoted 5 times

bdp123 2 years, 9 months ago

Selected Answer: A

<https://docs.aws.amazon.com/systems-manager/latest/userguide/session-manager.html>

upvoted 3 times

jahmad0730 2 years, 9 months ago

Selected Answer: A

Answer must be A

upvoted 3 times

jennyka76 2 years, 9 months ago

ANSWER - A

AWS SESSION MANAGER IS CORRECT LEAST EFFORTS TO ACCESS LINUX SYSTEM IN AWS CONSOLE AND YOU ARE ALREADY LOGIN TO AWS. SO NO NEED FOR THE TOKEN OR OTHER STUFF DONE IN THE BACKGROUND BY AWS. MAKES SENSE.

upvoted 3 times

cloudbusting 2 years, 9 months ago

Answer is A

upvoted 4 times

zTopic 2 years, 9 months ago

Selected Answer: A

Answer is A

upvoted 2 times

A company is using a fleet of Amazon EC2 instances to ingest data from on-premises data sources. The data is in JSON format and ingestion rates can be as high as 1 MB/s. When an EC2 instance is rebooted, the data in-flight is lost. The company's data science team wants to query ingested data in near-real time.

Which solution provides near-real-time data querying that is scalable with minimal data loss?

- A. Publish data to Amazon Kinesis Data Streams, Use Kinesis Data Analytics to query the data. **Most Voted**
- B. Publish data to Amazon Kinesis Data Firehose with Amazon Redshift as the destination. Use Amazon Redshift to query the data.
- C. Store ingested data in an EC2 instance store. Publish data to Amazon Kinesis Data Firehose with Amazon S3 as the destination. Use Amazon Athena to query the data.
- D. Store ingested data in an Amazon Elastic Block Store (Amazon EBS) volume. Publish data to Amazon ElastiCache for Redis. Subscribe to the Redis channel to query the data.

Correct Answer: A

Community vote distribution

A (71%)

B (29%)

Comments

LuckyAro **Highly Voted** 3 years, 3 months ago

Selected Answer: A

A: is the solution for the company's requirements. Publishing data to Amazon Kinesis Data Streams can support ingestion rates as high as 1 MB/s and provide real-time data processing. Kinesis Data Analytics can query the ingested data in real-time with low latency, and the solution can scale as needed to accommodate increases in ingestion rates or querying needs. This solution also ensures minimal data loss in the event of an EC2 instance reboot since Kinesis Data Streams has a persistent data store for up to 7 days by default.

upvoted 15 times

bogobob **Highly Voted** 2 years, 6 months ago

Selected Answer: B

The fact they specifically mention "near real-time" twice tells me the correct answer is KDF. On top of which its easier to setup and maintain. KDS is really only needed if you need real-time. Also using redshift will mean permanent data retention. The data in A could be lost after a year. Redshift queries are slow but you're still querying near real-time data

upvoted 8 times

Ernestokoro 2 years, 6 months ago

You are very correct. see supporting link <https://jayendrapatil.com/aws-kinesis-data-streams-vs-kinesis-firehose/#:~:text=vs%20Kine...-,Purpose,into%20AWS%20products%20for%20processing.>

upvoted 2 times

Kp002 **Most Recent** 1 year ago

Selected Answer: A

Publish data to Amazon Kinesis Data Streams. Use Kinesis Data Analytics to query the data.

- Kinesis Data Streams is designed for real-time data ingestion and can handle high throughput (1 MB/s is well within its capabilities).
- Data durability: Kinesis Data Streams stores data for up to 7 days (default is 24 hours), which helps prevent data loss during EC2 instance reboots.
- Kinesis Data Analytics allows you to run SQL queries on streaming data in near real-time, which satisfies the data science team's requirement.
- Scalable: Kinesis is designed to automatically scale with data volume, making it highly suitable for fluctuating ingestion

upvoted 1 times

Dantecito 1 year, 4 months ago

Selected Answer: A